

6-2

1.

t (hours)	4	7	12	15
$R(t)$ (liters/hour)	6.5	6.2	5.9	5.6

A tank contains 50 liters of oil at time $t = 4$ hours. Oil is being pumped into the tank at a rate $R(t)$ where $R(t)$ is measured in liters per hour, and t is measured in hours. Selected values of $R(t)$ are given in the table above. Using a right Riemann sum with three subintervals and data from the table, what is the approximation of the number of liters of oil that are in the tank at time $t = 15$ hours?

- (A) 64.9
(B) 68.2
(C) 114.9
(D) 116.6
(E) 118.2



6-2

2. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

x	0	1	2	3	4	5	6
$f'(x)$	4	3.5	2	0.8	1.7	5.8	7

The function f satisfies $f(0) = 20$. The first derivative of f satisfies the inequality $0 \leq f'(x) \leq 7$ for all x in the closed interval $[0, 6]$. Selected values of f' are shown in the table above. The function f has a continuous second derivative for all real numbers.

(a) Use a midpoint Riemann sum with three subintervals of equal length indicated by the data in the table to approximate the value of $f(6)$.

(b) Determine whether the actual value of $f(6)$ could be 70. Explain your reasoning.

(c) Evaluate $\int_2^4 f''(x) dx$.

(d) Find $\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10}$.

Part A

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_0^6 f'(x) dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x) dx$$

$$f(6) = f(0) + \int_0^6 f'(x) dx \approx 20 + 20.2 = 40.2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_0^6 f'(x) dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x) dx$$

$$f(6) = f(0) + \int_0^6 f'(x) dx \approx 20 + 20.2 = 40.2$$

6-2

Part B

1 point is earned for: integral bound

1 point is earned for: answer with reasoning

$$\text{Since } f'(x) \leq 7, \int_0^6 f'(x) dx \leq 6 \cdot 7 = 42.$$

$$f(6) - f(0) \leq 42 \Rightarrow f(6) \leq 20 + 42 = 62$$

Therefore, the actual value of $f(6)$ could not be 70.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: integral bound

1 point is earned for: answer with reasoning

$$\text{Since } f'(x) \leq 7, \int_0^6 f'(x) dx \leq 6 \cdot 7 = 42.$$

$$f(6) - f(0) \leq 42 \Rightarrow f(6) \leq 20 + 42 = 62$$

Therefore, the actual value of $f(6)$ could not be 70.

Part C

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_2^4 f''(x) dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_2^4 f''(x) dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$

Part D

6-2

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow 0} (f(x) - 20e^x) = 0$$

$$\lim_{x \rightarrow 0} (0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10} = \lim_{x \rightarrow 0} \frac{f'(x) - 20e^x}{0.5f'(x)} = \frac{4 - 20}{0.5(4)} = -8$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow 0} (f(x) - 20e^x) = 0$$

$$\lim_{x \rightarrow 0} (0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10} = \lim_{x \rightarrow 0} \frac{f'(x) - 20e^x}{0.5f'(x)} = \frac{4 - 20}{0.5(4)} = -8$$

6-2

3. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

x	0	1	2	3	4	5	6
$f'(x)$	4	3.5	2	0.8	1.7	5.8	7

The function f satisfies $f(0) = 20$. The first derivative of f satisfies the inequality $0 \leq f'(x) \leq 7$ for all x in the closed interval $[0, 6]$. Selected values of f' are shown in the table above. The function f has a continuous second derivative for all real numbers.

(a) Use a midpoint Riemann sum with three subintervals of equal length indicated by the data in the table to approximate the value of $f(6)$.

(b) Determine whether the actual value of $f(6)$ could be 70. Explain your reasoning.

(c) Evaluate $\int_2^4 f''(x) dx$.

(d) Find $\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10}$.

Part A

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_0^6 f'(x) dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x) dx$$

$$f(6) = f(0) + \int_0^6 f'(x) dx \approx 20 + 20.2 = 40.2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_0^6 f'(x) dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x) dx$$

$$f(6) = f(0) + \int_0^6 f'(x) dx \approx 20 + 20.2 = 40.2$$

6-2

Part B

1 point is earned for: integral bound

1 point is earned for: answer with reasoning

$$\text{Since } f'(x) \leq 7, \int_0^6 f'(x) dx \leq 6 \cdot 7 = 42.$$

$$f(6) - f(0) \leq 42 \Rightarrow f(6) \leq 20 + 42 = 62$$

Therefore, the actual value of $f(6)$ could not be 70.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: integral bound

1 point is earned for: answer with reasoning

$$\text{Since } f'(x) \leq 7, \int_0^6 f'(x) dx \leq 6 \cdot 7 = 42.$$

$$f(6) - f(0) \leq 42 \Rightarrow f(6) \leq 20 + 42 = 62$$

Therefore, the actual value of $f(6)$ could not be 70.

Part C

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_2^4 f''(x) dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_2^4 f''(x) dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$

Part D

6-2

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow 0} (f(x) - 20e^x) = 0$$

$$\lim_{x \rightarrow 0} (0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10} = \lim_{x \rightarrow 0} \frac{f'(x) - 20e^x}{0.5f'(x)} = \frac{4 - 20}{0.5(4)} = -8$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow 0} (f(x) - 20e^x) = 0$$

$$\lim_{x \rightarrow 0} (0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10} = \lim_{x \rightarrow 0} \frac{f'(x) - 20e^x}{0.5f'(x)} = \frac{4 - 20}{0.5(4)} = -8$$

6-2

4.

x	1	1.1	1.2	1.3	1.4
$f'(x)$	8	10	12	13	14.5

The function f is twice differentiable for $x > 0$ with $f(1) = 15$ and $f'(1) = 20$. Values of f' , the derivative of f , are given for selected values of x in the table above.

Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_1^{1.4} f'(x) dx$

. Use the approximation for $\int_1^{1.4} f'(x) dx$ to estimate the value of $f(1.4)$. Show the computations that lead to your answer.

Part B

One point is earned for the midpoint Riemann sum

One point is earned for the Fundamental Theorem of Calculus

One point is earned for the answer

$$\int_1^{1.4} f'(x) dx \approx (0.2)(10) + (0.2)(13) = 4.6$$

$$f(1.4) = f(1) + \int_1^{1.4} f'(x) dx$$

$$f(1.4) \approx 15 + 4.6 = 19.6$$



0	1	2	3
---	---	---	---

The student earns all of the following points:

One point is earned for the midpoint Riemann sum

One point is earned for the Fundamental Theorem of Calculus

One point is earned for the answer

$$\int_1^{1.4} f'(x) dx \approx (0.2)(10) + (0.2)(13) = 4.6$$

$$f(1.4) = f(1) + \int_1^{1.4} f'(x) dx$$

$$f(1.4) \approx 15 + 4.6 = 19.6$$

6-2

5.

Time (weeks)	0	2	6	10
Level	210	200	190	180

The table above gives the level of a person's cholesterol at different times during a 10-week treatment period.

What is the average level over this 10-week period obtained by using a trapezoidal approximation with the subintervals $[0, 2]$, $[2, 6]$, and $[6, 10]$?

(A) 188

(B) 193



(C) 195

(D) 198

6-2

6.



t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

(c) For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

(d) For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

Part A

1 point is earned for estimation

1 point is earned for interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for estimation

1 point is earned for interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

Part B

6-2

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

Part C

1 point is earned for left Riemann sum

1 point is earned for approximation

1 point is earned for underestimating with reason

$$\begin{aligned} \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\ &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\ &= \frac{1}{20} \cdot 1215.8 = 60.79 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing



0	1	2	3
---	---	---	---

The student earns all of the following points:

1 point is earned for left Riemann sum

1 point is earned for approximation

6-2

1 point is earned for underestimating with reason

$$\begin{aligned}\frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20}(4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\ &= \frac{1}{20}(4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\ &= \frac{1}{20} \cdot 1215.8 = 60.79\end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing

Part D

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\ &= 71.0 + 2.043155 = 73.043\end{aligned}$$



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\ &= 71.0 + 2.043155 = 73.043\end{aligned}$$

6-2

7.

t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \leq t \leq 2.4$, as shown in the table above. For $0 < t \leq 2.4$, $v(t) > 0$.

 Using correct units, interpret the meaning of $\int_0^{2.4} v(t) dt$ in the context of the problem. Approximate $\int_0^{2.4} v(t) dt$ using a midpoint Riemann sum with three subintervals of equal length and values from the table.

Part B

The response can earn up to 3 points:

1 point: For the correct interpretation

$\int_0^{2.4} v(t) dt$ is the total distance Ruth, traveled, in miles, from time $t = 0$ to time $t = 2.4$ hours.

1 point: For use of the midpoint Riemann sum

1 point: For the correct approximation

$\int_0^{2.4} v(t) dt \approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8) = 36.64$ miles



0	1	2	3
---	---	---	---

The response earns all three of the following points:

1 point: For the correct interpretation

$\int_0^{2.4} v(t) dt$ is the total distance Ruth, traveled, in miles, from time $t = 0$ to time $t = 2.4$ hours.

1 point: For use of the midpoint Riemann sum

1 point: For the correct approximation

$\int_0^{2.4} v(t) dt \approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8) = 36.64$ miles

6-2

8.



t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \leq t \leq 2.4$ hours, as shown in the table above. For $0 < t \leq 2.4$, $v(t) > 0$.

(a) Use the data in the table to approximate Ruth's acceleration at time $t = 1.4$ hours. Show the computations that lead to your answer. Indicate units of measure.

(b) Using correct units, interpret the meaning of $\int_0^{2.4} v(t) dt$ in the context of the problem. Approximate $\int_0^{2.4} v(t) dt$ using a midpoint Riemann sum with three subintervals of equal length and values from the table.

(c) For $0 \leq t \leq 2.4$ hours, Ruth's velocity can be modeled by the function g given by $g(t) = \frac{24t+5\sin(6t)}{t+0.7}$. According to the model, what was Ruth's average velocity during the time interval $0 \leq t \leq 2.4$?

(d) According to the model, $g(t) = \frac{24t+5\sin(6t)}{t+0.7}$, is Ruth's speed increasing or decreasing at time $t = 1.3$? Give a reason for your answer.

Part A

The response can earn up to 2 points:

1 point is earned for correct approximation

1 point is earned for correct units

$$a(1.4) \approx \frac{v(1.6)-v(1.2)}{1.6-1.2} = \frac{16.3-17.2}{1.6-1.2} = -2.25 \text{ miles/hr}^2$$



0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for correct approximation

1 point is earned for correct units

$$a(1.4) \approx \frac{v(1.6)-v(1.2)}{1.6-1.2} = \frac{16.3-17.2}{1.6-1.2} = -2.25 \text{ miles/hr}^2$$

Part B

The response can earn up to 3 points:

1 point is earned for the correct interpretation

$\int_0^{2.4} v(t) dt$ is the total distance Ruth traveled, in miles, from time $t = 0$ to time $t = 2.4$ hours.

1 point is earned for use of the midpoint Riemann sum

6-2

1 point is earned for the correct approximation

$$\int_0^{2.4} v(t) dt \approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8) \\ = 36.64 \text{ miles}$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point is earned for the correct interpretation

$$\int_0^{2.4} v(t) dt \text{ is the total distance Ruth traveled, in miles, from time } t = 0 \text{ to time } t = 2.4 \text{ hours.}$$

1 point is earned for use of the midpoint Riemann sum

1 point is earned for the correct approximation

$$\int_0^{2.4} v(t) dt \approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8) \\ = 36.64 \text{ miles}$$

Part C

The response can earn up to 2 points:

1 point is earned for the correct integral

$$\text{Average velocity} = \frac{1}{2.4} \int_0^{2.4} g(t) dt$$

1 point is earned for the correct answer

$$\text{Average velocity} = 14.064 \text{ miles/hr}$$



0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for the correct integral

$$\text{Average velocity} = \frac{1}{2.4} \int_0^{2.4} g(t) dt$$

1 point is earned for the correct answer

$$\text{Average velocity} = 14.064 \text{ miles/hr}$$

6-2

Part D

The response can earn up to 2 points:

2 points are earned for correct conclusion with reason

$$\text{Velocity} = g(1.3) = 18.096358 > 0$$

$$\text{Acceleration} = g'(1.3) = 3.761152 > 0$$

Ruth's speed is increasing at time $t = 1.3$ since velocity and acceleration have the same sign at this time.



0	1	2
---	---	---

The response earns all of the following points:

2 points are earned for correct conclusion with reason

$$\text{Velocity} = g(1.3) = 18.096358 > 0$$

$$\text{Acceleration} = g'(1.3) = 3.761152 > 0$$

Ruth's speed is increasing at time $t = 1.3$ since velocity and acceleration have the same sign at this time.

6-2

9. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

t (minutes)	0	1	5	6	8
$g(t)$ (cubic feet per minute)	12.8	15.1	20.5	18.3	22.7

Grain is being added to a silo. At time $t = 0$, the silo is empty. The rate at which grain is being added is modeled by the differentiable function g , where $g(t)$ is measured in cubic feet per minute for $0 \leq t \leq 8$ minutes. Selected values of $g(t)$ are given in the table above.

 (a) Using the data in the table, approximate $g'(3)$. Using correct units, interpret the meaning of $g'(3)$ in the context of the problem.

(b) Write an integral expression that represents the total amount of grain added to the silo from time $t = 0$ to time $t = 8$. Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate the integral.

(c) The grain in the silo is spoiling at a rate modeled by $w(t) = 32 \cdot \sqrt{\sin\left(\frac{\pi t}{74}\right)}$, where $w(t)$ is measured in cubic feet per minute for $0 \leq t \leq 8$ minutes. Using the result from part (b), approximate the amount of unspoiled grain remaining in the silo at time $t = 8$.

(d) Based on the model in part (c), is the amount of unspoiled grain in the silo increasing or decreasing at time $t = 6$? Show the work that leads to your answer.

Part A

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5) - g(1)}{5 - 1} = \frac{20.5 - 15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5) - g(1)}{5 - 1} = \frac{20.5 - 15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.

Part B

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

6-2

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t) dt$ cubic feet.

$$\begin{aligned}\int_0^8 g(t) dt &\approx g(1) \cdot (1 - 0) + g(5) \cdot (5 - 1) + g(6) \cdot (6 - 5) + g(8) \cdot (8 - 6) \\ &= 15.1 \cdot 1 + 20.5 \cdot 4 + 18.3 \cdot 1 + 22.7 \cdot 2 = 160.8\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t) dt$ cubic feet.

$$\begin{aligned}\int_0^8 g(t) dt &\approx g(1) \cdot (1 - 0) + g(5) \cdot (5 - 1) + g(6) \cdot (6 - 5) + g(8) \cdot (8 - 6) \\ &= 15.1 \cdot 1 + 20.5 \cdot 4 + 18.3 \cdot 1 + 22.7 \cdot 2 = 160.8\end{aligned}$$

Part C

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t) dt = 99.051497$$

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t) dt = 61.749$ (or 61.748) cubic feet.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t) dt = 99.051497$$

6-2

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t) dt = 61.749$ (or 61.748) cubic feet.

Part D

1 point is earned for: considers $g(6) - w(6)$

1 point is earned for: answer

$$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$$

Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: considers $g(6) - w(6)$

1 point is earned for: answer

$$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$$

Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.

6-2

10. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

t (minutes)	0	1	5	6	8
$g(t)$ (cubic feet per minute)	12.8	15.1	20.5	18.3	22.7

Grain is being added to a silo. At time $t = 0$, the silo is empty. The rate at which grain is being added is modeled by the differentiable function g , where $g(t)$ is measured in cubic feet per minute for $0 \leq t \leq 8$ minutes. Selected values of $g(t)$ are given in the table above.

 (a) Using the data in the table, approximate $g'(3)$. Using correct units, interpret the meaning of $g'(3)$ in the context of the problem.

(b) Write an integral expression that represents the total amount of grain added to the silo from time $t = 0$ to time $t = 8$. Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate the integral.

(c) The grain in the silo is spoiling at a rate modeled by $w(t) = 32 \cdot \sqrt{\sin\left(\frac{\pi t}{74}\right)}$, where $w(t)$ is measured in cubic feet per minute for $0 \leq t \leq 8$ minutes. Using the result from part (b), approximate the amount of unspoiled grain remaining in the silo at time $t = 8$.

(d) Based on the model in part (c), is the amount of unspoiled grain in the silo increasing or decreasing at time $t = 6$? Show the work that leads to your answer.

Part A

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5) - g(1)}{5 - 1} = \frac{20.5 - 15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5) - g(1)}{5 - 1} = \frac{20.5 - 15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.

Part B

1 point is earned for: integral expression

6-2

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t) dt$ cubic feet.

$$\begin{aligned}\int_0^8 g(t) dt &\approx g(1) \cdot (1 - 0) + g(5) \cdot (5 - 1) + g(6) \cdot (6 - 5) + g(8) \cdot (8 - 6) \\ &= 15.1 \cdot 1 + 20.5 \cdot 4 + 18.3 \cdot 1 + 22.7 \cdot 2 = 160.8\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t) dt$ cubic feet.

$$\begin{aligned}\int_0^8 g(t) dt &\approx g(1) \cdot (1 - 0) + g(5) \cdot (5 - 1) + g(6) \cdot (6 - 5) + g(8) \cdot (8 - 6) \\ &= 15.1 \cdot 1 + 20.5 \cdot 4 + 18.3 \cdot 1 + 22.7 \cdot 2 = 160.8\end{aligned}$$

Part C

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t) dt = 99.051497$$

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t) dt = 61.749$ (or 61.748) cubic feet.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t) dt = 99.051497$$

6-2

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t) \, dt = 61.749$ (or 61.748) cubic feet.

Part D

1 point is earned for: considers $g(6) - w(6)$

1 point is earned for: answer

$$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$$

Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: considers $g(6) - w(6)$

1 point is earned for: answer

$$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$$

Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.

6-2

t (minutes)	0	4	8	12	16
$H(t)$ ($^{\circ}\text{C}$)	65	68	73	80	91

The temperature, in degrees Celsius ($^{\circ}\text{C}$), of an oven being heated is modeled by an increasing differentiable function H of time t , where t is measured in minutes. The table above gives the temperature as recorded every 4 minutes over a 16-minute period.

11. Write an integral expression in terms of H for the average temperature of the oven between time $t = 0$ and time $t = 16$. Estimate the average temperature of the oven using a left Riemann sum with four subintervals of equal length. Show the computations that lead to your answer.

Part B

1 point is earned for $\frac{1}{16} \int_0^{16} H(t) dt$

1 point is earned for left Riemann sum

1 point is earned for the answer

Average temperature is $\frac{1}{16} \int_0^{16} H(t) dt$

$$\frac{1}{16} \int_0^{16} H(t) dt = 4 \cdot (65 + 68 + 73 + 80)$$

$$\text{Average temperature} = \frac{4 \cdot 286}{16} = 71.5^{\circ}\text{C}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{1}{16} \int_0^{16} H(t) dt$

1 point is earned for left Riemann sum

1 point is earned for the answer

Average temperature is $\frac{1}{16} \int_0^{16} H(t) dt$

$$\frac{1}{16} \int_0^{16} H(t) dt = 4 \cdot (65 + 68 + 73 + 80)$$

6-2

$$\text{Average temperature} = \frac{4 \cdot 286}{16} = 71.5^\circ \text{C}$$

12. Is your approximation in part (b) an underestimate or an overestimate of the average temperature? Give a reason for your answer.
-

Part C

1 point is earned for the answer with reason

The left Riemann sum approximation is an underestimate of the integral because the graph of H is increasing. Dividing by 16 will not change the inequality, so 71.5°C is an underestimate of the average temperature.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with reason

The left Riemann sum approximation is an underestimate of the integral because the graph of H is increasing. Dividing by 16 will not change the inequality, so 71.5°C is an underestimate of the average temperature.

6-2

13.

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

The twice-differentiable function W models the volume of water in a reservoir at time t , where $W(t)$ is measured in (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \leq t \leq 30$ days. At time $t = 30$, the reservoir contains 125 gicaliters of water.

Use a left Riemann sum, with the three subintervals indicated by the data in the table, to approximate $\int_0^{30} W'(t) dt$. Use this approximation to estimate the volume of water $W(t)$, in gicaliters, in the reservoir at time $t = 0$. Show the computations that lead to your answer.

Part B

The response can earn up to 3 points:

1 point: For the left Riemann sum

1 point: For the approximation

1 point: For the correct answer

$$\int_0^{30} W'(t) dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$$

$$W(0) = W(30) - \int_0^{30} W'(t) dt = 125 - 22.4 = 102.6$$



0	1	2	3
---	---	---	---

The response earns all three of the following points:

1 point: For the left Riemann sum

1 point: For the approximation

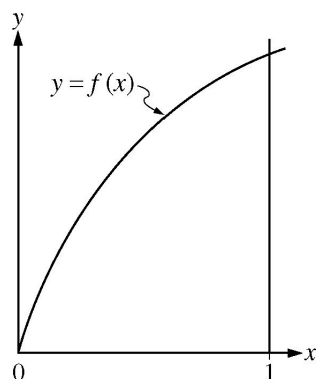
1 point: For the correct answer

$$\int_0^{30} W'(t) dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$$

$$W(0) = W(30) - \int_0^{30} W'(t) dt = 125 - 22.4 = 102.6$$

6-2

14.



1. A left Riemann sum, a right Riemann sum, and a trapezoidal sum are used to approximate the value of $\int_0^1 f(x) dx$, each using the same number of sub intervals. The graph of the function f is shown in the figure above. Which of the sums give an underestimate of the value of $\int_0^1 f(x) dx$?

I. Left sum

II. Right sum

III. Trapezoidal sum

(A) I only

(B) II only

(C) III only

(D) I and III only

(E) II and III only



15. What is the approximation of $\int_2^{14} \sqrt{1 + \sin x} dx$ found by using a right Riemann sum with 2 subintervals of equal length?

(A) $6\sqrt{1 + \sin 2} + 6\sqrt{1 + \sin 8}$

(B) $6\sqrt{1 + \sin 2} + 6\sqrt{1 + \sin 14}$

(C) $6\sqrt{1 + \sin 5} + 6\sqrt{1 + \sin 11}$

(D) $6\sqrt{1 + \sin 8} + 6\sqrt{1 + \sin 14}$

**Answer D**

Correct. A right Riemann sum on the interval $[2, 14]$ uses the right endpoints of the two subintervals $[2, 8]$ and $[8, 14]$ of length $\Delta x = \frac{14-2}{2} = 6$. These endpoints are at $x = 8$ and $x = 14$. The right Riemann sum is therefore $f(8)\Delta x + f(14)\Delta x = 6\sqrt{1 + \sin 8} + 6\sqrt{1 + \sin 14}$.

6-2

16. Let f be the function given by $f(x) = 9^x$. If four subintervals of equal length are used, what is the value of the right Riemann sum approximation for $\int_0^2 f(x) dx$?

(A) 20
(B) 40
(C) 60
(D) 80
(E) 120



17.

x	0	0.5	1	1.5	2	2.5	3
$f(x)$	0	4	10	18	28	40	54

The table above gives selected values for a continuous function f . If f is increasing over the closed interval $[0, 3]$, which of the following could be the value of $\int_0^3 f(x) dx$?

(A) 50
(B) 62
(C) 77
(D) 100
(E) 154



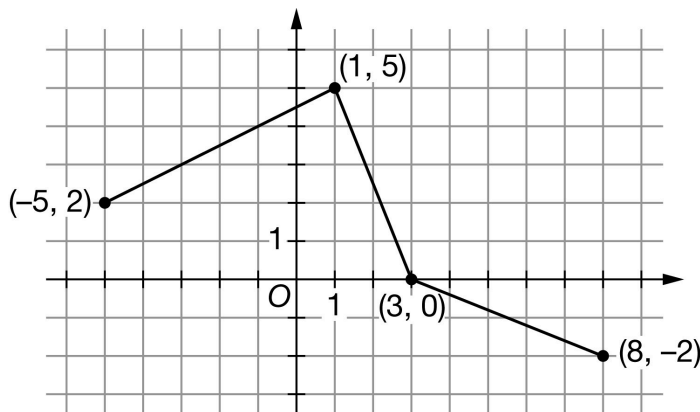
6-2

18. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the function g on the interval $-5 \leq x \leq 8$ consists of three line segments, as shown in the figure above. Let m be the function defined by $m(x) = \int_1^x g(t) dt$ for $-5 \leq x \leq 8$.

- For the function m , is $x = 3$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute maximum value of m on the interval $-5 \leq x \leq 8$. Justify your answer.
- On what open intervals contained in $-5 < x < 8$ is the graph of m both increasing and concave down? Give a reason for your answer.
- Find the value of $m''(1)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 1} \frac{m(x)}{x+1}$. Show the computations that lead to your answer.
- Let k be the function defined by $k(x) = m(x^2)$. Find the value of $k'(1)$ or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local maximum or relative maximum. We also need to see a connection to g . The obvious is $m' = g$. A response of local maximum with the explanation: " g (or the graph) changes from positive to negative" is an implied connection. A response of local maximum with explanation: " m' changes from positive to negative" without the connection of $m' = g$ is not acceptable.

Simply saying " m changes from increasing to decreasing" is not acceptable.

In this situation, the Second Derivative Test $m''(3) < 0$ is an incorrect statement and the response with this reason does not earn the point.

6-2



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 3$ is the location of a relative maximum of m because $m'(x) = g(x)$ changes from positive to negative there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = 3$ is the only critical point. (g is positive on $(-5, 3)$ and negative on $(3, 8)$ is another way to state this.)

Note: Stating that $x = 3$ is the only relative maximum critical point is not enough.

The second point is for arguing that $x = 3$ then produces the absolute maximum. Note that the response does not have to repeat the justification of $x = 3$ being a relative maximum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = 3$. This consideration is more than just the statement $x = 3$. The response must use $x = 3$ and an area calculation in trying to find $m(3)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as difference of $g(x)$ values.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

Common errors that show an attempt to find area:

· $m(-5) = -9$ (finding the area of the triangle and forgetting the rectangle)

· $m(8) = 10$ (adding the triangle areas). These can be used to earn the second point.

6-2

A correct argument that eliminates $x = -5$ by stating $m(-5)$ is negative will earn the second point as long as $x = 8$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = 3$, and $x = 8$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no calculation mistakes in intermediate work.

NOTE: A coordinate point is not an acceptable answer.

✓

0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute maximum
- ☐ Answer with justification
- ☐ Considers $x = 3$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = 3$ is the only point on the interval $-5 \leq x \leq 8$ where $m'(x) = g(x) = 0$, $x = 3$ is the only critical point of m on the interval $-5 \leq x \leq 8$.

Because $x = 3$ is also the location of a relative maximum for m , $m(3)$ is the absolute maximum for m in $-5 \leq x \leq 8$.

$$m(3) = \int_1^3 g(t) dt = 5$$

Therefore, the absolute maximum value of m on $-5 \leq x \leq 8$ is $m(3) = 5$.

Alternate solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$x = 3$ is the only critical point in the interval because it is the only point where $m'(x) = g(x) = 0$.

$$m(3) = \int_1^3 g(t) dt = 5$$

$$m(-5) = \int_1^{-5} g(t) dt = - \int_{-5}^1 g(t) dt = -21$$

$$m(8) = \int_1^8 g(t) dt = 0$$

Therefore, the absolute maximum value of m on $-5 \leq x \leq 8$ is $m(3) = 5$.

6-2

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(1, 3)$ or inequality $1 < x < 3$.

We will allow $[1, 3]$, $(1, 3]$, $[1, 3)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons m' is positive and decreasing or m' is positive and m'' is negative, will earn the second point if the connection $m' = g$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function m , then the second point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

m is increasing where $m' = g$ is positive.

m is concave down where $m' = g$ is decreasing.

The graph of m is increasing and concave down on the open interval $(1, 3)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $m''(x) = g'(x)$.

We are not allowing an import of a previous connection between m'' and g' .

The explicit connection, $m''(x) = g'(x)$, earns the first point.

It is not assumed that a response which only presents $m'(x) = g(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to g' , the differentiability of g , or the characteristic of the graph of g .

The second point is for the answer with explanation.

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving g' as the argument of the limit,

6-2

· or some correct verbal description of the characteristic of the graph of g at $x = 1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $g'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: g' does not exist because g (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of g . The phrase $m''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $m''(1)$ does not exist because there is a cusp on g is acceptable and earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $g'(x)$
- ☐ Answer with explanation

Solution:

$$m''(x) = g'(x)$$

Because $\lim_{x \rightarrow 1^-} \frac{g(x) - g(1)}{x - 1} = \frac{1}{2}$ and $\lim_{x \rightarrow 1^+} \frac{g(x) - g(1)}{x - 1} = -\frac{5}{2}$, $g'(1) = m''(1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of m does not need to be stated.

An answer of $\frac{m(1)}{2}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{2}$.

Copy/calculation errors in the denominator will not earn the point. Disregard notation errors.



0	1
---	---

The student response accurately includes the criteria below.

6-2

☐ Answer
Solution:

m is continuous. $\Rightarrow \lim_{x \rightarrow 1} m(x) = m(1)$

$$\lim_{x \rightarrow 1} \frac{m(x)}{x+1} = \frac{m(1)}{1+1} = \frac{0}{2} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $k : m'(x^2) \cdot 2x$.

This point is earned for any of $m'(x^2) \cdot 2x$, $g(x^2) \cdot 2x$, $g(1) \cdot 2$, or $5 \cdot 2$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$m'(x^2) = g(x^2), m'(1) = g(1), \text{ or } m'(1) = 5.$$

This point would be earned with a response of $g(1) \cdot 2$, or $5 \cdot 2$, even with no preceding work.

The third point is for either $5 \cdot 2$ or 10. Note, the product need not be simplified.

An answer of 10 with no supporting work earns no points in part (f).



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = m'(x^2) \cdot 2x = g(x^2) \cdot 2x$$

$$k'(1) = g(1) \cdot 2 = 5 \cdot 2 = 10$$

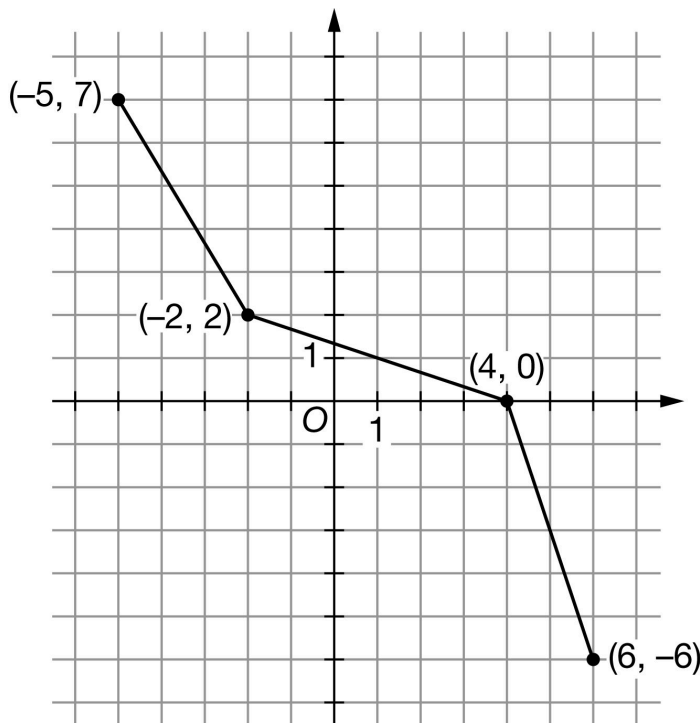
6-2

19. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the function f on the interval $-5 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let h be the function defined by $h(x) = \int_{-2}^x f(t) \, dt$ for $-5 \leq x \leq 6$.

- For the function h , is $x = 4$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute maximum value of h on the interval $-5 \leq x \leq 6$. Justify your answer.
- On what open intervals contained in $-5 < x < 6$ is the graph of h both increasing and concave down? Give a reason for your answer.
- Find the value of $h''(-2)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow -2} \frac{h(x)}{x-2}$. Show the computations that lead to your answer.
- Let k be the function defined by $k(x) = h(x^2)$. Find the value of $k'(\sqrt{5})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{5}$ can be keyboarded as sqrt(5))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-2

The response must include the wording local maximum or relative maximum. We also need to see a connection to h . The obvious is $h' = f$. A response of local maximum with the explanation: " f (or the graph) changes from positive to negative" is an implied connection. A response of local maximum with explanation: " h' changes from positive to negative," without the connection of $h' = f$ will not earn the point.

Simply saying " h . changes from increasing to decreasing" is not enough to earn the point.

In this situation, appealing to the Second Derivative Test by claiming that $h''(4) < 0$ is an incorrect statement, so a response with this reason does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 4$ is the location of a relative maximum of h because $h'(x) = f(x)$ changes from positive to negative at this point.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = 4$ is the only critical point. (f is positive on $(-5, 4)$ and negative on $(4, 6)$ is another way to state this.)

Note: Stating that $x = 4$ is the only relative maximum critical point is not enough.

The second point is for arguing that $x = 4$ then produces the absolute maximum. Note that the response does not have to repeat the justification of $x = 4$ being a relative maximum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct value of the maximum.

The presentation of a coordinate point $(4, 6)$ as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = 4$. This consideration is more than just the statement $x = 4$. The response must use $x = 4$ and an area calculation in trying to find $h(4)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $f(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $f(x)$ values.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find areas for $h(-5)$ and $h(6)$.

6-2

The response cannot simply write the integrals with the correct upper limits. Attempts at both areas must be made.

Using the y -values of the coordinate points from $f(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $f(x)$ values.

Common errors that show an attempt to find area:

· $h(6) = 12$ (adding the triangle areas)

· $Ah(-5) = -\frac{15}{2}$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = -5$ by stating $h(-5)$ is negative will earn the second point as long as $x = 6$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = 4$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

A coordinate point $(4, 6)$ is not an acceptable answer.

✓

0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute maximum
- ☐ Answer with justification
- ☐ Considers $x = 4$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = 4$ is the only point on the interval $-5 \leq x \leq 6$ where $h'(x) = f(x) = 0$, $x = 4$ is the only critical point of h on the interval $-5 \leq x \leq 6$.

Because $x = 4$ is also the location of a relative maximum of h , $h(4)$ is the absolute maximum for h on $-5 \leq x \leq 6$.

$$h(4) = \int_{-2}^4 f(t) dt = 6$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 6$ is $h(4) = 6$.

Alternate solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$x = 4$ is the only critical point in the interval because it is the only point where $h'(x) = f(x) = 0$.

6-2

$$h(4) = \int_{-2}^4 f(t) dt = 6$$

$$h(-5) = \int_{-2}^{-5} f(t) dt = -\int_{-5}^{-2} f(t) dt = -13.5$$

$$h(6) = \int_{-2}^6 f(t) dt = 0$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 6$ is $h(4) = 6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(-5, 4)$ or inequality $-5 < x < 4$.

We will allow $[-5, 4]$, $(-5, 4]$, $[-5, 4)$, or an equivalent inequality.

Presenting $(-5, -2) \cup (-2, 4)$ with correct reasoning earns one out of the two points. Any accepted version (open parentheses or brackets) earns the point. The “ $\cup$ ” is not necessary. Be careful, one bracket on either -2 is actually the entire interval and is eligible for both points.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons “ h' is positive and decreasing” or “ h' is positive and h'' is negative” will earn the second point if the connection $h' = f$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function h , then the second point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is increasing where $h' = f$ is positive.

h is concave down where $h' = f$ is decreasing.

The graph of h is both increasing and concave down on the open interval $(-5, 4)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $h''(x) = f'(x)$.

We are not allowing an import of a previous connection between h'' and f' .

6-2

The explicit connection, $h''(x) = f'(x)$, earns the first point.

It is not assumed that a response which only presents $h'(x) = f(x)$ has made the connection between h'' and f' .

The second point is for the answer with explanation.

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appear in these scoring guidelines,
- limits involving h' as the argument of the limit,
- or some correct verbal description of the characteristic of the graph of f at $x = -2$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h''(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: f' does not exist because f (or "the graph") has a/is a (blank). Fill in the blank with corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of f . The phrase $h''(-2)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $h''(-2)$ does not exist because there is a cusp on f is acceptable and earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $f'(x)$
- ☐ Answer with explanation

Solution:

$$h''(x) = f'(x)$$

Because $\lim_{x \rightarrow -2^-} \frac{f(x) - f(-2)}{x - (-2)} = -\frac{5}{3}$ and $\lim_{x \rightarrow -2^+} \frac{f(x) - f(-2)}{x - (-2)} = -\frac{1}{3}$, $f'(-2) = h''(-2)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of h does not need to be stated.

An answer of $\frac{h(-2)}{-4}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

6-2

The minimal answer is $\frac{0}{-4}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard limit notation errors.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

h is continuous. $\Rightarrow \lim_{x \rightarrow -2} h(x) = h(-2)$

$$\lim_{x \rightarrow -2} \frac{h(x)}{x-2} = \frac{h(-2)}{-2-2} = \frac{0}{-4} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $k : h'(x^2) \cdot 2x$.

This point is earned for any of $h'(x^2) \cdot 2x$, $f(x^2) \cdot 2x$, $f(5) \cdot 2\sqrt{5}$, or $-3 \cdot 2\sqrt{5}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but could still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$h'(x^2) = f(x^2)$, $h'(5) = f(5)$, or $h'(5) = -3$.

This point would be earned with a response of $f(5) \cdot 2\sqrt{5}$ or $-3 \cdot 2\sqrt{5}$ even with no preceding work.

The third point is for either $-3 \cdot 2\sqrt{5}$ or $-6\sqrt{5}$. Note, the product need not be simplified.

An answer of $-6\sqrt{5}$ with no supporting work earns no points in part (f).



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

6-2

$$k'(x) = h'(x^2) \cdot 2x = f(x^2) \cdot 2x$$

$$k'(\sqrt{5}) = f(5) \cdot 2\sqrt{5} = -3 \cdot 2\sqrt{5} = -6\sqrt{5}$$

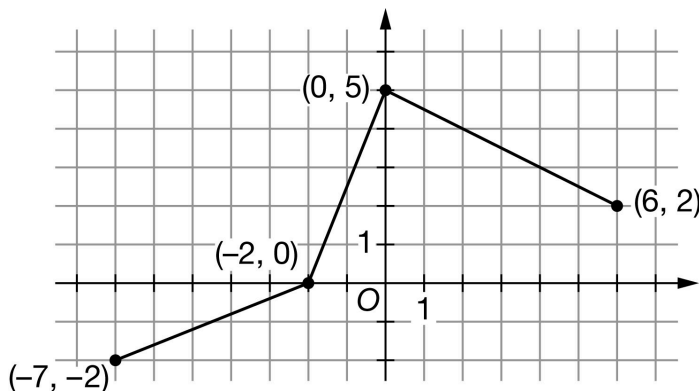
6-2

20. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The graph of the function h on the interval $-7 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let r be the function defined by $r(x) = \int_0^x h(t) \, dt$ for $-7 \leq x \leq 6$.

- For the function r , is $x = -2$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute minimum value of r on the interval $-7 \leq x \leq 6$. Justify your answer.
- On what open intervals contained in $-7 < x < 6$ is the graph of r both increasing and concave down? Give a reason for your answer.
- Find the value of $r''(-2)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 0} \frac{r(x)}{x-4}$. Show the computations that lead to your answer.
- Let v be the function defined by $v(x) = r(x^2)$. Find the value of $v'(\sqrt{3})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{3}$ can be keyboarded as sqrt(3))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to h . The obvious is $r' = h$. A response of local minimum with the explanation: " h (or the graph) changes from negative to positive" is an implied connection. A response of local minimum with explanation: " r' changes from negative to positive," without the connection of $r' = h$ is not acceptable.

Simply saying " r changes from decreasing to increasing" is not acceptable.

In this situation, the Second Derivative Test $r''(2) > 0$ is an incorrect statement, and a response with this reason does not earn the point.

6-2



0

1

The student response accurately includes the criteria below.

☐

Answer with reason

Solution:

$x = -2$ is the location of a relative minimum of r because $r'(x) = h(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -2$ is the only critical point. (h is negative on $(-7, -2)$ and positive on $(-2, 6)$ is another way to state this.)

Note: Stating that $x = -2$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -2$ then produces the absolute minimum. Note that the response does not have to repeat the justification of $x = -2$ being a relative minimum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = -2$. This consideration is more than just the statement $x = -2$. The response must use $x = -2$ and an area calculation in trying to find $r(-2)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as difference of $h(x)$ values.

A common error that shows an attempt to find area is $r(-2) = 5$. This earns the first point.

The second point is for consideration of the endpoints along with an attempt to find area. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as the difference of $h(x)$ values.

Common errors that show an attempt to find area:

6-2

· $r(-7) = 10$ (adding the triangle areas)

· $r(6) = 9$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = 6$ by stating $r(6)$ is positive will earn the second point as long as $x = -7$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -7$, $x = -2$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

Note: A coordinate point is not an acceptable answer.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute minimum
- ☐ Answer with justification
- ☐ Considers $x = -2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = -2$ is the only point on the interval $-7 \leq x \leq 6$ where $r'(x) = h(x) = 0$, $x = -2$ is the only critical point of r on the interval $-7 \leq x \leq 6$.

Because $x = -2$ is also the location of a relative minimum for r , $r(-2)$ is the absolute minimum for r on $-7 \leq x \leq 6$.

$$r(-2) = \int_0^{-2} h(t) dt = - \int_{-2}^0 h(t) dt = -5$$

Therefore, the absolute minimum value of r on $-7 \leq x \leq 6$ is $r(-2) = -5$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -2$ is the only critical point in the interval because it is the only point where $r'(x) = h(x) = 0$.

$$r(-2) = \int_0^{-2} h(t) dt = - \int_{-2}^0 h(t) dt = -5$$

$$r(-7) = \int_0^{-7} h(t) dt = - \int_{-7}^0 h(t) dt = 0$$

$$r(6) = \int_0^6 h(t) dt = 21$$

6-2

Therefore, the absolute minimum value of r on $-7 \leq x \leq 6$ is $r(-2) = -5$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(0, 6)$ or inequality $0 < x < 6$.

We will allow $[0, 6]$, $(0, 6]$, $[0, 6)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons r' is positive and decreasing or r' is positive and r'' is negative will earn the second point if the connection $r' = h$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function r , then the second point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

r is increasing where $r' = h$ is positive.

r is concave down where $r' = h$ is decreasing.

The graph of r is both increasing and concave down on the open interval $(0, 6)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a consideration of $r''(x) = h'(x)$.

We are not allowing an import of a previous connection between r'' and h' .

The explicit connection $r''(x) = h'(x)$ earns the first point.

It is not assumed that a response which presents $r'(x) = h(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to h' , the differentiability of h , or the characteristic of the graph of h .

The second point is for the answer with explanation

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,

6-2

· limits involving h' as the argument of the limit, or some correct verbal description of the characteristic of the graph of h at $x = -2$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: h' does not exist because h (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of h . The phrase $r''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $r''(1)$ does not exist because there is a cusp on h is acceptable and earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'(x)$
- ☐ Answer with explanation

Solution:

Because $\lim_{x \rightarrow -2^-} \frac{h(x) - h(-2)}{x - (-2)} = \frac{2}{5}$ and $\lim_{x \rightarrow -2^+} \frac{h(x) - h(-2)}{x - (-2)} = \frac{5}{2}$, $h'(-2) = r''(-2)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of r does not need to be stated.

An answer of $\frac{r(0)}{-4}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{-4}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard notation errors



0	1
---	---

The student response accurately includes the criteria below.

6-2

☐ Answer
Solution:

r is continuous. $\Rightarrow \lim_{x \rightarrow 0} r(x) = r(0)$

$$\lim_{x \rightarrow 0} \frac{r(x)}{x-4} = \frac{r(0)}{0-4} = \frac{0}{-4} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $v : r'(x^2) \cdot 2x$.

This point is earned for any of $r'(x^2) \cdot 2x$, $h(x^2) \cdot 2x$, $h(3) \cdot 2\sqrt{3}$, or $\frac{7}{2} \cdot 2\sqrt{3}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$r'(x^2) = h(x^2), r'(3) = h(3), \text{ or } r'(3) = \frac{7}{2}.$$

This point would be earned with a response of $h(3) \cdot 2\sqrt{3}$, or $\frac{7}{2} \cdot 2\sqrt{3}$, even with no preceding work.

The third point is for either $\frac{7}{2} \cdot 2\sqrt{3}$ or $7\sqrt{3}$. Note, the product need not be simplified.

An answer of $7\sqrt{3}$ with no supporting work earns no points in part (f).



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
☐ Fundamental Theorem of Calculus
☐ Answer

Solution:

$$v'(x) = r'(x^2) \cdot 2x = h(x^2) \cdot 2x$$

$$v'(\sqrt{3}) = h(3) \cdot 2\sqrt{3} = \frac{7}{2} \cdot 2\sqrt{3} = 7\sqrt{3}$$

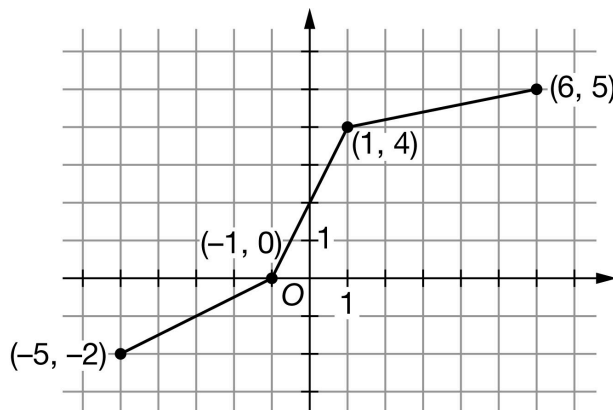
6-2

21. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The graph of the function h on the interval $-5 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let p be the function defined by $p(x) = \int_1^x h(t) \, dt$ for $-5 \leq x \leq 6$.

- For the function p , is $x = -1$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute minimum value of p on the interval $-5 \leq x \leq 6$. Justify your answer.
- On what open intervals contained in $-5 < x < 6$ is the graph of p both decreasing and concave up? Give a reason for your answer.
- Find the value of $p''(1)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 1} \frac{p(x)}{x+4}$. Show the computations that lead to your answer.
- Let k be the function defined by $k(x) = p(x^2)$. Find the value of $k'(\sqrt{2})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{2}$ can be keyboarded as `sqrt(2)`)

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to h . The obvious is $p' = h$. A response of local minimum with the explanation: " h (or the graph) changes from negative to positive" is an implied connection. A response of local minimum with explanation: " p' changes from negative to positive," without the connection of $p' = h$ is not acceptable.

Simply saying " p changes from decreasing to increasing" is not acceptable.

In this situation, the Second Derivative Test $p''(-1) > 0$ is an incorrect statement and a response with this reason would not earn the point.

6-2



0

1

The student response accurately includes the criteria below.

☐

Answer with reason

Solution:

$x = -1$ is the location of a relative minimum of p because $p'(x) = h(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -1$ is the only critical point. (h is negative on $(-5, -1)$ and positive on $(-1, 6)$ is another way to state this.)

Note: Stating that $x = -1$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -1$ then produces the absolute minimum. Note that they do not have to repeat the justification of $x = -1$ being a relative minimum from part (a). If this is not in (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = -1$. This consideration is more than just the statement $x = -1$. The response must use $x = -1$ and an area calculation in trying to find $p(-1)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $h(x)$ values.

A common error that shows an attempt to find area is $k(-1) = 4$. This earns the first point.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as the difference of $h(x)$ values.

Common errors that show an attempt to find area:

$\cdot p(-5) = 8$ (adding the triangle areas)

6-2

$p(6) = \frac{5}{2}$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = 6$ by stating $p(6)$ is positive will earn the second point as long as $x = -5$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = -1$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no calculation mistakes in intermediate work.

A coordinate point is not an acceptable answer.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute minimum
- ☐ Answer with justification
- ☐ Considers $x = -1$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = -1$ is the only point on the interval $-5 \leq x \leq 6$ where $p'(x) = h(x) = 0$, $x = -1$ is the only critical point of p on the interval $-5 \leq x \leq 6$.

Because $x = -1$ is also the location of a relative minimum for p , $p(-1)$ is the absolute minimum for p on $-5 \leq x \leq 6$.

$$p(-1) = \int_1^{-1} h(t) dt = - \int_{-1}^1 h(t) dt = -4$$

Therefore, the absolute minimum value of p on $-5 \leq x \leq 6$ is $p(-1) = -4$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -1$ is the only critical point in the interval because it is the only point where $p'(x) = h(x) = 0$.

$$p(-1) = \int_1^{-1} h(t) dt = - \int_{-1}^1 h(t) dt = -4$$

$$p(-5) = \int_1^{-5} h(t) dt = - \int_{-5}^1 h(t) dt = 0$$

$$p(6) = \int_1^6 h(t) dt = 22.5$$

Therefore, the absolute minimum value of p on $-5 \leq x \leq 6$ is $p(-1) = -4$.

6-2

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the interval $(-5, -1)$ or inequality $-5 < x < -1$.

We will allow $[-5, -1]$, $(-5, -1]$, $[-5, -1)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons p' is negative and increasing or p' is negative and p'' is positive will earn the second point if the connection $p' = h$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function p , then the second point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

p is decreasing where $p' = h$ is negative.

p is concave up where $p' = h$ is increasing.

The graph of p is both decreasing and concave up on the open interval $(-5, -1)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $p''(x) = h'(x)$.

We are not allowing an import of a previous connection between p'' and h' .

The explicit connection, $p'' = h'$, earns the first point.

It is not assumed that a response which only presents $p' = h$ has made that connection.

The implicit connection will be made in the explanation with a connection to h' , the differentiability of h , or the characteristic of the graph of h .

The second point is for the answer with explanation

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving h' as the argument of the limit,

6-2

· or some correct verbal description of the characteristic of the graph of h at $x = 1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: h' does not exist because h (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of h . The phrase $p''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $p''(1)$ does not exist because there is a cusp on h is acceptable and earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'(x)$
- ☐ Answer with explanation

Solution:

$$p'(x) = h'(x)$$

Because $\lim_{x \rightarrow 1^-} \frac{h(x) - h(1)}{x - 1} = 2$, and $\lim_{x \rightarrow 1^+} \frac{h(x) - h(1)}{x - 1} = \frac{1}{5}$, $h'(1) = p'(1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of p does not need to be stated.

An answer of $\frac{p(1)}{5}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{5}$.

Copy/calculation errors in the denominator will not earn the point. Disregard notation errors



0	1
---	---

The student response accurately includes the criteria below.

Answer

6-2

**Solution:**

$$p \text{ is continuous. } \Rightarrow \lim_{x \rightarrow 1} p(x) = p(1) \lim_{x \rightarrow 1} \frac{p(x)}{x+4} = \frac{p(1)}{1+4} = \frac{0}{5} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of k : $p'(x^2) \cdot 2x$.

This point is earned for any of $p'(x^2) \cdot 2x$, $h(x^2) \cdot 2x$, $h(2) \cdot 2\sqrt{2}$, or $\frac{21}{5} \cdot 2\sqrt{2}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$p'(x^2) = h(x^2), p'(2) = h(2), \text{ or } p'(2) = \frac{21}{5}.$$

This point would be earned with a response of $h(2) \cdot 2\sqrt{2}$, or $\frac{21}{5} \cdot 2\sqrt{2}$, even with no preceding work.

The third point is for either $\frac{21}{5} \cdot 2\sqrt{2}$ or $\frac{42}{5}\sqrt{2}$. Note, the product need not be simplified.

An answer of $\frac{42}{5}\sqrt{2}$ with no supporting work earns no points in part (f).



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = p'(x^2) \cdot 2x = h(x^2) \cdot 2x$$

$$k'(\sqrt{2}) = h(2) \cdot 2\sqrt{2} = \frac{21}{5} \cdot 2\sqrt{2} = \left(\frac{42}{5}\right)\sqrt{2}$$

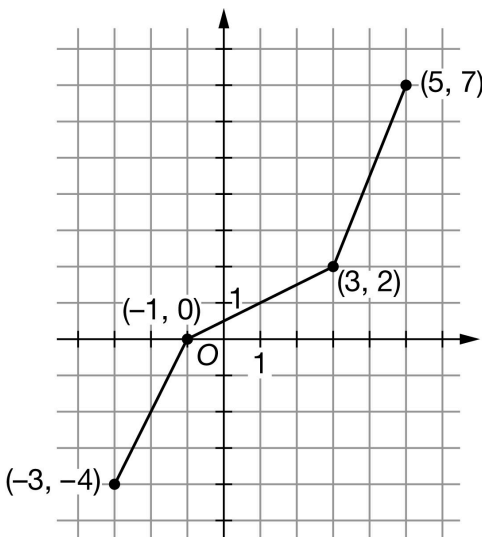
6-2

22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the function g on the interval $-3 \leq x \leq 5$ consists of three line segments, as shown in the figure above. Let k be the function defined by $k(x) = \int_3^x g(t) \, dt$ for $-3 \leq x \leq 5$.

- For the function k , is $x = -1$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute minimum value of k on the interval $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals contained in $-3 < x < 5$ is the graph of k both increasing and concave up? Give a reason for your answer.
- Find the value of $k''(-1)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 3} \frac{k(x)}{x+3}$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = k(x^2)$. Find the value of $h'(\sqrt{3})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{3}$ can be keyboarded as sqrt(3))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to g . The obvious is $k' = g$. A response of local minimum with the explanation: “ g (or the graph) changes from negative to positive” is an implied connection. A response of local minimum with explanation: “ k' changes from negative to positive,” without the connection of $k' = g$ is not acceptable.

6-2

Simply saying “ k changes from decreasing to increasing” is not acceptable.

In this situation, the Second Derivative Test $k''(-1) > 0$ is an incorrect statement and a response with this reason does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -1$ is the location of a relative minimum of k because $k'(x) = g(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -1$ is the only critical point. (g is negative on $(-3, -1)$ and positive on $(-1, 5)$ is another way to state this.)

Note: Stating that $x = -1$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -1$ then produces the absolute minimum. Note that the response does not have to repeat the justification of $x = -1$ being a relative minimum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = -1$. This consideration is more than just the statement $x = -1$. The response must use $x = -1$ AND an area calculation in trying to find $k(-1)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

A common error that shows an attempt to find area is $k(-1) = 4$. This earns the first point.

The second point is for consideration of the endpoints. The words “must check the endpoints” is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

6-2

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

Common errors that show an attempt to find area:

- $k(-3) = 8$ (adding the triangle areas)
- $k(5) = 5$ (finding the area of the triangle and forgetting the rectangle)
- These can be used to earn the second point.

A correct argument that eliminates $x = 5$ by stating $k(5)$ is positive will earn the second point as long as $x = -3$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -3$, $x = -1$, and $x = 5$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

A coordinate point is not an acceptable answer.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute minimum
- ☐ Answer with justification
- ☐ Considers $x = -1$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = -1$ is the only point on the interval $-3 \leq x \leq 5$ where $k'(x) = g(x) = 0$, $x = -1$ is the only critical point of k on the interval $-3 \leq x \leq 5$.

Because $x = -1$ is also the location of a relative minimum for k , $k(-1)$ is the absolute minimum for k on $-3 \leq x \leq 5$.

$$k(-1) = \int_3^{-1} g(t)dt = - \int_{-1}^3 g(t)dt = -4$$

Therefore, the absolute minimum value of k on $-3 \leq x \leq 5$ is $k(-1) = -4$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -1$ is the only critical point in the interval because it is the only point where $k'(x) = g(x) = 0$.

$$k(-1) = \int_3^{-1} g(t)dt = - \int_{-1}^3 g(t)dt = -4$$

6-2

$$k(-3) = \int_3^{-3} g(t) dt = - \int_{-3}^3 g(t) dt = 0$$

$$k(5) = \int_3^5 g(t) dt = 9$$

Therefore, the absolute minimum value of k on $-3 \leq x \leq 5$ is $k(-1) = -4$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the interval $(-1, 5)$ or inequality $-1 < x < 5$.

We will allow $[-1, 5]$, $(-1, 5]$, $[-1, 5)$, or an equivalent inequality.

Presenting $(-1, 3) \cup (3, 5)$ with correct reasoning earns one out of the two points. Any accepted version (open parentheses or brackets) earns the point. The “ $\cup$ ” is not necessary. Be careful, one bracket on either 3 is actually the entire interval and is eligible for both points.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons “ k' is positive and increasing” or “ k' is positive and k'' is positive” will earn the second point if the connection $k' = g$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function k , then the second point is not earned.

The interval $(-1, 7)$ with the correct justification will earn one of the two points. (The response took the y -coordinate of the right-hand point of the interval $(5, 7)$). This is considered a copy error. This is the only copy error that is allowed.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

k is increasing where $k' = g$ is positive.

k is concave up where $k' = g$ is increasing.

The graph of k is both increasing and concave up on the open interval $(-1, 5)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $k''(x) = g'(x)$.

We are not allowing an import of a previous connection between k'' and g' .

6-2

The explicit connection, $k''(x) = g'(x)$ earns the first point.

It is not assumed that a response which only presents $k'(x) = g(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to g' , the differentiability of g , or the characteristic of the graph of g .

The second point is for the answer with explanation.

This must include “Does not exist” and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving g' as the argument of the limit,
- or some correct verbal description of the characteristic of the graph of g at $x = -1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $g'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: g' does not exist because g (or “the graph”) has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will NOT accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of g . The phrase $k''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $k''(1)$ does not exist because there is a cusp on g is acceptable and earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $g'(x)$
- ☐ Answer with explanation

Solution:

Because $\lim_{x \rightarrow -1^-} \frac{g(x) - g(-1)}{x - (-1)} = 2$ and $\lim_{x \rightarrow -1^+} \frac{g(x) - g(-1)}{x - (-1)} = \frac{1}{2}$, $g''(-1) = k''(-1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of k does not need to be stated.

An answer of $\frac{k(3)}{6}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

6-2

The minimal answer is $\frac{0}{6}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard notation errors.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

k is continuous. $\Rightarrow \lim_{x \rightarrow 3} k(x) = k(3)$

$$\lim_{x \rightarrow 3} \frac{k(x)}{x+3} = \frac{k(3)}{3+3} = \frac{0}{6} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $h : k'(x^2) \cdot 2x$.

This point is earned for any of $k'(x^2) \cdot 2x$, $g(x^2) \cdot 2x$, $g(3) \cdot 2\sqrt{3}$, or $2 \cdot 2\sqrt{3}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$k'(x^2) = g(x^2)$, $k'(3) = g(3)$, or $k'(3) = 2$.

This point would be earned with a response of $g(3) \cdot 2\sqrt{3}$, or $2 \cdot 2\sqrt{3}$, even with no preceding work.

The third point is for either $2 \cdot 2\sqrt{3}$, or $4\sqrt{3}$. Note, the product need not be simplified.

An answer of $4\sqrt{3}$ with no supporting work earns no points in part (f).



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

6-2

$$h'(x) = k'(x^2) \cdot 2x = g(x^2) \cdot 2x$$

$$h'(\sqrt{3}) = g(3) \cdot 2\sqrt{3} = 2 \cdot 2\sqrt{3} = 4\sqrt{3}$$

23.

t (seconds)	0	3	5	8	12
$k(t)$ (feet per second)	0	5	10	20	24

Kathleen skates on a straight track. She starts from rest at the starting line at time $t = 0$. For $0 < t \leq 12$ seconds, Kathleen's velocity k , measured in feet per second, is differentiable and increasing. Values of $k(t)$ at various times t are given in the table above.

Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate $\int_0^{12} k(t) dt$. Indicate units of measure. Is this approximation an overestimate or an underestimate for the value of $\int_0^{12} k(t) dt$? Explain your reasoning.

General

1 point is earned for: right Riemann sum

1 point is earned for: approximation with units

1 point is earned for: overestimate with reason

$$\int_0^{12} k(t) dt \approx (5)(3) + (10)(2) + (20)(3) + (24)(4) = 191 \text{ feet}$$

This approximation is an overestimate since a right Riemann sum is used and the function k is increasing.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: right Riemann sum

1 point is earned for: approximation with units

1 point is earned for: overestimate with reason

$$\int_0^{12} k(t) dt \approx (5)(3) + (10)(2) + (20)(3) + (24)(4) = 191 \text{ feet}$$

This approximation is an overestimate since a right Riemann sum is used and the function k is increasing.

6-2

24.

t (hours)	0	2	7	9
$R(t)$ (tons per hour)	15	9	5	4

On a certain day, the rate at which material is deposited at a recycling center is modeled by the function R , where $R(t)$ is measured in tons per hour and t is the number of hours since the center opened. Using a trapezoidal sum with the three subintervals indicated by the data in the table, what is the approximate number of tons of material deposited in the first 9 hours since the center opened?

(A) 68



(B) 70.5

(C) 85

(D) 136

Answer A

This option is correct. The trapezoidal sum is the average of the left and right Riemann sums. The three intervals are of length 2, 5, and 2, respectively. Taking the average of the left and right endpoint values on each interval and multiplying by the length of the interval gives the following trapezoidal sum: $\frac{2}{2}(15 + 9) + \frac{5}{2}(9 + 5) + \frac{2}{2}(5 + 4) = 68$

25.

t	0	0.5	2	3
$v(t)$	20	60	40	30

The table above gives the velocity $v(t)$, in miles per hour, of a truck at selected times t , in hours.

Using a trapezoidal sum with the three subintervals indicated by the table, what is the approximate distance, in miles, the truck traveled from time $t = 0$ to $t = 3$?

(A) 140

(B) 130



(C) 125

(D) 120

6-2

26.

x	0	1	2	3	4	5	6
$f(x)$	0	5	2	-1	-2	0	3

The function f is continuous on the closed interval $[0, 6]$ and has values as shown in the table above. Using the intervals $[0, 2]$, $[2, 4]$, and $[4, 6]$, what is the approximation of $\int_0^6 f(x) dx$ obtained from a midpoint Riemann sum?

- (A) 0
(B) 3
(C) 4
(D) 6
(E) 8



27.

x	0	25	30	50
$f(x)$	4	6	8	12

The values of a continuous function f for selected values of x are given in the table above. What is the value of the left Riemann sum approximation to $\int_0^{50} f(x) dx$ using the subintervals $[0, 25]$, $[25, 30]$, and $[30, 50]$?

- (A) 290
(B) 360
(C) 380
(D) 390
(E) 430



6-2

28.

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

The twice-differentiable function W models the volume of water in a reservoir at time t , where $W(t)$ is measured in gigaleters (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \leq t \leq 30$ days. At time $t=30$, the reservoir contains 125 gigaleters of water.

Use a left Riemann sum, with the three subintervals indicated by the data in the table, to approximate $\int_0^{30} W'(t) dt$. Use this approximation to estimate the volume of water $W(t)$, in gigaleters, in the reservoir at time $t=0$. Show the computations that lead to your answer.

Part B

One point is earned for the left Riemann sum

One point is earned for the approximation

One point is earned for the answer

$$\int_0^{30} W'(t) dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$$

$$W(0) = W(30) - \int_0^{30} W'(t) dt = 125 - 22.4 = 102.6$$



0	1	2	3
---	---	---	---

The student earns all of the following points:

One point is earned for the left Riemann sum

One point is earned for the approximation

One point is earned for the answer

$$\int_0^{30} W'(t) dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$$

$$W(0) = W(30) - \int_0^{30} W'(t) dt = 125 - 22.4 = 102.6$$

6-2

29.

x	0	2	4	6
$f(x)$	4	k	8	12

The function f is continuous on the closed interval $[0,6]$ and has the values given in the table above. The trapezoidal approximation for $\int_0^6 f(x)dx$ found with 3 subintervals of equal length is 52. What is the value of k ?

- (A) 2
 (B) 6
 (C) 7
 (D) 10
 (E) 14



30.

t (minutes)	0	4	7	9
$r(t)$ (gallons per minute)	9	6	4	3

Water is flowing into a tank at the rate $r(t)$, where $r(t)$ is measured in gallons per minute and t is measured in minutes. The tank contains 15 gallons of water at time $t = 0$. Values of $r(t)$ for selected values of t are given in the table above. Using a trapezoidal sum with the three intervals indicated by the table, what is the approximation of the number of gallons of water in the tank at time $t = 9$?

- (A) 52
 (B) 57
 (C) 67
 (D) 77
 (E) 79



31.

t (minutes)	0	5	10	15
$R(t)$ (people per minute)	100	100	75	55

During an evacuation drill, people leave a building at a rate of $R(t)$ people per minute, where t is the number of minutes since the start of the drill. Selected values of $R(t)$ are shown in the table above.

Using a right Riemann sum with three subintervals and data from the table, what is the approximation of the number of people who leave the building during the first 15 minutes of the evacuation drill?

- (A) 230
 (B) 1150
 (C) 1375
 (D) 2075



6-2

32.

t (hours)	0	2	7	9
$R(t)$ (tons per hour)	15	9	5	4

On a certain day, the rate at which material is deposited at a recycling center is modeled by the function R , where $R(t)$ is measured in tons per hour and t is the number of hours since the center opened. Using a trapezoidal sum with the three subintervals indicated by the data in the table, what is the approximate number of tons of material deposited in the first 9 hours since the center opened?

(A) 68



(B) 70.5

(C) 85

(D) 136

Answer A

The trapezoidal sum is the average of the left and right Riemann sums. The three intervals are of length 2, 5, and 2, respectively. Taking the average of the left and right endpoint values on each interval and multiplying by the length of the interval gives the following trapezoidal sum:

33.

t (minutes)	0	5	10	15
$R(t)$ (people per minute)	100	100	75	55

During an evacuation drill, people leave a building at a rate of $R(t)$ people per minute, where t is the number of minutes since the start of the drill. Selected values of $R(t)$ are shown in the table above.

Using a right Riemann sum with three subintervals and data from the table, what is the approximation of the number of people who leave the building during the first 15 minutes of the evacuation drill?

(A) 230

(B) 1150



(C) 1375

(D) 2075

6-2

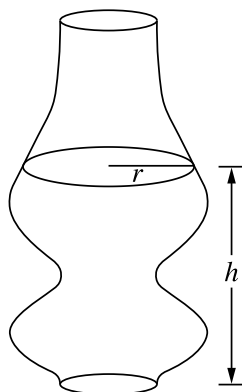
34.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



h (centimeters)	0	6.2	13	20
$r(h)$ (centimeters)	2.5	2.7	4.3	2.6

Juice is sold in 20-centimeter-tall bottles with horizontal cross sections parallel to the base that are circles, as shown in the figure. The radius of the circular cross section at height h above the base of the bottle is given by a differentiable function r , where h and $r(h)$ are measured in centimeters. Selected values of $r(h)$ are given in the table shown.

- (a) Approximate $r'(3.1)$ using the average rate of change of r over the interval $0 \leq h \leq 6.2$. Show the computations that lead to your answer. Indicate units of measure.

- (b) Interpret the meaning of $\frac{1}{20} \int_0^{20} r(h) \, dh$ in the context of the problem. Use a trapezoidal sum with the three subintervals $[0, 6.2]$, $[6.2, 13]$, and $[13, 20]$ to approximate the value of $\frac{1}{20} \int_0^{20} r(h) \, dh$.

- (c) The radius of the circular cross section at height h above the base of the bottle can also be modeled by the function f , where h and $f(h)$ are measured in centimeters. Let $f(h) = 3\sqrt{2 - 1.25 \cos\left(\frac{\pi}{100}(h - 20)^2\right)}$ for $0 \leq h \leq 20$. Based on the model, at a height of $h = 12$ centimeters, is the radius increasing or decreasing as h increases? Give a reason for your answer.

- (d) Using the model f defined in part (c), find the volume of the bottle.

Part A

6-2

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must provide a difference and a quotient. A response of only $\frac{2.7-2.5}{6.2}$ or $\frac{0.2}{6.2-0}$ earns the first point.

To earn the second point a response must have $\frac{\text{cm}}{\text{cm}}$ (or the equivalent) attached to a numerical value. These units do not reduce to 1 because the numerator, cm, refers to radius of the bottle and the denominator, cm, refers to height of the bottle.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Units

Solution:

$$r'(3.1) \approx \frac{r(6.2) - r(0)}{6.2 - 0} = \frac{2.7 - 2.5}{6.2}$$

0.032 centimeter per centimeter

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must provide “average,” “radius,” and “the interval from 0 to 20” (or equivalent such as “throughout the bottle” or “from the bottom to the top”). Units are not required to earn the first point.

To earn the second point, at least five of the six factors in the trapezoidal sum must be correct.

If there is any error in the trapezoidal sum, the response does not earn the third point.

A response of $\frac{1}{20} \left(6.2 \cdot \frac{2.5+2.7}{2} + 6.8 \cdot \frac{2.7+4.3}{2} + 7 \cdot \frac{4.3+2.6}{2} \right)$ earns the second and third points, unless there is a subsequent error in simplification, in which case the response would earn only the second point.

A response that is missing the factor of $\frac{1}{2}$ in the trapezoidal sum does not earn the second and third points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Interpretation
- ☐ Form of trapezoidal sum
- ☐ Approximation

Solution:

6-2

$\frac{1}{20} \int_0^{20} r(h) \, dh$ is the average value of the radius of the juice bottle, in centimeters, over the interval $0 \leq h \leq 20$ centimeters.

$$\begin{aligned} & \frac{1}{20} \int_0^{20} r(h) \, dh \\ & \approx \frac{1}{20} \left(6.2 \cdot \frac{r(0)+r(6.2)}{2} + 6.8 \cdot \frac{r(6.2)+r(13)}{2} + 7 \cdot \frac{r(13)+r(20)}{2} \right) \\ & = \frac{1}{20} \left(6.2 \cdot \frac{2.5+2.7}{2} + 6.8 \cdot \frac{2.7+4.3}{2} + 7 \cdot \frac{4.3+2.6}{2} \right) \\ & = \frac{1}{20} \cdot 64.07 = 3.2035 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the second point, a response does not need to provide a value for $f'(12)$; the sign of $f'(12)$ is sufficient.

Any presented value of $f'(12)$ must be correct for the digits presented, up to three decimal places.

If a response presents an incorrect sign or value for $f'(12)$, the response does not earn the second point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $f'(12) = -0.0006660598$. If a response presents this value, it earns the first point but is not eligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $f'(12)$
- ☐ Answer with reason

Solution:

$$f'(12) = -0.535902$$

At a height of 12 centimeters, the radius is decreasing as h increases because $f'(12) < 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a definite integral with an integrand of $(f(h))^2$, regardless of any subsequent errors in presentation of $f(h)$. A response that *begins* with an error in copying $f(h)$ into an integrand of the form $(f(h))^2$ does not earn the first point, but is eligible to earn the second point for a correct answer (1007.112 or 1007.111).

An answer of 320.574π rounds and truncates to 1007.028, and therefore does not earn the second point. Similarly, an answer of 320.573π truncates to 1007.109 and rounds to 1007.110, and therefore does not earn the second point. However, answers of 320.5736π or 320.5737π do round and truncate to the correct answer and earn the second point.

6-2

Degree mode: $\pi \int_0^{20} (f(h))^2 dh = 427.508$ (or 427.507) or 136.0799156π . A response that did not earn the second point in part (c) because of degree mode is eligible to earn the answer point in part (d) for this value. Furthermore, in degree mode 136.079π rounds to 427.505 and truncates to 427.504, and therefore does not earn the second point. However, in degree mode 136.080π is accurate to three decimals and is eligible for the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integrand
- ☐ Answer

Solution:

$$\text{Volume} = \pi \int_0^{20} (f(h))^2 dh$$

$$= 1007.112 \text{ (or } 1007.111) \text{ cubic centimeters}$$

6-2

35. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	2	5	9	11	12
$L(t)$ (cars per hour)	15	40	24	68	18

The rate at which cars enter a parking lot is modeled by $E(t) = 30 + 5(t - 2)(t - 5)e^{-0.2t}$. The rate at which cars leave the parking lot is modeled by the differentiable function L . Selected values of $L(t)$ are given in the table above. Both $E(t)$ and $L(t)$ are measured in cars per hour, and time t is measured in hours after 5 A.M. ($t = 0$). Both functions are defined for $0 \leq t \leq 12$.

-  (a) What is the rate of change of $E(t)$ at time $t = 7$? Indicate units of measure.
- (b) How many cars enter the parking lot from time $t = 0$ to time $t = 12$? Give your answer to the nearest whole number.
- (c) Use a trapezoidal sum with the four subintervals indicated by the data in the table to approximate $\int_2^{12} L(t) dt$. Using correct units, explain the meaning of $\int_2^{12} L(t) dt$ in the context of this problem.
- (d) For $0 \leq t < 6$, 5 dollars are collected from each car entering the parking lot. For $6 \leq t \leq 12$, 8 dollars are collected from each car entering the parking lot. How many dollars are collected from the cars entering the parking lot from time $t = 0$ to time $t = 12$? Give your answer to the nearest whole dollar.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ answer with units

Solution:

$$E'(7) = 6.164924$$

6-2

The rate of change of $E(t)$ at time $t = 7$ is 6.165 (or 6.164) cars per hour per hour.

Part B

The first point is earned with $E(t)$ or $30 + 5(t - 2)(t - 5)e^{-0.2t}$ written as the integrand. Limits are part of the second point.

It is possible to earn the second point with a single copy error of a constant in the integrand.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ integrand
- ☐ answer

Solution:

$$\int_0^{12} E(t) dt = 520.070489$$

To the nearest whole number, 520 cars enter the parking lot from time $t = 0$ to time $t = 12$.

Part C

The first point is earned for showing the method of a trapezoidal sum, so the sum of three products must be given. This point can be earned with a maximum of one error in either a coefficient or a value extracted from the table.

The second point can be earned concurrently with the first point. Simplification is not required. Functions values from the table must be substituted.

The third point requires that the explanation includes (1) units (cars), (2) the time interval ($2 \leq t \leq 12$ or 7 A.M. to 5 P.M.), (3) a meaning that expresses the (approximate) number of cars leaving the parking lot.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ trapezoidal sum
- ☐ approximation
- ☐ explanation

Solution:

6-2

$$\begin{aligned}
 \int_2^{12} L(t) \, dt &\approx (5-2) \cdot \frac{L(2)+L(5)}{2} + (9-5) \cdot \frac{L(5)+L(9)}{2} \\
 &\quad + (11-9) \cdot \frac{L(9)+L(11)}{2} + (12-11) \cdot \frac{L(11)+L(12)}{2} \\
 &= 3 \cdot \frac{15+40}{2} + 4 \cdot \frac{40+24}{2} + 2 \cdot \frac{24+68}{2} + 1 \cdot \frac{68+18}{2} \\
 &= 345.5
 \end{aligned}$$

$\int_2^{12} L(t) \, dt$ is the number of cars that leave the parking lot in the 10 hours between 7 A.M. ($t = 2$) and 5 P.M. ($t = 12$).

Part D

The first point is earned with two definite integrals, both having $E(t)$ as the integrand.

The third point may be earned with an answer correctly rounded or truncated to 1, 2, or 3 decimal places.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3
---	---	---	---



The student response accurately includes **all three** of the criteria below.

- ☐ integrand
- ☐ limits and constants
- ☐ answer

Solution:

$$5 \int_0^6 E(t) \, dt + 8 \int_6^{12} E(t) \, dt = 3530.1396$$

To the nearest dollar, 3530 dollars are collected from time $t = 0$ to time $t = 12$.

6-2

36. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	60	90	120	135	150
$f(t)$ (gallons per second)	0	0.1	0.15	0.1	0.05	0

A customer at a gas station is pumping gasoline into a gas tank. The rate of flow of gasoline is modeled by a differentiable function f , where $f(t)$ is measured in gallons per second and t is measured in seconds since pumping began. Selected values of $f(t)$ are given in the table.

(a) Using correct units, interpret the meaning of $\int_{60}^{135} f(t) \, dt$ in the context of the problem. Use a right Riemann sum with the three subintervals $[60, 90]$, $[90, 120]$, and $[120, 135]$ to approximate the value of $\int_{60}^{135} f(t) \, dt$.

(b) Must there exist a value of c , for $60 < c < 120$, such that $f'(c) = 0$? Justify your answer.

(c) The rate of flow of gasoline, in gallons per second, can also be modeled by $g(t) = \left(\frac{t}{500}\right) \cos\left(\left(\frac{t}{120}\right)^2\right)$ for $0 \leq t \leq 150$. Using this model, find the average rate of flow of gasoline over the time interval $0 \leq t \leq 150$. Show the setup for your calculations.

(d) Using the model g defined in part (c), find the value of $g'(140)$. Interpret the meaning of your answer in the context of the problem.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the response must reference gallons of gasoline added/pumped and the time interval $t = 60$ to $t = 135$.

To earn the second point at least five of the six factors in the Riemann sum must be correct.

If there is any error in the Riemann sum, the response does not earn the third point.

A response of $(0.15)(30) + (0.1)(30) + (0.05)(15)$ earns both the second and third points, unless there is a subsequent error in simplification, in which case the response would earn only the second point.

6-2

A response that presents a correct value with accompanying work that shows the three products in the Riemann sum but does not show all six of the factors and/or the sum process, does not earn the second point but does earn the third point. For example, responses of either $4.5 + 3.0 + 0.75$ or $(0.15)(30), 0.1(30), 0.05(15) \rightarrow 8.25$ earn the third point but not the second.

A response of $f(90)(90 - 60) + f(120)(120 - 90) + f(135)(135 - 120) = 8.25$ earns both the second and the third points.

A response that presents an answer of only 8.25 does not earn either the second or third point.

A response that provides a completely correct left Riemann sum with accompanying work, $f(60)(30) + f(90)(30) + f(120)(15) = 9$, or $(0.1)(30) + (0.15)(30) + (0.1)(15)$ earns 1 of the last 2 points. A response with any errors or missing factors in a left Riemann sum earns neither of the last 2 points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Interpretation with units
- ☐ Form of Riemann sum
- ☐ Answer

Solution:

$\int_{60}^{135} f(t) dt$ represents the total number of gallons of gasoline pumped into the gas tank from time $t = 60$ seconds to time $t = 135$ seconds.

$$\begin{aligned} & \int_{60}^{135} f(t) dt \\ & \approx f(90)(90 - 60) + f(120)(120 - 90) + f(135)(135 - 120) \\ & = (0.15)(30) + (0.1)(30) + (0.05)(15) = 8.25 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must present either $f(120) - f(60) = 0$, $0.1 - 0.1 = 0$ (perhaps as the numerator of a quotient), or $f(60) = f(120)$.

To earn the second point a response must:

- have earned the first point,
- state that f is continuous because f is differentiable (or equivalent), and
- answer “yes” in some way.

A response may reference either the Mean Value Theorem or Rolle’s Theorem.

A response that references the Intermediate Value Theorem cannot earn the second point.

6-2



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $f(120) - f(60) = 0$
- ☐ Answer with justification

Solution:

f is differentiable. $\Rightarrow f$ is continuous on $[60, 120]$.

$$\frac{f(120) - f(60)}{120 - 60} = \frac{0.1 - 0.1}{60} = 0$$

By the Mean Value Theorem, there must exist a c , for $60 < c < 120$, such that $f'(c) = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The exact value of $\frac{1}{150} \int_0^{150} g(t) dt$ is $\frac{12}{125} \sin\left(\frac{25}{16}\right)$.

A response may present the average value formula in single or multiple steps. For example, the following response earns both points: $\int_0^{150} g(t) dt = 14.399504$ so the average rate is 0.0959967.

A response that presents the average value formula in multiple steps but provides incorrect or incomplete communication (e.g., $\int_0^{150} g(t) dt = \frac{14.399504}{150} = 0.0959967$) earns 1 out of 2 points.

A response of $\int_0^{150} g(t) dt = 0.0959967$ does not earn either point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $\frac{1}{150} \int_0^{150} g(t) dt = 0.149981$ or 0.002618.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Average value formula
- ☐ Answer

Solution:

6-2

$$\frac{1}{150-0} \int_0^{150} g(t) dt = 0.0959967$$

The average rate of flow of gasoline, in gallons per second, is 0.096 (or 0.095).

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The exact value of $g'(140)$ is $\frac{1}{500} \cos\left(\frac{49}{36}\right) - \frac{49}{9000} \sin\left(\frac{49}{36}\right)$.

The value of $g'(140)$ may appear only in the interpretation.

To be eligible for the second point a response must present some numerical value for $g'(140)$.

To earn the second point the interpretation must include “the rate of flow of gasoline is changing at a rate of [the declared value of $g'(140)$]” and “at $t = 140$ ” (or equivalent).

An interpretation of “decreasing at a rate of -0.005 ” or “increasing at a rate of ” does not earn the second point.

Degree mode: In degree mode, $g'(140) = 0.001997$ or 0.00187 .



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $g'(140)$
- ☐ Interpretation

Solution:

$$g'(140) \approx -0.004908$$

$$g'(140) = -0.005 \text{ (or } -0.004)$$

The rate at which gasoline is flowing into the tank is decreasing at a rate of 0.005 (or 0.004) gallon per second per second at time $t = 140$ seconds.

6-2

37. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (weeks)	2	4	5	8	10
$L'(t)$ (feet per week)	1.0	0.8	0.7	0.4	0.2

The length of a vine during a 12-hour period is given by a twice-differentiable function L , where $L(t)$ is measured in feet and t is measured in weeks for $0 \leq t \leq 12$. The graph of L is concave down on the interval $0 \leq t \leq 12$. Selected values of the derivative of L , $L'(t)$, are given in the table above. At time $t = 4$, the length of the vine is 5 feet.

(a) Use the tangent line approximation for L at time $t = 4$ to estimate $L(4.3)$, the length of the vine at time $t = 4.3$. Is the approximation an overestimate or an underestimate for $L(4.3)$? Give a reason for your answer.

(b) Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\int_2^{10} L'(t) \, dt$. Indicate units of measure.

(c) Is the approximation in part (b) an overestimate or an underestimate for $\int_2^{10} L'(t) \, dt$? Give a reason for your answer.

(d) The sum $\sum_{k=1}^n L'\left(\frac{10k}{n}\right) \frac{10}{n}$ is a right Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

For the first point, the approximation must show the form of a correct tangent line equation and evaluation of that equation at $t = 4.3$. Numerical answers do not need to be simplified, and units are not required to earn the approximation point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ underestimate with reason

Solution:

6-2

$$L(4.3) \approx L(4) + L'(4)(4.3 - 4) = 5 + 0.8(0.3) = 5.24 \text{ feet}$$

Since the graph of L is concave down on $4 \leq t \leq 4.3$, the tangent line lies above the graph of L on this interval. Therefore, the approximation is an overestimate for $L(4.3)$.

Part B

To earn the first point, the response must present the form of a left Riemann sum. To earn the second point, values for L' must be substituted into the sum. Therefore, $2 \cdot 1.0 + 1 \cdot 0.8 + 3 \cdot 0.7 + 2 \cdot 0.4$ earns the first and second points. Numerical answers do not need to be simplified for the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ left Riemann sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned} \int_2^{10} L'(t) dt &\approx (4 - 2)L'(2) + (5 - 4)L'(4) + (8 - 5)L'(5) + (10 - 8)L'(8) \\ &= 2 \cdot 1.0 + 1 \cdot 0.8 + 3 \cdot 0.7 + 2 \cdot 0.4 = 5.7 \text{ feet} \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes overestimate with reason that references the behavior of L'

Solution:

Since the graph of L is concave down on $0 \leq t \leq 12$, L' is decreasing on $2 \leq t \leq 10$. Therefore, the left Riemann sum approximation is an overestimate for $\int_2^{10} L'(t) dt$.

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length 10. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-2



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\Delta t = \frac{10}{n}$
- ☐ limits of integration
- ☐ integrand

Solution:

For this Riemann sum, $\Delta t = \frac{10}{n}$. Thus, the integral is over an interval of length 10. This interval can be taken to be $0 \leq t \leq 10$.

The right endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{10k}{n}$ for k from 1 to n .

Therefore, $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n L' \left(\frac{10k}{n} \right) \frac{10}{n} \right) = \int_0^{10} L' (t) dt.$

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+10} L' (t - a) dt.$

6-2

38. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	2	4	5	8	10	14
$h'(t)$ (feet per hour)	0.3	0.6	0.7	1.4	2.3	7.7

The height of water in a storage tank during a 15-hour period is given by a twice-differentiable function h , where $h(t)$ is measured in feet and t is measured in hours since midnight for $0 \leq t \leq 15$. The graph of h is concave up on the interval $0 \leq t \leq 15$. Selected values of the derivative of h , $h'(t)$, are given in the table above. At time $t = 2$, the height of the water is 7 feet.

(a) Use the tangent line approximation for h at time $t = 2$ to estimate $h(2.8)$, the height of the water at time $t = 2.8$. Is the approximation an overestimate or an underestimate for $h(2.8)$? Give a reason for your answer.

(b) Use a right Riemann sum with the five subintervals indicated by the data in the table to approximate $\int_2^{14} h'(t) \, dt$. Indicate units of measure.

(c) Is the approximation in part (b) an overestimate or underestimate for $\int_2^{14} h'(t) \, dt$? Give a reason for your answer.

(d) The sum $\sum_{k=1}^n h' \left(\frac{15}{n}(k-1) \right) \frac{15}{n}$ is a left Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

For the first point, the approximation must show the form of a correct tangent line equation and evaluation of that equation at $t = 2.8$.

Numerical answers do not need to be simplified, and units are not required to earn the approximation point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ underestimate with reason

6-2

Solution:

$$h(2.8) \approx h(2) + h'(2)(2.8 - 2) = 7 + 0.3(0.8) = 7.24 \text{ feet}$$

Since the graph of h is concave up on $2 \leq t \leq 2.8$, the tangent line lies under the graph of h on this interval. Therefore, the approximation is an underestimate for $h(2.8)$.

Part B

To earn the first point, the response must present the form of a right Riemann sum.

To earn the second point, values for h' must be substituted into the sum. Therefore, $2 \cdot 0.6 + 1 \cdot 0.7 + 3 \cdot 1.4 + 2 \cdot 2.3 + 4 \cdot 7.7$ earns the first and second points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3
---	---	---	---



The student response accurately includes all three of the criteria below.

- ☐ right Riemann sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned} \int_2^{14} h'(t) dt &\approx (4 - 2)h'(4) + (5 - 4)h'(5) + (8 - 5)h'(8) + (10 - 8)h'(10) + (14 - 10)h'(14) \\ &= 2 \cdot 0.6 + 1 \cdot 0.7 + 3 \cdot 1.4 + 2 \cdot 2.3 + 4 \cdot 7.7 = 41.5 \text{ feet} \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1
---	---



The student response accurately includes the criteria below.

- ☐ overestimate with reason that references the behavior of h'

Solution:

Since the graph of h is concave up on $0 \leq t \leq 15$, h' is increasing on $2 \leq t \leq 14$. Therefore, the right Riemann sum approximation is an overestimate for $\int_2^{14} h'(t) dt$.

Part D

6-2

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length 15. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\Delta t = \frac{15}{n}$
- ☐ limits of integration
- ☐ integrand

Solution:

For this Riemann sum, $\Delta t = \frac{15}{n}$.

Thus, the integral is over an interval of length 15. This interval can be taken to be $0 \leq t \leq 15$.

The left endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{15}{n}(k-1)$ for k from 1 to n .

Therefore, $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n h' \left(\frac{15}{n}(k-1) \right) \frac{15}{n} \right) = \int_0^{15} h'(t) dt$.

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+15} h'(t-a) dt$.

6-2

39. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
$R(t)$ (gallons per hour)	16	12	7	3	0

The rate at which oil leaks from a drum is modeled by the twice-differentiable function R , where $R(t)$ is measured in gallons per hour and t is measured in hours for $0 \leq t \leq 2$. Values of $R(t)$ are given in the table above for selected values of t .

(a) Use the data in the table to find an approximation for $R'\left(\frac{3}{4}\right)$. Show the computations that lead to your answer. Indicate units of measure.

(b) Use a trapezoidal sum with the four subintervals indicated by the data in the table to approximate $\int_0^2 R(t) \, dt$. Indicate units of measure.

(c) Use the data in the table to evaluate $\int_0^1 R'(t) \, dt$.

(d) The sum $\sum_{k=1}^n R\left(\frac{1}{2} + \frac{k}{n}\right) \frac{1}{n}$ is a right Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

Numerical answers do not need to be simplified for the approximation point. The approximation must include a substitution of the function values from the table into the difference quotient.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ units

Solution:

6-2

$$R' \left(\frac{3}{4} \right) \approx \frac{R(1) - R\left(\frac{1}{2}\right)}{1 - \frac{1}{2}} = \frac{7 - 12}{\frac{1}{2}} = -10$$

gallons per hour per hour

Part B

To earn the first point, the response must present the form of a trapezoidal sum. Therefore,

$\frac{1}{2} \left(\frac{1}{2}(16 + 12) + \frac{1}{2}(12 + 7) + \frac{1}{2}(7 + 3) + \frac{1}{2}(3 + 0) \right)$ earns the first and second points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ trapezoidal sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned} \int_0^1 R(t) \, dt &\approx \frac{1}{2} \left(\frac{1}{2} \left(R(0) + R\left(\frac{1}{2}\right) \right) + \frac{1}{2} \left(R\left(\frac{1}{2}\right) + R(1) \right) + \frac{1}{2} \left(R(1) + R\left(\frac{3}{2}\right) \right) + \frac{1}{2} \left(R\left(\frac{3}{2}\right) + R(2) \right) \right) \\ &= \frac{1}{2} \left(\frac{1}{2}(16 + 12) + \frac{1}{2}(12 + 7) + \frac{1}{2}(7 + 3) + \frac{1}{2}(3 + 0) \right) \\ &= \frac{1}{4}(28 + 19 + 10 + 3) = 15 \text{ gallons} \end{aligned}$$

Part C

The response should include supporting work for the numerical answer of -9 .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct value.

Solution:

$$\int_0^1 R'(t) \, dt = R(1) - R(0) = 7 - 16 = -9$$

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length 1. A response that produces the correct definite integral earns all 3 points.

6-2

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\Delta t = \frac{1}{n}$
- ☐ limits of integration
- ☐ integrand

Solution:

For this Riemann sum, $\Delta t = \frac{1}{n}$. Thus, the integral is over an interval of length 1. This interval can be taken to be $\left[\frac{1}{2}, \frac{3}{2}\right]$.

The right endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{1}{2} + \frac{k}{n}$ for k from 1 to n .

Therefore,
$$\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n R\left(\frac{1}{2} + \frac{k}{n}\right) \cdot \frac{1}{n} \right) = \int_{\frac{1}{2}}^{\frac{3}{2}} R(t) \, dt.$$

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+1} R\left(\frac{1}{2} - a + t\right) \, dt$.

6-2

40. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	$\frac{1}{3}$	$\frac{2}{3}$	1
$R(t)$ (gallons per hour)	11	8	5	0

The rate at which water leaks from a container is modeled by the twice-differentiable function R , where $R(t)$ is measured in gallons per hour and t is measured in hours for $0 \leq t \leq 1$. Values of $R(t)$ are given in the table above for selected values of t .

(a) Use the data in the table to find an approximation for $R'(\frac{1}{2})$. Show the computations that lead to your answer. Indicate units of measure.

(b) Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate $\int_0^1 R(t) \, dt$. Indicate units of measure.

(c) Use the data in the table to evaluate $\int_0^{\frac{1}{3}} R'(t) \, dt$.

(d) The sum $\sum_{k=1}^n R\left(\frac{1}{4} + \frac{k}{2n}\right) \frac{1}{2n}$ is a right Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

Numerical answers do not need to be simplified for the approximation point. The approximation must include a substitution of the function values from the table into the difference quotient.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ units

Solution:

6-2

$$R' \left(\frac{1}{2} \right) \approx \frac{R \left(\frac{2}{3} \right) - R \left(\frac{1}{3} \right)}{\frac{2}{3} - \frac{1}{3}} = \frac{5 - 8}{\frac{1}{3}} = -9 \text{ gallons per hour per hour}$$

Part B

To earn the first point, the response must present the form of a left Riemann sum. Therefore, $\frac{1}{3}(11 + 8 + 5)$ earns the first and second points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ left Riemann sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned} \int_0^1 R(t) \, dt &\approx \frac{1}{3} \left(R(0) + R\left(\frac{1}{3}\right) + R\left(\frac{2}{3}\right) \right) \\ &= \frac{1}{3}(11 + 8 + 5) = 8 \text{ gallons} \end{aligned}$$

Part C

The response should include supporting work for the numerical answer of -3 .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct value.

Solution:

$$\int_0^{\frac{1}{3}} R'(t) \, dt = R\left(\frac{1}{3}\right) - R(0) = 8 - 11 = -3$$

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length $\frac{1}{2}$. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-2



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\Delta t = \frac{1}{2n}$
- ☐ limits of integration
- ☐ integrand

Solution:

For this Riemann sum, $\Delta t = \frac{1}{2n}$. Thus, the integral is over an interval of length $\frac{1}{2}$. This interval can be taken to be $\left[\frac{1}{4}, \frac{3}{4}\right]$.

The right endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{1}{4} + \frac{k}{2n}$ for k from 1 to n .

Therefore, $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n R\left(\frac{1}{4} + \frac{k}{2n}\right) \cdot \frac{1}{2n} \right) = \int_{\frac{1}{4}}^{\frac{3}{4}} R(t) \, dt.$

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+\frac{1}{2}} R\left(\frac{1}{4} - a + t\right) \, dt.$

6-2

41. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (years)	0	10	22	30	40
$A(t)$ (hundred pounds per year)	163	142	124	118	113

The amount of nitrogen in a particular river is affected by two sources. Nitrogen from the first source enters the river at a rate modeled by the decreasing, differentiable function A . Selected values of $A(t)$ are shown in the table above. Nitrogen from the second source enters the river at a rate modeled by the function B defined by $B(t) = 20 + \left(\frac{95}{1+\sqrt[3]{t}}\right)^{1.25}$. Both $A(t)$ and $B(t)$ are measured in hundreds of pounds of nitrogen per year, and t is measured in years for $0 \leq t \leq 40$.

- (a) Using a right Riemann sum with the four subintervals indicated by the data in the table, approximate the total amount of nitrogen from the first source that enters the river over the time interval $0 \leq t \leq 40$. Is this approximation an overestimate or an underestimate? Give a reason for your answer.
- (b) Find $B'(39)$. Using correct units, interpret the meaning of $B'(39)$ in the context of the problem.
- (c) Based on model B , find the average rate, in hundreds of pounds per year, at which nitrogen from the second source enters the river during the time interval $5 \leq t \leq 35$.
- (d) Using model B and your approximation from part (a), find an approximation for the total combined amount of nitrogen from the two sources that enters the river over the time interval $0 \leq t \leq 40$. Give your answer to the nearest hundred pounds.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a sum of four products with at most one error among the 8 factors. If there is an error in the setup, the second point is not earned.

A response of $10 \cdot A(10) + 12 \cdot A(22) + 8 \cdot A(30) + 10 \cdot A(40)$ would earn the first point, but not yet the second. The first two points are earned for $10 \cdot 142 + 12 \cdot 124 + 8 \cdot 118 + 10 \cdot 113$.

Without the products, the sum $1420 + 1488 + 944 + 1130$ by itself would earn the second point but not the first point.

6-2

Once the first point is earned it cannot be lost by incorrectly using values from the table or performing some type of operation on the Riemann sum.

A completely correct setup and approximation using a left Riemann sum

$(10 \cdot 163 + 12 \cdot 142 + 8 \cdot 124 + 10 \cdot 118 = 5506)$ will earn one of the first two points and is not eligible for the third point. Any error in the setup or evaluation using a left Riemann sum will earn no points.

Units are not scored in this part, except an approximation of 498,200 must clearly be labeled as *pounds* to earn the second point.

We will interpret the word “function” to mean A , unless otherwise prescribed.

The response must indicate which function is decreasing. A response of “underestimate because it (or the graph, or the curve) is decreasing” will not earn the third point.

The third point will not be earned if the response concludes the approximation is an underestimate because the table values are decreasing.

The third point will not be earned if incorrect statements are made in addition to saying A is decreasing.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Right Riemann sum setup
- ☐ Approximation
- ☐ Underestimate with reason

Solution:

$$\begin{aligned} \int_0^{40} A(t) \, dt &\approx (10 - 0) \cdot A(10) + (22 - 10) \cdot A(22) \\ &\quad + (30 - 22) \cdot A(30) + (40 - 30) \cdot A(40) \\ &= 10 \cdot 142 + 12 \cdot 124 + 8 \cdot 118 + 10 \cdot 113 = 4982 \end{aligned}$$

Approximately 4982 hundred pounds of nitrogen from the first source enters the river over the time interval $0 \leq t \leq 40$.

The approximation is an underestimate because a right Riemann sum is used and A is decreasing.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The value of $B'(39)$ may appear in the interpretation. No work is required in order to earn the first point.

In the presence of a correct value for $B'(39)$, any symbolic attempt at differentiation will be ignored.

Any unevaluated expression equivalent to $1.25 \left(\frac{95}{1 + \sqrt[3]{39}} \right)^{0.25} \left(\frac{-95}{\left(1 + \sqrt[3]{39}\right)^2} \right) \left(\frac{1}{3} (39)^{-2/3} \right)$ will earn the first point.

The second point cannot be earned if no value for $B'(39)$ is presented.

6-2

The units (hundred pounds per year per year) may be attached to the value of B' (39) and if so, they do not need to be repeated in the interpretation.

If units attached to B' (39) do not agree with units in the interpretation, read the units in the interpretation.

The interpretation must include time $t = 39$, rate of change of the rate at which nitrogen enters the river (or that the rate that nitrogen enters the river is decreasing at the stated rate), and units of hundred pounds per year per year.

The second point cannot be earned with an incorrect statement such as "... is decreasing at a rate of -0.385 ..." or the statement "... is increasing at a rate of -0.385 ..."

Use of the term "acceleration" of the amount of nitrogen coming from the second source will not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Value
- ☐ Interpretation

Solution:

$$B'(39) = -0.384963$$

At time $t = 39$ years, the rate at which nitrogen from the second source enters the river was decreasing at a rate of 0.385 (or 0.384) hundred pounds per year per year.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for presenting the definite integral $\int_5^{35} B(t) dt$.

Dividing by 30 will be scored in the second point.

Once the first point is earned, it cannot be lost.

If the first point is not earned because of a copy error in the integrand, the response can earn the second point for the correct answer.

An unsupported answer of 80.063 does not earn the second point.

Units are not scored in this part, except an answer of 8006.3 must clearly be labeled as *pounds per year* in order to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

6-2

- ☐ Definite integral
- ☐ Answer

Solution:

$$\frac{1}{35-5} \int_5^{35} B(t) dt = 80.063385$$

During the time interval $5 \leq t \leq 35$, the average rate at which nitrogen from the second source enters the river was 80.063 hundred pounds per year.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

For any $c > 0$, the presence of $c \int_0^{40} B(t) dt$ will earn the first point.

A copy error in the integrand imported from part (c) will earn the first point. Any new copy error will not earn the first point. In either case the second point can be earned if the correct answer is given.

Read for correctness or consistency with any positive value of $\int_0^{40} A(t) dt$ imported from part (a).

The second point is not earned for a correct value with no supporting work.

The second point is earned for an answer that is one of the four values 8368, 8369, 8368.35, or 8368.351 (with supporting work).

If one of the values 8368, 8369, 8368.35, or 8368.351 is presented but then rounded to 8400 (e.g., $8368 \approx 8400$), the response will earn the second point. However, a response of just 8400, without one of these four values present, will not earn the second point.

Units are not scored in this part, except an answer of 836,800 (or 836,900) must clearly be labeled as *pounds* in order to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $\int_0^{40} B(t) dt$
- ☐ Answer

Solution:

$$\text{Total} = \int_0^{40} A(t) dt + \int_0^{40} B(t) dt \approx 4982 + 3386.350701 \approx 8368$$

The total combined amount of nitrogen from the two sources that enters the river over the time interval $0 \leq t \leq 40$ is approximately 8368 hundred pounds.

6-2

42. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

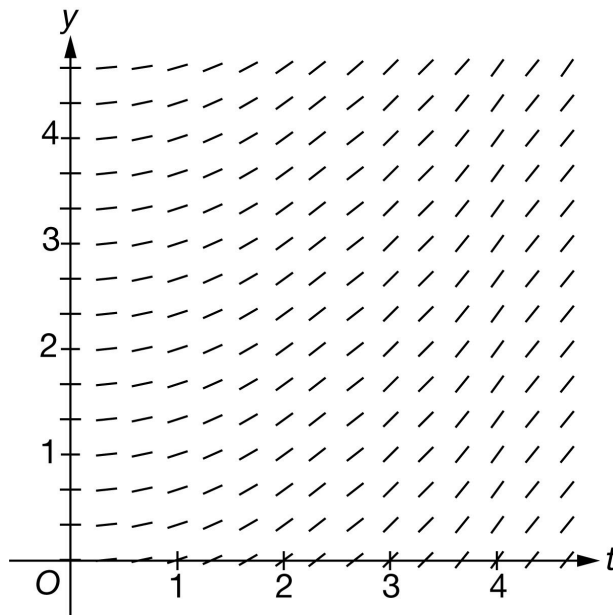
t (days)	0	4	10	18
$M(t)$ (number of 1-cent coins)	0	200	540	900

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 18$, the total number of coins in the container is modeled by an increasing, twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

- Using correct units, approximate $M'(10)$ using the average rate of change of M over the interval $4 \leq t \leq 18$.
- Explain the meaning of $M'(10)$ in the context of the problem.
- It is known that $M'(4) = 48$. Write an equation for the line tangent to the graph of $y = M(t)$ at $t = 4$.
- Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{18} M(t) \, dt$. Show the computations that lead to your answer.
- Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{18} M(t) \, dt$? Give a reason for your answer.
- Find the value of $\int_0^6 M'(3t) \, dt$. Show the computations that lead to your answer.
- For $0 \leq t \leq 18$, the total number of coins collected can also be modeled by the differentiable function N , where N is a function of t . Assume the weight of each coin is 2.5 grams. Write an expression for the rate of change, in grams per day, of the total weight of the coins at $t = 9$ days.

6-2

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\begin{aligned} & \cdot \frac{900-200}{18-4} \\ & \cdot \frac{M(18)-M(4)}{14} = \frac{700}{14} \\ & \cdot \frac{M(18)-M(4)}{14} = 50 \end{aligned}$$

Examples that do not earn the first point include:

$$\begin{aligned} & \cdot \frac{M(18)-M(4)}{14} \\ & \cdot \frac{M(18)-M(4)}{18-4} \end{aligned}$$

Minimal responses:

$$\begin{aligned} & \cdot \frac{900-200}{14} \text{ or } \frac{700}{18-4} \text{ earns first point} \\ & \cdot \text{Neither } \frac{700}{14} \text{ nor } 50, \text{ with no supporting work, will earn this point} \end{aligned}$$

The second point is for correct units

The correct units must be attached to a numerical value: either correct or incorrect.

6-2

The units may be written using words or a combination of words and symbols. For example:

- Coins per day
- Pennies per day
- Cents per day
- Coins/day

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$M'(10) \approx \frac{M(18) - M(4)}{18 - 4} = \frac{900 - 200}{14} = 50 \text{ coins per day}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of change
- Including the context of coin collection
- Referring to the specific point in time, $t = 10$

Students are not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “increasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “The number of coins is increasing at a rate of 50 coins per day...” is okay.

Acceptable references to a specific point in time include (but are not limited to):

- On day 10 and after 10 days
- At $t = 10$
- At $t = 10$ weeks or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from day 10 on,” “during the first 10 days,” or “for $t = 0$ to $t = 10$ ”), the point is not earned.

Examples that earn the point include:

6-2

- $M'(10)$ is the rate of change of the number of coins per day at time $t = 10$
- The rate of change of number of coins is 50 coins per day on day 10
- $M'(10)$ is the rate of change in the number of coins at time $t = 10$ in coins per *month*
- 50 is the rate of increase in the number of coins per day on day 10

Examples that do not earn the point include:

- $M'(10)$ is the average rate of change of the total number of coins per day on day 10
- $M'(10)$ is the total number of coins per day per day on day 10
- The rate of change is 50 coins per day (Missing “at $t = 10$ ”)
- $M'(10)$ is the slope of the tangent line at $t = 10$
- Rate of change of M at time $t = 10$
- Coins per day increases on day 10



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$M'(10)$ is the rate at which the number of 1-cent coins in the container is changing, in coins per day, on day 10.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable. Some examples include

- $y - 200 = 48(t - 4)$, $y - 200 = 48(x - 4)$
- $y = 48(t - 4) + 200$, $y = 48(x - 4) + 200$
- $y = 48t + 8$, $y = 48x + 8$

The key calculus concept being assessed in this part of the problem is that the value of $M'(4)$ is the slope of the tangent line. In that spirit, we are accepting both $y = \dots$ and $M(t) = \dots$, but not $M(4) = \dots$.

- $M(t) = 48(t - 4) + 200$ earns the point
- However, $M(4) = 48(t - 4) + 200$ does not earn the point

6-2



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = M'(4)(t - 4) + M(4)$$

$$y = 48(t - 4) + 200$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

· Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (d).

The second point is for the correct numerical result.

· In order to earn the second point, there must be supporting calculations.

· The numerical values for $M(4)$, $M(10)$, and $M(18)$ must be pulled from the table.

· Any simplification shown must be correct.

Use the following as an illustration to score both points. (In most cases, the expression on the left side represents the entirety of relevant work shown for the response.)

· A response of $200 \cdot (4 - 0) + 540 \cdot (10 - 4) + 900 \cdot (18 - 10)$ earns both points.

· A response of $4M(4) + 6M(10) + 8M(18)$ earns the first point. Eligible for second point if more work is on the page.

· A response of $4M(4) + 6M(10) + 8M(18) = 11,240$ earns both points.

· A response of $4 \cdot 200 + 6 \cdot 540 + 8 \cdot 900$ earns both points.

· A response of $4 \cdot 200 + 10 \cdot 540 + 8 \cdot 900$ contains one numerical error and earns the first point, but is not eligible for the second point.

· A response of $4 \cdot 200 + 6 \cdot 540 + 8 \cdot 900 = 10,240$ earns the first point for left side of equation but does not earn second point due to incorrect simplification.

6-2

· A response of $800 + 3240 + 7200$ or $800 + 3240 + 7200 = 11,240$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$4 \cdot 200 + 6 \cdot 540 + 8 \cdot 900$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{18} M(t) dt \approx M(4)(4-0) + M(10)(10-4) + M(18)(18-10)$$

$$= 200 \cdot 4 + 540 \cdot 6 + 900 \cdot 8 = 11,240$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned when the response states that the approximation is an overestimate because M is increasing.

The response does not need an explicit reference to the approximation in part (d), nor does there need to be an answer given in part (d).

The following statements are sufficient for earning this point:

- Overestimate because M is increasing
- Overestimate because the function is increasing

Note: At this point, there is only one function being discussed in this problem so we will assume “the function” is M .

When a response includes additional information about concavity, interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function M , the point is not earned.

Examples that earn the point:

- The right Riemann sum overestimates because the function is increasing.
- The sum is an overestimate because $M(t)$ is increasing.
- The sum is an overestimate because $M'(t) > 0$.

Examples that do not earn the point:

6-2

- The sum is an overestimate because $M'(t)$ is increasing. ($M'(t)$ increasing is not known.)
- The right Riemann sum overestimates because the graph is increasing.
- The right Riemann sum overestimates because the function is increasing and concave up.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because M is an increasing function, and this is a right Riemann sum, the answer from part (d) is an overestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that need to be seen:

- Uses $u = 3t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{3}$
- Uses the correct limits for the substitution (0 and 18). These limits may only be seen when evaluating the antiderivative. These don't need to be seen as limits on a definite integral.

· Either of the expressions below will earn the substitution point without any additional supporting work: $\frac{1}{3} M(3t) \Big|_0^6$ or $\frac{1}{3} M(u) \Big|_0^{18}$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{18}^0 M'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 6$ or $b = 18$. No explanation or additional supporting work is necessary.

- $kM(3t) \Big|_0^b$
- $k(M(b) - M(0))$

6-2

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 300.

Values from the table must be used. $\frac{1}{3}[M(18) - M(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{3}(900 - 0)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{3}(900)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 \, dt$.

When $t = 0$, $u = 0$.

When $t = 6$, $u = 18$.

$$\begin{aligned} \int_0^6 M'(3t) \, dt &= \frac{1}{3} \int_0^{18} M'(u) \, du \\ &= \frac{1}{3}(M(18) - M(0)) \\ &= \frac{1}{3}(900 - 0) = 300 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 9$. No supporting work or justification is required.

The correct expression of $2.5M'(9)$ or $2.5N'(9)$ earns this point.

No justifications or explanations are required.

Any incorrect statement such as, $M'(9) = 2.5N'(9)$, does not earn this point.

6-2



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

An expression for the rate of change, in grams per day, of the total weight of the coins at $t = 9$ days is $2.5N'(9)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = 0.4y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field., but it must explicitly reference one and implicitly reference the other.

Acceptable responses include:

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point $(4, 0)$, the slope of the segment should be 0, but it is positive.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point. For example, merely stating either $\frac{dy}{dt}$ is positive, or that the slope field is positive discretely does not earn the point.

- “The values for y are greater than or equal to zero so $\frac{dy}{dt} \geq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are positive” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is increasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0	1
---	---

6-2

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-2

43. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	6	13	18
$M(t)$ (coins per day)	42	38	35	30

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 18$, the rate at which coins are being added to the container is modeled by a decreasing, twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

(a) Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{18} M(t) \, dt$. Show the computations that lead to your answer.

(b) Is the approximation from part (a) an overestimate or an underestimate for $\int_0^{18} M(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $M'(13)$ using the average rate of change of M over the interval $6 \leq t \leq 18$.

(d) Explain the meaning of $M'(13)$ in the context of the problem.

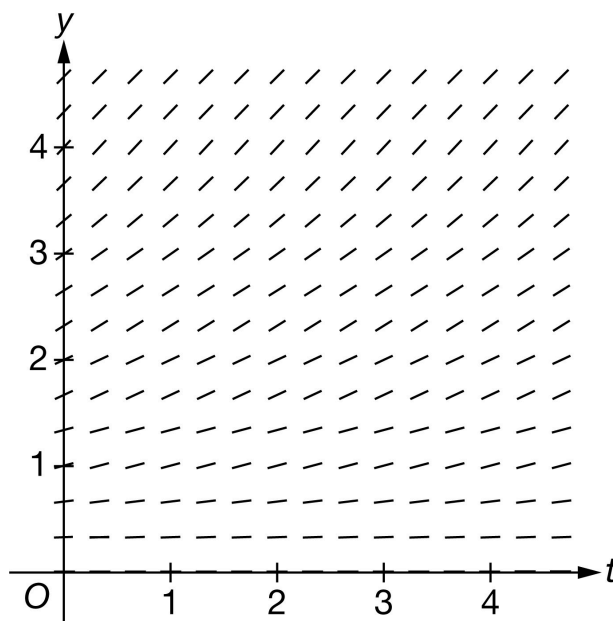
(e) It is known that $M'(6) = -0.5$. Write an equation for the line tangent to the graph of $y = M(t)$ at $t = 6$.

(f) Find the value of $\int_0^{18} M'\left(\frac{t}{3}\right) \, dt$. Show the computations that lead to your answer.

(g) A second class is also collecting 1-cent coins. For $0 \leq t \leq 18$, the total number of coins collected by the second class can be modeled by the decreasing and differentiable function G , where G is a function of t . Assume the weight of each coin is 3 grams. Write an expression for the rate of change, in grams per day, of the total weight of the coins for the second class at $t = 8$ days.

6-2

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = G'(t)$ for the function G defined in part (g).

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

38	6	228
35	7	245
30	5	+150
		623

or

$(6 - 0)(38)$	$= 228$
$(13 - 6)(35)$	$= 245$
$(18 - 13)(30)$	$= 150$
	623

Implicit responses such as those above are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (a).

The second point is for the correct numerical result.

6-2

In order to earn the second point, there must be supporting calculations.

The numerical values for $M(6)$, $M(13)$, and $M(18)$ must be pulled from the table

Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression represents the entirety of relevant work shown for the response.)

- A response of $38 \cdot (6 - 0) + 35 \cdot (13 - 6) + 30 \cdot (18 - 13)$ earns both points.
- A response of $6 \cdot M(6) + 7 \cdot M(13) + 5 \cdot M(18)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $6 \cdot M(6) + 7 \cdot M(13) + 5 \cdot M(18) = 623$ earns both points.
- A response of $6 \cdot 38 + 7 \cdot 35 + 5 \cdot 30$ earns both points.
- A response of $6 \cdot 38 + 7 \cdot 35 + 6 \cdot 30$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $6 \cdot 38 + 7 \cdot 35 + 5 \cdot 30 = 641$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $228 + 245 + 150$ or $228 + 245 + 150 = 623$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points. $6 \cdot 38 + 7 \cdot 35 + 5 \cdot 30$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{18} M(t) dt \approx M(6)(6-0) + M(13)(13-6) + M(18)(18-13)$$

$$= 38 \cdot 6 + 35 \cdot 7 + 30 \cdot 5 = 623$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because M is decreasing.

The response does not need an explicit reference to part (a) or to a right Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

6-2

The following examples earn the point:

- Underestimate because M is decreasing
- Underestimate because $M'(t) < 0$
- Underestimate because the graph of M is decreasing (we know they are talking about the given function M)
- Underestimate because the function is decreasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Underestimate because M is concave down/up
- Underestimate because the graph/curve is decreasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Underestimate because the table [or table values] is/are decreasing. The behavior of the values shown in the table is not the same as the behavior of the function M . The table does not tell us what is happening to M between the given points.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because M is a decreasing function, and this is a right Riemann sum, this is an underestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $M(18)$ and $M(6)$ must be pulled from the table.

Decimal approximation is not expected, but if presented, then the decimal value must be correct to three decimal places, either rounded or truncated. We do not read anything past the third decimal place.

- If the first three places match -0.666 or -0.667 then the point is earned.
- It does not matter whether any subsequent decimal places match the correct value or not. (For example, the value -0.6672 will earn the point because the first three decimal places match the correct rounded value.)
- Responses with fewer than three decimal places (-0.66 , -0.67 , -0.7 , etc.) will not earn the point

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

6-2

Can be written using words or a combination of words and symbols. For example:

- Coins per day per day or Coins/day/day
- Coins per day² or Coins/day²
- Coins per day squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter “C” is not a standard abbreviation for “coins.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points. $\frac{30-38}{12}$ coins/day²



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$M'(13) \approx \frac{M(18)-M(6)}{18-6} = \frac{30-38}{12} = -\frac{2}{3} \text{ coin per day per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of a rate
- Including the context of coin collection
- Explicitly referring to the specified point in time ($t = 13$)

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use $-\frac{2}{3}$ as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of $-\frac{2}{3}$ ” is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)

6-2

- “... decreasing at a rate of $\frac{2}{3}$ ” is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)
- “... decreasing at a rate of $-\frac{2}{3}$ ” is not OK and the point is not earned.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

- The rate of change in the rate at which coins are changing
- The rate of change in coins is changing (or decreasing) at a rate of ...
- The rate at which the rate of change of coins is changing

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “coins per day” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

“Rate of change of M ” is not enough without describing what M represents in this context.

Context

A key component of the context for this explanation is “coins.”

Explicit reference to $t = 13$

Acceptable reference to specific point in time include (but are not limited to):

- On day 13 (where day could be replaced with any other unit of time)
- At $t = 13$ (with or without units of time)
- After 13 days (where days could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from day 13 on” or “during the first 13 days”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐

Explanation

Solution:

$M'(13)$ is the rate at which the rate of change of the number of 1-cent coins in the container is changing, in coins per day per day, on day 13.

Part E

6-2

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

- $y = -0.5(t - 6) + 38$

- $y - 38 = -0.5(t - 6)$

- $y = -0.5t + 41$

The key calculus concept being assessed in this part of the problem is that the value of $M'(6)$ is the slope of the tangent line. In that spirit:

- The response will not be penalized for using x instead of t .
- We are accepting $M(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = M'(6)(t - 6) + M(6)$$

$$y = -0.5(t - 6) + 38$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that we need to see:

- Uses $u = \frac{t}{3}$, which may be done implicitly
- Uses the correct constant of 3
- Uses the correct limits for the substitution (0 and 6). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $3M\left(\frac{t}{3}\right)\bigg|_0^{18}$ or $3M(u)\bigg|_0^6$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa)

6-2

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\int_6^0 M'(u) \, du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 6$ or $b = 18$. No explanation or additional supporting work is necessary.

o $kM\left(\frac{t}{3}\right)\Big|_0^b$

o $k(M(b) - M(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is -12 .

Values from the table must be used. $3(M(6) - M(0))$ is not enough to earn the answer point.

The expression $3(38 - 42)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $3(38 - 42)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{3}$. Then $du = \frac{dt}{3}$.

When $t = 0$, $u = 0$.

When $t = 18$, $u = 6$.

$$\int_0^{18} M'\left(\frac{t}{3}\right) dt = 3 \int_0^6 M'(u) du$$

$$= 3(M(6) - M(0))$$

$$= 3(38 - 42) = -12$$

6-2

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 8$ earns the point. It may be written as $3 \cdot G'(8)$ or $3 \cdot \frac{dG}{dt} \Big|_{t=8}$.

The expression $3 \cdot G'(t)$ is not sufficient because it does not use $t = 8$.

The response must use G (not M or any other function name) in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

An expression for the rate of change, in grams per day, of the total weight of the coins for the second class at $t = 8$ days is $3G'(8)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation $\frac{dy}{dt} = G'(t)$ and the given slope field.

In order to earn the point, the response must state that the values of the differential equation are negative while the slopes in the given slope field are positive or 0 (nonnegative).

Some ways to say that the values of differential equation are negative:

- G is decreasing
- $G'(t) < 0$
- G has negative slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has nonnegative slopes.
- The slopes shown are positive or 0.
- The slope field contains positive slopes.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

6-2

- The slope field is increasing/positive.
- The slope field looks like it represents an increasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ G is a decreasing function, so the signs of the slopes in the slope field should be negative” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” positive or “all” positive.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

G is a decreasing function, so $\frac{dy}{dt} = G'(t) \leq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

6-2

44. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	6	18
$M(t)$ (coins per day)	40	54	46

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 18$, the rate at which coins are being added to the container, in coins per day, is modeled by a twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $M'(6)$ using the average rate of change of M over the interval $0 \leq t \leq 18$.

(b) Explain the meaning of $M'(6)$ in the context of the problem.

(c) The function $C(t)$ models the number of coins in the container at time t days, where $C'(t) = M(t)$. It is known that $C(6) = 300$. Write an equation for the line tangent to the graph of $y = C(t)$ at $t = 6$.

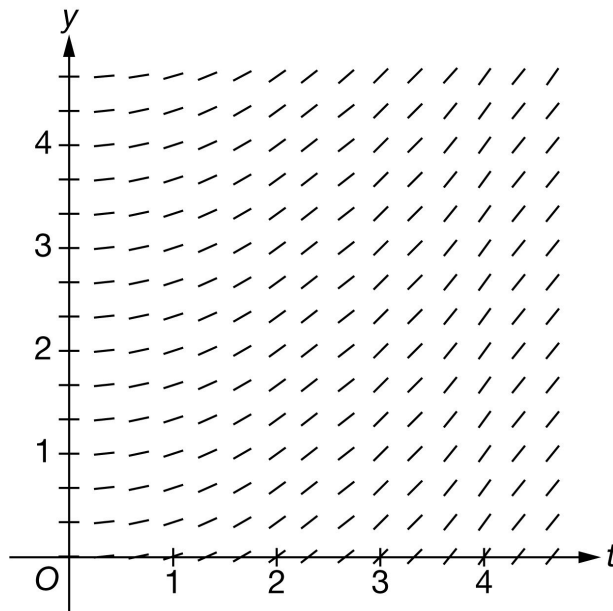
(d) Find the value of $\int_0^6 M'(3t) \, dt$. Show the calculations that lead to your answer.

(e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{18} M(t) \, dt$. Show the calculations that lead to your answer.

(f) A second class is also collecting 1-cent coins. For $0 \leq t \leq 18$, the rate at which coins are being collected by the second class can be modeled by the differentiable function B , where B is a function of t . Assume the weight of each coin is 3 grams. Write an expression in terms of M and B for the rate of change, in grams per day, of the total weight of the coins collected by both classes at $t = 7$ days.

6-2

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = 0.3y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{46-40}{18}$ coins per day per day.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

6-2

$$M'(6) \approx \frac{M(18) - M(0)}{18 - 0} = \frac{46 - 40}{18} = \frac{1}{3} \text{ coin per day per day}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of coin collection
- Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

- $M'(6)$ is the average (or an approximation of, or an estimate of) rate of change of
- $M'(6)$ is the acceleration (or rate of change of velocity) of

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $M'(6)$ is the rate of increase of the rate of change of the number of coins at 6 days. (earns the point)
- Example: $M'(6)$ means the number of coins per day is increasing by 6 coins/day/day. (does not earn the point)

Acceptable references to time include (but are not limited to):

- On day 6
- At $t = 6$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 6$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “from day 0 to day 6”) the response will not earn the point.

Examples that earn the point include:

- $M'(6)$ represents the rate at which the rate of change of the number of coins is changing on day 6.
- $M'(6)$ is the rate of change in the rate of change of the number of coins at 6 days.
- $M'(6)$ represents the rate at which the rate of change of the number of coins is increasing at $t = 6$.
- $M'(6)$ equals the answer from part (a) means the rate of change in the number of coins per day is increasing at a rate of (answer from part (a)) at $t = 6$.
- $M'(6)$ represents the rate at which the rate of change of the number of coins is decreasing on the sixth day.

Examples that do not earn the point include:

- $M'(6)$ represents the rate at which the number of coins is changing on day 6.

6-2

- $M'(6)$ is the rate of change in the number of coins per day at $t = 6$ days
- The rate of change in the number of coins per day is increasing at $t = 6$.
- The rate of change in the rate of change of the number of coins per day is increasing at $t = 6$.
- $M'(6) = -\frac{1}{3}$ means that the rate of change in the number of coins is decreasing at a rate of $-\frac{1}{3}$ at $t = 6$. (double negative).



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$M'(6)$ is the rate at which the rate of change of the number of 1-cent coins in the container is changing, in coins per day per day, on day 6.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

- $y = 54(t - 6) + 300$
- $y - 300 = 54(t - 6)$
- $y = 54t - 24$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $C(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = C'(6)(t - 6) + C(6)$$

$$y = 54(t - 6) + 300$$

Part D

6-2

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required to award this point:

- Coefficient of $\frac{1}{3}$ correctly applied to an integral or an antiderivative
- Evidence that the correct limits of 0 to 18 will be or have been applied

Examples that will earn the first point (these are just a few):

$$\cdot \frac{1}{3} \int_0^{18} M'(u) \, du$$

$$\cdot \frac{1}{3} M(u) \Big|_0^{18} \text{ or } \frac{1}{3} M(3t) \Big|_0^6$$

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\cdot \int_0^6 M'(3t) \, dt = \frac{1}{3} \int_0^6 M'(u) \, du = \frac{1}{3} M(3t) \Big|_0^6 \text{ earns the first point, but will not earn the third point.}$$

$$\cdot \int_0^6 M'(3t) \, dt = \frac{1}{3} \int_0^6 M'(u) \, du = \frac{1}{3} [M(6) - M(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b M'(u) \, du = M(u) \Big|_a^b = M(b) - M(a)$$

This point can be earned with either $kM(3t) \Big|_a^b$, $kM(t) \Big|_a^b$, $k(M(b) - M(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 6$ or $b = 18$.

A response can earn this point without earning the first point.

$$\frac{1}{3} [M(18) - M(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{3} [M(18) - M(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\cdot \int_0^6 M'(3t) \, dt = \frac{1}{3} \int M'(u) \, du = \frac{1}{3} M(3t) \Big|_0^6 = \dots \text{ is eligible for the third point.}$$

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{3}(46 - 40)$.

6-2



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 dt$.

When $t = 0$, $u = 0$.

When $t = 6$, $u = 18$.

$$\begin{aligned}\int_0^6 M'(3t) dt &= \frac{1}{3} \int_0^{18} M'(u) du \\ &= \frac{1}{3}(M(18) - M(0)) \\ &= \frac{1}{3}(46 - 40) = 2\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the form of a trapezoidal sum.

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point (e.g., $\frac{1}{2}[6(M(0) - M(6)) + 12(M(6) - M(18))]$ earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one incorrect numerical values.

$\frac{1}{2}(6)[M(0) + 2M(6) + M(18)]$ is equivalent to $\frac{1}{2}[6(M(0) + M(6)) + 6(M(6) + M(18))]$, thus has only one incorrect numerical value (the second interval width is 6 instead of 12), so can earn the point.

6-2

$\cdot \frac{1}{2}(9)[M(0) + 2M(6) + M(18)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without the first if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn only the first point:

$\cdot k \left[\left(\frac{40+54}{2} \right) \cdot 6 + \left(\frac{54+46}{2} \cdot 12 \right) \right]$ for $k > 0$ earns the first but not the second point.

The minimum response needed to earn both points: $\left(\frac{94}{2} \right) \cdot 6 + \left(\frac{100}{2} \right) \cdot 12$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\int_0^{18} M(t) dt \approx \left(\frac{M(0)+M(6)}{2} \right) (6-0) + \left(\frac{M(6)+M(18)}{2} \right) (18-6)$$

$$= \left(\frac{40+54}{2} \right) \cdot 6 + \left(\frac{54+46}{2} \right) \cdot 12 = 882$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 3 as a factor of the expression, and/or the correct sum $B(7) + M(7)$.

If the sum is incorrect, then the constant factor of 3 must be applied either to a sum that involves only $B(t)$ and $M(t)$, $B'(t)$ and $M'(t)$, or integrals of $B(t)$ and $M(t)$, or to one of the correct terms of the sum $B(7)$ and $M(7)$, with no additional incorrect terms.

Examples that earn the point include:

$$\cdot 3 [M(t) + B(t)]$$

$$\cdot 3 \int [M(t) + B(t)] dt$$

6-2

$$\cdot 3 [M' + B']$$

$$\cdot 3B(7)$$

Examples that do not earn the point include:

$$\cdot 3 [M(t) + B'(t)]$$

$$\cdot 3 \int M(t) dt$$

$$\cdot 3M' + B'$$

$$\cdot 3B(7) + t$$

Special case: $3M(7) + B(7)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[B(7) + M(7)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$3B(7) + 3M(7)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn just the first point. For example, $3B(7) + 3M(7) + 3(2)$ earns only the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

An expression for the rate of change, in grams per day, of the total weight of the coins collected by both classes at time $t = 7$ days is $3B(7) + 3M(7)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y as t changes.”)

6-2

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(1, 2)$ the slope shown on the slope field is 1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-2

45. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

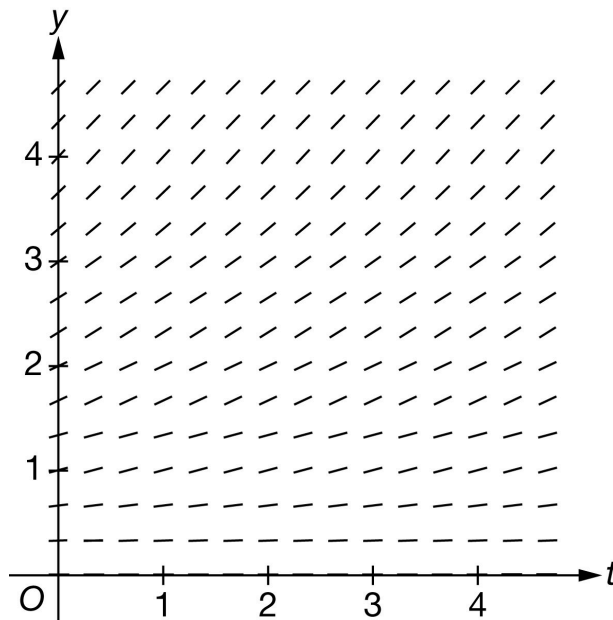
t (minutes)	0	10	20	30	40
$N(t)$ (number of 5-cent coins remaining)	1200	1000	750	400	0

A child has collected 5-cent coins and is preparing them to deposit in the bank by counting them into bags. The child begins with 1200 coins. For time t minutes, $0 \leq t \leq 40$, the total number of coins remaining to prepare is modeled by a decreasing, twice-differentiable function N . Values of $N(t)$ at various times t are shown in the table above.

- (a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate $\int_0^{40} N(t) \, dt$. Show the computations that lead to your answer.
- (b) Find the value of $\int_0^{10} N'(2t) \, dt$. Show the computations that lead to your answer.
- (c) Using correct units, approximate $N'(20)$ using the average rate of change of N over the interval $10 \leq t \leq 30$.
- (d) Explain the meaning of $N'(20)$ in the context of the problem.
- (e) It is known that $N'(10) = -20$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 10$.
- (f) It is known that $N''(t) < 0$ for $10 \leq t \leq 20$. If $N(17)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $N(17)$? Give a reason for your answer.
- (g) For $0 \leq t \leq 40$, the total number of 5-cent coins remaining can also be modeled by the differentiable function G , where G is a function of t . Assume the weight of each coin is 6 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the 5-cent coins remaining at $t = 37$ minutes.

6-2

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 20 \cdot (N(10) + N(30)) \text{ or } 20 \cdot N(10) + 20 \cdot N(30)$$

$$\cdot (20 - 0) \cdot 1000 + (40 - 20) \cdot 400$$

$$\cdot 20 \cdot (1000 + 400) \text{ or } 20 \cdot 1000 + 20 \cdot 400$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{20}{b} \cdot (N(10) + N(30))$, for any nonzero value of b would earn the point. Note: The value $\frac{20}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (N(10) + N(30))$ for $a \neq 20$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 20 \cdot (1000 + 400)$$

$$\cdot 20 \cdot 1000 + 20 \cdot 400$$

6-2



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint sum
- ☐ Answer

Solution:

$$\int_0^{40} N(t) dt \approx N(10)(20 - 0) + N(30)(40 - 20),$$

$$= 1000 \cdot 20 + 400 \cdot 20 = 28,000$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 2t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{2}$

Uses the correct limits for their substitution.

- The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .
- Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int N'(2t) dt$ or $\int N'(u) du$.

The point is earned for any of the following:

- The equation $\int N'(2t) dt = N(2t)$ or $\int N'(u) du = N(u)$
- $\frac{1}{2}N(2t)|_0^{10}$ or $\frac{1}{2}N(u)|_0^{20}$
- $N(20) - N(0)$, $N(20)$, or $N(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b N'(u) du = N'(b) - N'(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of N' is N .

Third Point (Answer)

6-2

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$N(20)$ and $N(0)$ must be evaluated; the expression $\frac{1}{2}(N(20) - N(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{2}(750 - 1200)$ or $\frac{1}{2}(750) - \frac{1}{2}(1200)$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

$$\cdot \frac{1}{2} N(2t) \Big|_0^{10}$$

$$\cdot \frac{1}{2} (N(20) - N(0))$$

Responses of only the following would earn only the Second point:

$$\cdot N(20) - N(0)$$

$$\cdot N(10) - N(0)$$

$$\cdot \int_0^{10} N'(u) \, du = N(10) - N(0)$$

$$\cdot N(2t) \Big|_0^{10} \text{ or } N(u) \Big|_0^{20}$$

The following responses would earn no points:

$$\cdot -225 \text{ in the absence of one of the first two points.}$$

$$\cdot \int_0^{10} N'(2t) \, dt = N(10) - N(0)$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2 \, dt$.

When $t = 0$, $u = 0$.

6-2

When $t = 10$, $u = 20$.

$$\int_0^{10} N'(2t) dt = \frac{1}{2} \int_0^{20} N'(u) du$$

$$= \frac{1}{2}(N(20) - N(0))$$

$$= \frac{1}{2}(750 - 1200) = -225$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{400-1000}{20}$ or $\frac{-600}{30-10}$.

Second point (Units)

The units must be attached to a value (either correct or incorrect).

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(20) \approx \frac{N(30) - N(10)}{30 - 10} = \frac{400 - 1000}{20} = -30 \text{ coins per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

- indicate the rate of change
- the specific context of the number of coins remaining
- the specific point in time, $t = 20$.

Responses do not need to include correct units or any units.

6-2

The word “decreasing” may be used in place of the word “changing”.

The response is not earned for:

· “the rate of coins remaining at minute 20 ” as this does not indicate a change in the rate of deposit.

· “decreasing at a rate of -30 coins per minute”

“the rate of change of N ” without specifying that N is the number of coins remaining, does not earn the point.

Acceptable ways to provide the specific point in time:

· At minute 20 or after 20 minutes.

· At $t = 20$ (missing the unit of time)

· At 20 seconds (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 20 minutes” or “from minute 20 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(20)$ is the rate at which the number of 5-cent coins remaining is changing, in coins per minute, at time $t = 20$ minutes.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $N(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

· $y = 1000 - 20(t - 10)$

· $y - 1000 = 20(t - 10)$

· $y = -20t + 200 + 1000$

· $y = -20t + 1200$



0	1
---	---

6-2

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = N'(10)(t - 10) + N(10)$$

$$y = -20(t - 10) + 1000$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point the response must include $N''(t) < 0$ and overestimate.

“ $N(t)$ is concave down” can be used instead of $N''(t) < 0$.

$N'(t)$ by itself is not sufficient and does not earn this point

“Overestimate because it is concave down” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states $N''(t) < 0$ and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $N'(t)$ is concave down, overestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because $N''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $6 \cdot G'(37)$ earns this point.

Only expression $6 \cdot N'(37)$ also earns this point.

No justifications or explanations are required.

Any incorrect statement such as, $G'(37) = 6 \cdot N'(37)$, does not earn this point.

6-2



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the 5-cent coins remaining at $t = 37$ minutes is $6G'(37)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

- “ $\frac{dy}{dt}$ is negative, not positive”. We will infer that positive is referring to the slopes shown in the slope field.
- “This would be correct for a positive derivative, but $\frac{dy}{dt}$ is negative.” We will read “This” as “the slope field” in this sentence.
- “All of the slopes in the slope field are positive, not negative.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are negative.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

- “ $\frac{dy}{dt}$ is negative” lacks any reference to the slope field and so will not earn the point.
- “All the slopes in the graph are positive” lacks any reference to the differential equation, and so will not earn the point.
- Vague answers such as “The slope field has positive points” or “The field is increasing” will not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

6-2

When $y > 0$, $\frac{dy}{dt} < 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

6-2

46. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

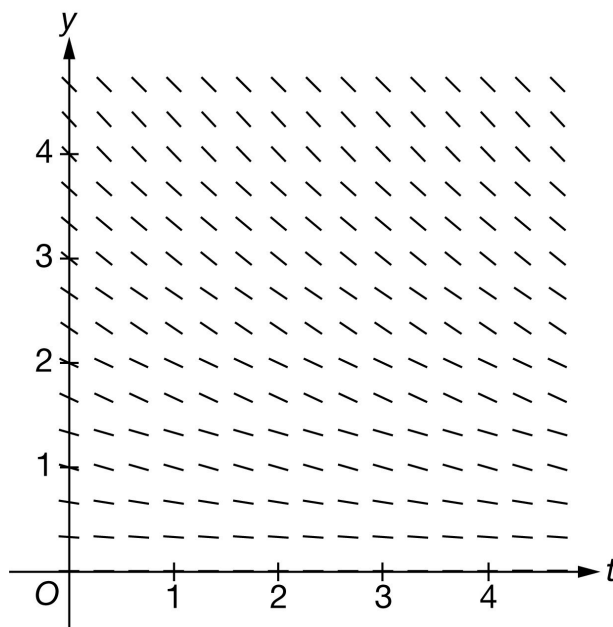
t (minutes)	0	10	35	40
$N(t)$ (coins per minute)	-8	-11	-13	-15

A child has collected 5-cent coins and is preparing them to deposit in the bank by counting them into bags. For time t minutes, $0 \leq t \leq 40$, the rate of the total number of coins remaining to prepare, in 5-cent coins per minute, is modeled by a decreasing, twice-differentiable function N . Values of $N(t)$ at various times t are shown in the table above.

- (a) Use a left Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{40} N(t) \, dt$. Show the calculations that lead to your answer.
- (b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{40} N(t) \, dt$? Give a reason for your answer.
- (c) Using correct units, approximate $N'(10)$ using the average rate of change of N over the interval $0 \leq t \leq 35$.
- (d) Explain the meaning of $N'(10)$ in the context of the problem.
- (e) It is known that $N'(35) = -0.2$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 35$.
- (f) Find the value of $\int_0^{40} N'\left(\frac{t}{4}\right) \, dt$. Show the calculations that lead to your answer.
- (g) A second child is also preparing 5-cent coins for deposit. For $0 \leq t \leq 40$, the total number of 5-cent coins prepared by the second child for deposit can be modeled by the increasing, differentiable function F , where F is a function of t . Assume the weight of each coin is 5 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the 5-cent coins remaining for the second child at $t = 23$ minutes.

6-2

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = F'(t)$ for the function F defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

10	(-8)	-80
25	(-11)	-275
5	(-13)	+(-65)
		-420

or

$(10 - 0)(-8)$	$= -80$
$(35 - 10)(-11)$	$= -275$
$(40 - 35)(-13)$	$= -65$
	-420

Implicit responses such as those above are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

6-2

In order to earn the second point, there must be supporting calculations.

The numerical values for $N(0)$, $N(10)$, and $N(35)$ must be pulled from the table.

Any simplification shown must be correct.

Use the following as an illustration of how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $(-8) \cdot (10 - 0) + (-11) \cdot (35 - 10) + (-13) \cdot (40 - 35)$ earns both points.
- A response of $10 \cdot N(0) + 25 \cdot N(10) + 5 \cdot N(35)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $10 \cdot N(0) + 25 \cdot N(10) + 5 \cdot N(35) = -420$ earns both points.
- A response of $10 \cdot (-8) + 25 \cdot (-11) + 5 \cdot (-13)$ earns both points.
- A response of $10 \cdot (-8) + 35 \cdot (-11) + 5 \cdot (-13)$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $10 \cdot (-8) + 25 \cdot (-11) + 5 \cdot (-13) = -400$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $(-80) + (-275) + (-65)$ or $(-80) + (-275) + (-65) = -420$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$10 \cdot (-8) + 25 \cdot (-11) + 5 \cdot (-13)$$

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{40} N(t) dt \approx N(0)(10 - 0) + N(10)(35 - 10) + N(35)(40 - 35)$$

$$= (-8) \cdot 10 + (-11) \cdot 25 + (-13) \cdot 5 = -420$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because N is decreasing.

6-2

The response does not need an explicit reference to part (a) or to a left Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following table contains some examples of what WILL or will NOT earn the point.

The following examples earn the point:

- Underestimate because N is decreasing
- Underestimate because $N'(t) < 0$
- Underestimate because the graph of N is decreasing (we know they are talking about the given function N)
- Underestimate because the function is decreasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Underestimate because N is concave down/up
- Underestimate because the graph/curve is decreasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Underestimate because the table [or table values] is/are decreasing. The behavior of the values shown in the table is not the same as the behavior of the function N . The table does not tell us what is happening to N between the given points.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because N is a decreasing function, and this is a left Riemann sum, this is an overestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $N(35)$ and $N(0)$ must be pulled from the table.

Decimal approximation is not expected, but if presented, then the decimal value must be correct to three decimal places, either rounded or truncated. We do not read anything past the third decimal place.

- If the first three places match -0.142 or -0.143 then the point is earned.
- It does not matter whether any subsequent decimal places match the correct value or not. (For example, the value -0.1424 will earn the point because the first three decimal places match the correct value.)
- Responses with fewer than 3 decimal places (-0.14 , -0.1 , etc.) will not earn the point

6-2

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Coins per minute per minute or Coins/min/min
- Coins per min^2 or Coins/ min^2
- Coins per minute squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter “C” is not a standard abbreviation for “Coins.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to

earn both points. $\frac{-13+8}{35}$ coins/ min^2 or $\frac{-13-(-8)}{35}$ coins/ min^2



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(10) \approx \frac{N(35)-N(0)}{35-0} = \frac{-13-(-8)}{35} = -\frac{1}{7} \text{ coin per minute per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of a rate
- Including the context of coin preparation (coins remaining)
- Explicitly referring to the specified point in time ($t = 10$)

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

6-2

The following examples use $\frac{-1}{7}$ as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of $\frac{-1}{7}$ ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)
- “... decreasing at a rate of $\frac{-1}{7}$ ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)
- “... decreasing at a rate of $\frac{-1}{7}$ ” is not OK and the point is not earned.

Note: the context is complicated. $N(t)$ is describing the rate of change of the remaining unprepared coins. The values in the table indicate that the number of unprepared coins is decreasing more rapidly as time passes. Common sense might tell us that this equates to the idea that the rate at which coins are prepared is increasing as time passes. Explaining the meaning using “coins prepared” rather than “coins remaining” will not prevent the response from earning this point.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

- The rate of change in the rate at which coins are changing
- The rate of change in coins is changing/increasing/decreasing at a rate of ...
- The rate at which the rate of change of coins is changing

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “coins per minute” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

“Rate of change of N ” is not enough without describing what N represents in this context.

Context

A key component of the context for this explanation is “coins.”

Any misstatement about coins prepared vs. coins remaining will not prevent the response from earning this point.

Explicit reference to $t = 10$

Acceptable references to a specific point in time include (but are not limited to):

- At minute 10 (where minute could be replaced with any other unit of time)
- At $t = 10$ (with or without units of time)
- After 10 minutes (where minutes could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from minute 10 on” or “during the first 10 minutes”), this point will not be earned.



0

1

6-2

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(10)$ is the rate at which the rate of change in the number of 5-cent coins remaining is changing, in coins per minute per minute, at minute 10.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

· $y = -0.2(t - 35) - 13$

· $y + 13 = -0.2(t - 35)$

· $y = -0.2t - 6$

The key calculus concept being assessed in this part of the problem is that the value of $N'(35)$ is the slope of the tangent line. In that spirit:

- The response will not be penalized for using x instead of t .
- We are accepting $N(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$y = N'(35)(t - 35) + N(35)$

$y = -0.2(t - 35) - 13$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = \frac{t}{4}$, which may be done implicitly
- Uses the correct constant of 4

6-2

· Uses the correct limits for the substitution (0 and 10). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $4N\left(\frac{t}{4}\right)\bigg|_0^{40}$ or $4N(u)\bigg|_0^{10}$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

· Limits for t attached to an integral with u as the variable of integration (or vice versa)

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\int_{10}^0 N'(u) du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 10$ or $b = 40$. No explanation or additional supporting work is necessary.

· $kN\left(\frac{t}{4}\right)\bigg|_0^b$

· $k(N(b) - N(0))$

This point can be earned without the substitution point.

The third point is for the correct numerical value, which is -12 .

Values from the table must be used. $4(N(10) - N(0))$ is not enough to earn the answer point.

The expression $4(-11 - (-8))$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they do not need to be correct to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $4(-11 - (-8))$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{4}$. Then $du = \frac{dt}{4}$.

6-2

When $t = 0$, $u = 0$.

When $t = 40$, $u = 10$.

$$\begin{aligned}\int_0^{40} N' \left(\frac{t}{4} \right) dt &= 4 \int_0^{10} N' (u) du \\ &= 4 (N (10) - N (0)) \\ &= 4 (-11 - (-8)) = -12\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy for this part: This question was difficult due to the fact that the new function introduced here represents the total number of coins prepared by the second child, but the question asks about the total weight of the coins remaining.

The key calculus concept being assessed in this part of the problem is to understand that the rate of change in the total weight of the coins (either prepared or remaining) at time $t = 23$ could be written as an expression involving a constant multiple of $F' (23)$. In that spirit, either of the following expressions will earn the point: $5F' (23)$ or $-5F' (23)$.

Notations such as $5 \cdot \frac{dF}{dt} \Big|_{t=23}$ may also be used and earn the point.

The expression must use the time $t = 23$, so $5F' (t)$ is not sufficient to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

Let $C (0)$ be the total number of coins available to be prepared at time $t = 0$. Let $R (t)$ be a model for the number of coins remaining at time t with $R (t) = C (0) - F (t)$. The rate of change of the number of coins remaining at time t , in grams per minute, is $R' (t) = -F' (t)$. Therefore, the rate of change of the total weight of the 5 -cent coins remaining for the second child at $t = 23$ minutes, in grams per minute, is $-5F' (23)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation $\frac{dy}{dt} = F' (t)$ and the given slope field.

In order to earn the point, the response must:

State that the values of the differential equation are positive while the slopes in the given slope field are negative or 0 (non-positive).

Some ways to say that the values of differential equation are positive:

6-2

- F is increasing
- $F'(t) > 0$
- F has positive slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has nonpositive slopes.
- The slopes shown are negative or 0.
- The slope field contains negative slopes.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is decreasing/negative.
- The slope field looks like it represents a decreasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ F is an increasing function, so the signs of the slopes in the slope field should be positive” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” negative or “all” negative.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

F is an increasing function, so $\frac{dy}{dt} = F'(t) \geq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

6-2

47. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	10	35	40
$N(t)$ (number of 5-cent coins)	1000	810	200	0

A child has collected 5-cent coins and is preparing them to deposit in the bank by counting them into bags. The child begins with 1000 coins. For time t minutes, $0 \leq t \leq 40$, the total number of coins remaining to prepare is modeled by a decreasing, twice-differentiable function N . Values of $N(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $N'(35)$ using the average rate of change of N over the interval $10 \leq t \leq 40$.

(b) Explain the meaning of $N'(35)$ in the context of the problem.

(c) It is known that $N'(10) = -20$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 10$.

(d) Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{40} N(t) \, dt$. Show the computations that lead to your answer.

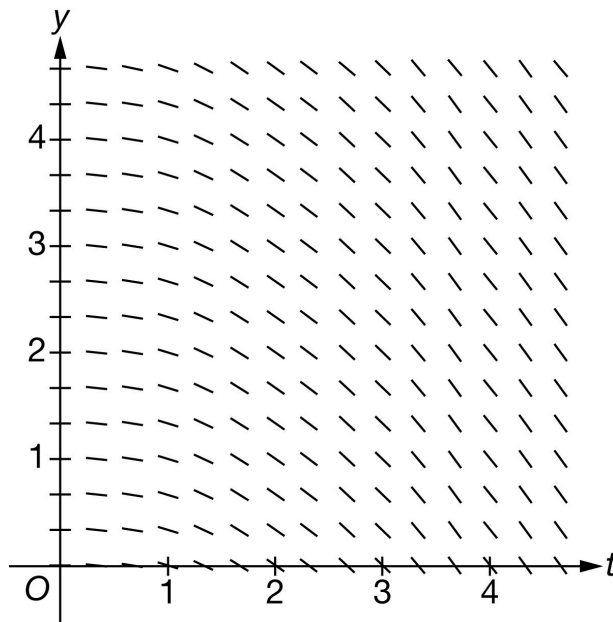
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{40} N(t) \, dt$? Give a reason for your answer.

(f) Find the value of $\int_0^{10} N'(4t) \, dt$. Show the computations that lead to your answer.

(g) For $0 \leq t \leq 40$, the total number of 5-cent coins remaining can also be modeled by the differentiable function F , where F is a function of t . Assume the weight of each coin is 5 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the 5-cent coins remaining at $t = 11$ minutes.

6-2

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.3y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\begin{aligned} & \cdot \frac{0-810}{40-10} \\ & \cdot \frac{N(40)-N(10)}{30} = \frac{-810}{30} \\ & \cdot \frac{N(40)-N(10)}{30} = -27 \end{aligned}$$

An example that does not earn the first point is $\frac{N(40)-N(10)}{30}$.

Minimal response that earns the first point: $\frac{0-810}{40-10}$ or $\frac{0-810}{30}$

· Neither $\frac{-810}{30}$ nor -27 , with no supporting work, will earn the point.

The second point is for correct units

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

- Coins per minute
- Nickels per minute,

6-2

- 5-cents per minute
- Coins/minute

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(35) \approx \frac{N(40) - N(10)}{40 - 10} = \frac{0 - 810}{30} = -27 \text{ coins per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a *rate of change*
- Including the context of coin collection
- Referring to the specific point in time, $t = 35$

The response is not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “decreasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “the number of coins is decreasing at a rate of 27 coins per day...” is okay, but “the number of coins is decreasing at a rate of -27 coins per day...” would not earn the point.

Acceptable references to a specific point in time include (but are not limited to):

- On minute 35 and after 35 minutes
- At $t = 35$
- At $t = 35$ hours or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from minute 35 on,” “during the first 35 minutes,” or “for $t = 0$ to $t = 35$ ”), the point is not earned.

Examples that earn the point include:

- $N'(35)$ is the rate of change of the number of coins per minute at time $t = 35$
- The rate of change of number of coins is -27 coins per minute on minute 35

6-2

- $N'(35)$ is the rate of change in the number of coins at time $t = 35$ in coins per minute
- 27 is the rate of decrease in the number of coins per minute on minute 35

Examples that do not earn the point include:

- $N'(35)$ is the average rate of change of the total number of coins per minute on minute 35
- $N'(35)$ is the total number of coins per minute on minute 35
- The rate of change is -27 coins per minute (Missing “at $t = 35$ ”)
- $N'(35)$ is the slope of the tangent line at $t = 35$
- Rate of change of N at time $t = 35$
- Coins per minute decreases at minute 35



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(35)$ is the rate at which the number of 5-cent coins remaining is changing, in coins per minute, at time $t = 35$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

- $y = -20(t - 10) + 810$, $y = -20(x - 10) + 810$
- $y - 810 = -20(t - 10)$, $y - 810 = -20(x - 10)$
- $y = -20t + 1010$, $y = -20x + 1010$

The key calculus concept being assessed in this part of the problem is that the value of $N'(10)$ is the slope of the tangent line. In that spirit, we are accepting $N'(t) = \dots$ in lieu of $y = \dots$, but we are not accepting $N'(10) = \dots$

Note that $N(t) = -20(t - 10) + 810$ earns the point, but $N(10) = -20(t - 10) + 810$ does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

6-2

Solution:

$$y = N'(10)(t - 10) + N(10)$$

$$y = -20(t - 10) + 810$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

- A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

- Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (d).

The second point is for the correct numerical result.

- In order to earn the second point, there must be supporting calculations.

- The numerical values for $N(10)$, $N(35)$, and $N(40)$ must be pulled from the table

- Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $810 \cdot (10 - 0) + 200 \cdot (35 - 10) + 0 \cdot (40 - 35)$ earns both points.

- A response of $10 \cdot N(10) + 25 \cdot N(35) + 5 \cdot N(40)$ earns the first point. Eligible for second point if more work is on the page.

- A response of $10 \cdot N(10) + 25 \cdot N(35) + 5 \cdot N(40) = 13,100$ earns both points.

- A response of $10 \cdot 810 + 25 \cdot 200 + 5 \cdot 0$ earns both points.

- A response of $10 \cdot 810 + 15 \cdot 200 + 5 \cdot 0$ contains one numerical error. Earns the first point. Not eligible for the second point.

- A response of $10 \cdot 810 + 25 \cdot 200 + 5 \cdot 0 = 12,100$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.

- A response of $8100 + 5000 + 0$ or $8100 + 5000 + 0 = 13,100$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$10 \cdot 810 + 25 \cdot 200 + 5 \cdot 0$$

6-2



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{40} N(t) dt \approx N(10)(10 - 0) + N(35)(35 - 10) + N(40)(40 - 35)$$

$$= 810 \cdot 10 + 200 \cdot 25 + 0 \cdot 5 = 13,100$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because N is decreasing.

The response does not need an explicit reference in part (d). The following statements are sufficient for earning this point:

- Underestimate because N is decreasing.
- Underestimate because the function is decreasing

Note: At this point, there is only one function being discussed in this problem, so we will assume “the function” is N .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function N , the point is not earned.

Examples that earn the point:

- The right Riemann sum underestimates because the function is decreasing.
- The sum is an underestimate because $N(t)$ is decreasing.
- The sum is an underestimate because $N'(t) < 0$.

Examples that do not earn the point:

- The sum is an underestimate because $N'(t)$ is decreasing. ($N'(t)$ decreasing is not known)
- The right Riemann sum underestimates because the graph is decreasing.
- The right Riemann sum underestimates because the function is decreasing and concave up.



0	1
---	---

6-2

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because N is a decreasing function, and this is a right Riemann sum, the answer from part (d) is an underestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = 4t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{4}$
- Uses the correct limits for the substitution (0 and 40). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

· Either of the expressions below will earn the substitution point without any additional supporting work: $\frac{1}{4} N(4t) \Big|_0^{10}$ or $\frac{1}{4} N(u) \Big|_0^{40}$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{40}^0 N'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 10$ or $b = 40$. No explanation or additional supporting work is necessary.

- o $kN(4t) \Big|_0^b$
- o $k(N(b) - N(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is -250 .

Values from the table must be used. $\frac{1}{4}[N(40) - N(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{4}(0 - 1000)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

6-2

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{4}(-1000)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4 \, dt$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 40$.

$$\begin{aligned} \int_0^{10} N'(4t) \, dt &= \frac{1}{4} \int_0^{40} N'(u) \, du \\ &= \frac{1}{4}(N(40) - N(0)) \\ &= \frac{1}{4}(0 - 1000) = -250 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 11$. No supporting work or justification is required.

The correct expression of just $5F'(11)$ or $5N'(11)$ earns this point.

No justifications or explanations are required.

Any incorrect statement, such as $N'(11) = 5F'(11)$, does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the 5-cent coins remaining at $t = 11$ minutes is $5F'(11)$.

6-2

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = -0.3y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field., but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point (4, 0), the slope of the segment should be 0, but it is negative.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to one of the two things being compared will not earn the point. For example, merely stating either $\frac{dy}{dt}$ is negative, or that the slope field is negative does not earn the point.

- “The values for y are less than or equal to zero so $\frac{dy}{dt} \leq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are negative” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is decreasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-2

48. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	10	40
$N(t)$ (coins per minute)	-10	-16	-8

A child has collected 5-cent coins and is preparing them to deposit in the bank by putting them into bags. For time t minutes, $0 \leq t \leq 40$, the rate of change in the number of coins remaining to prepare, in coins per minute, is modeled by a twice-differentiable function N . Values of $N(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $N'(10)$ using the average rate of change of N over the interval $0 \leq t \leq 40$.

(b) Explain the meaning of $N'(10)$ in the context of the problem.

(c) The function $C(t)$ models the number of coins remaining to be put into bags at time t minutes, where $C'(t) = N(t)$. It is known that $C(10) = 400$. Write an equation for the line tangent to the graph of $y = C(t)$ at $t = 10$.

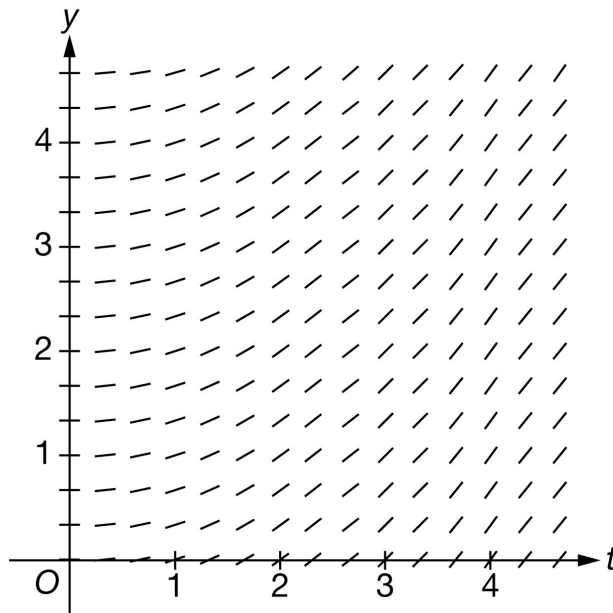
(d) Find the value of $\int_0^{10} N'(4t) \, dt$. Show the calculations that lead to your answer.

(e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{40} N(t) \, dt$. Show the calculations that lead to your answer.

(f) A second child is preparing another deposit of 5-cent coins. For $0 \leq t \leq 40$, the rate of change in the number of coins remaining for the second child to prepare, in coins per minute, can be modeled by the differentiable function F , where F is a function of t . Assume the weight of each coin is 5 grams. Write an expression in terms of N and F for the rate of change, in grams per minute, in the combined weight of the 5-cent coins remaining for the two children to prepare at $t = 11$ minutes.

6-2

(g) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to 3 decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{-8+10}{40}$ coins per minute per minute



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

6-2

$$N'(10) \approx \frac{N(40) - N(0)}{40 - 0} = \frac{-8 - (-10)}{40} = \frac{1}{20} \text{ coin per minute per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of coin collection
- Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

- $N'(10)$ is the average (or an approximation of, or an estimate of) rate of change of
- $N'(10)$ is the acceleration (or rate of change of velocity) of

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $N'(10)$ is the rate of increase of the rate of change of the number of coins at 10 minutes. (earns the point)
- Example: $N'(10)$ means the number of coins per minute is increasing by $\frac{1}{20}$ coins/min/min (does not earn the point)

Acceptable references to time include (but are not limited to):

- At the tenth minute
- At $t = 10$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 10$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “over the first ten minutes”) the response will not earn the point.

Examples that earn the point include:

- $N'(10)$ represents the rate at which the rate of change of the number of coins is changing at minute 10.
- $N'(10)$ is the rate of change in the rate of change of the number of coins at 10 minutes.
- $N'(10)$ represents the rate at which the rate of change of the number of coins is increasing at $t = 10$.
- $N'(10)$ equals (answer from part (a)) means the rate of change in the number of coins per minute is increasing at a rate of (answer from part (a)) at $t = 10$.
- $N'(10)$ represents the rate at which the rate of change of the number of coins is decreasing after 10 10 minutes.

Examples that do not earn the point include:

- $N'(10)$ represents the rate at which the number of coins is changing at minute 10.

6-2

- $N'(10)$ is the rate of change in the number of coins per minute at 10 minutes.
- The rate of change in the number of coins per minute is increasing at $t = 10$.
- The rate of change in the rate of change of the number of coins per minute is increasing at $t = 10$.
- $N'(10) = -\frac{1}{20}$ means that the rate of change in the number of coins is decreasing at a rate of $-\frac{1}{20}$ after 10 minutes. (double negative)



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(10)$ is the rate at which the rate of change of the number of 5-cent coins remaining is changing, in coins per minute per minute, at time $t = 10$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

· $y = -16(t - 10) + 400$

· $y - 400 = -16(t - 10)$

· $y = -16t + 560$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $C(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = C'(10)(t - 10) + C(10)$$

$$y = -16(t - 10) + 400$$

6-2

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required to award this point:

- Coefficient of $\frac{1}{4}$ correctly applied to an integral or an antiderivative
- Evidence that the correct limits of 0 to 40 will be or have been applied

Examples that will earn the first point (these are just a few):

$$\cdot \frac{1}{4} \int_0^{40} N'(u) \, du$$

$$\cdot \frac{1}{4} N(u) \Big|_0^{40} \text{ or } \frac{1}{4} N(4t) \Big|_0^{10}$$

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\cdot \int_0^{10} N'(4t) \, dt = \frac{1}{4} \int_0^{10} N'(u) \, du = \frac{1}{4} N(4t) \Big|_0^{10} \text{ earns the first point, but will not earn the third point.}$$

$$\cdot \int_0^{10} N'(4t) \, dt = \frac{1}{4} \int_0^{10} N'(u) \, du = \frac{1}{4} [N(10) - N(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b N'(u) \, du = N(u) \Big|_a^b = N(b) - N(a)$$

This point can be earned with either $kN(4t) \Big|_a^b$, $kN(t) \Big|_a^b$, or $k(N(b) - N(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 10$ or $b = 40$.

A response can earn this point without earning the first point.

$$\frac{1}{4} [N(40) - N(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{4} [N(40) - N(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\cdot \int_0^{10} N'(4t) \, dt = \frac{1}{4} \int N'(u) \, du = \frac{1}{4} N(4t) \Big|_0^{10} = \dots \text{ is eligible for the third point.}$$

Eligibility: Must have earned the first two points.

6-2

The minimum response needed to earn all three points is: $\frac{1}{4}(-8 + 10)$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4 \, dt$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 40$.

$$\int_0^{10} N'(4t) \, dt = \frac{1}{4} \int_0^{40} N'(u) \, du$$

$$= \frac{1}{4}(N(40) - N(0))$$

$$= \frac{1}{4}(-8 - (-10)) = \frac{1}{2}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a trapezoidal sum

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point

(e.g. $\frac{1}{2}[10(N(0) - N(10)) + 30(N(10) - N(40))]$ earns no points.)

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one incorrect numerical value.

$\frac{1}{2}(10)[N(0) + 2N(10) + N(40)]$ is equivalent to $\frac{1}{2}[10(N(0) + N(10)) + 10(N(10) + N(40))]$

$\frac{1}{2}[10(N(0) + N(10)) + 10(N(10) + N(40))]$, thus has only one incorrect numerical value (the second interval width is 10

6-2

instead of 30), so can earn the point.

$\cdot \frac{1}{2}(20) [N(0) + 2N(10) + N(40)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first point, if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn the first, but not the second point:

$\cdot k \left[\left(\frac{-10-16}{2} \right) \cdot 10 + \left(\frac{-16-8}{2} \cdot 30 \right) \right]$ for $k > 0$ earns only the first point.

The minimum response needed to earn both points: $\left(\frac{-26}{2} \right) \cdot 10 + \left(\frac{-24}{2} \right) \cdot 30$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\int_0^{40} N(t) dt \approx \left(\frac{N(0)+N(10)}{2} \right) (10-0) + \left(\frac{N(10)+N(40)}{2} \right) (40-10)$$

$$= \left(\frac{-10-16}{2} \right) \cdot 10 + \left(\frac{-16-8}{2} \right) \cdot 30 = -490$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 5 as a factor of the expression, and/or the correct sum $N(11) + F(11)$.

If the sum is incorrect, then the constant factor of 5 must be applied either to a sum that involves only $N(t)$ and $F(t)$, $N'(t)$ and $F'(t)$, or integrals of $N(t)$ and $F(t)$ or to one of the correct terms of the sum $N(11)$ or $F(11)$, with no additional incorrect terms.

Examples that earn the point include:

$$\cdot 5 [N(t) + F(t)]$$

$$\cdot 5 \int [N(t) + F(t)] dt$$

6-2

$$\cdot 5 [N' + F']$$

$$\cdot 5N (11)$$

Examples that do not earn the point include:

$$\cdot 5 [N (t) + F' (t)]$$

$$\cdot 5 \int N (t) dt$$

$$\cdot 5N' + F'$$

$$\cdot 5N (11) + t$$

Special case: $5N (11) + F (11)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k [N (11) + F (11)]$, for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$5 [N (11) + F (11)]$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn the first point only. For example: $5 [N (11) + F (11)] + 5 (2)$ earns only the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

The rate of change, in grams per minute, in the combined weight of the 5-cent coins remaining for the two children to prepare at $t = 11$ minutes is $5N (11) + 5F (11)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y , as t changes.”)

6-2

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(1, 2)$ the slope shown on the slope field is 1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-2

49. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	3	6	9	12
$S(t)$ (number of boxes of snacks remaining)	600	500	440	240	20

A student organization is selling boxes of snacks to raise money for a school trip. The student organization begins with 600 boxes of snacks and sells them over a twelve-day period. For time t days, $0 \leq t \leq 12$, the total number of boxes of snacks remaining to sell is modeled by a twice-differentiable function S . Values of $S(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate $\int_0^{12} S(t) \, dt$. Show the computations that lead to your answer.

(b) Find the value of $\int_0^3 S'(3t) \, dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $S'(9)$ using the average rate of change of S over the interval $6 \leq t \leq 12$.

(d) Explain the meaning of $S'(9)$ in the context of the problem.

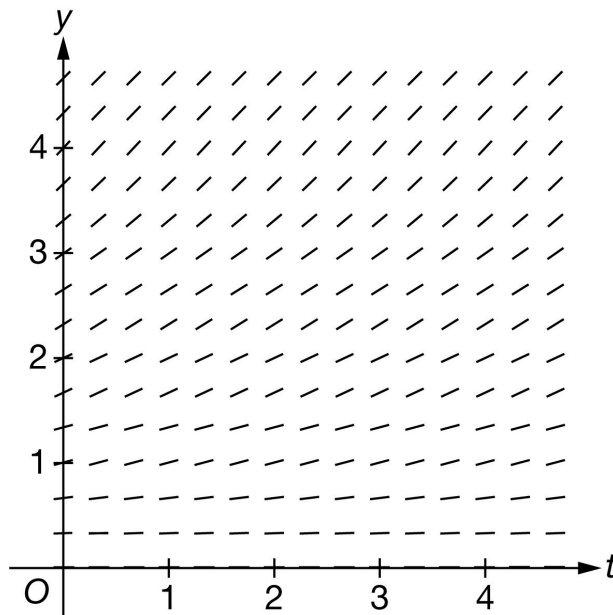
(e) It is known that $S'(6) = -40$. Write an equation for the line tangent to the graph of $y = S(t)$ at $t = 6$.

(f) It is known that $S''(t) < 0$ for $6 \leq t \leq 9$. If $S(8)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $S(8)$? Give a reason for your answer.

(g) For $0 \leq t \leq 12$, the total number of boxes of snacks remaining can be modeled by the differentiable function Q , where Q is a function of t . Assume the weight of each box is 8 ounces. Write an expression for the rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 5$ days.

6-2

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 6 \cdot (S(3) + S(9)) \text{ or } 6 \cdot S(3) + 6 \cdot S(9)$$

$$\cdot (6 - 0) \cdot 500 + (12 - 6) \cdot 240$$

$$\cdot 6 \cdot (500 + 240) \text{ or } 6 \cdot 500 + 6 \cdot 240$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{6}{b} \cdot (S(3) + S(9))$, for any nonzero value of b would earn the point. Note: The value $\frac{6}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (S(3) + S(9))$, for $a \neq 6$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 6 \cdot (500 + 240)$$

$$\cdot 6 \cdot 500 + 6 \cdot 240$$

6-2



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint sum
- ☐ Answer

Solution:

$$\int_0^{12} S(t) dt \approx S(3)(6-0) + S(9)(12-6)$$

$$= 500 \cdot 6 + 240 \cdot 6 = 4440$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 3t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{3}$

Uses the correct limits for their substitution.

- The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .
- Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int S'(3t) dt$ or $\int S'(u) du$.

The point is earned for any of the following:

- The equation $\int S'(3t) dt = S(3t)$ or $\int S'(u) du = S(u)$
- $\frac{1}{3}S(3t)\big|_0^3$ or $\frac{1}{3}S(u)\big|_0^9$
- $S(9) - S(0)$, $S(9)$, or $S(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b S'(u) du = S'(b) - S'(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of S' is S .

Third Point (Answer)

6-2

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits but will still earn the third point for the correct answer.

$S(9)$ and $S(0)$ must be evaluated; the expression $\frac{1}{3}(S(9) - S(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{3}(240 - 600)$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

· $\frac{1}{3}S(3t)\Big|_0^3$

· $\frac{1}{3}(S(9) - S(0))$

Responses of only the following would earn only the second point:

· $S(9) - S(0)$

· $S(3) - S(0)$

· $\int_0^9 S'(u) \, du = S(9) - S(0)$

· $S(3t)\Big|_0^3$ or $S(u)\Big|_0^9$

The following response would earn no points: -120 in the absence of one of the first two points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 \, dt$.

When $t = 0$, $u = 0$.

When $t = 3$, $u = 9$.

6-2

$$\int_0^3 S'(3t) dt = \frac{1}{3} \int_0^9 S'(u) du$$

$$= \frac{1}{3}(S(9) - S(0))$$

$$= \frac{1}{3}(240 - 600) = -120$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{20-440}{6}$ or $\frac{-420}{12-6}$.

Second point (Units)

The units must be attached to a correct or incorrect value.

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$S'(9) \approx \frac{S(12) - S(6)}{12 - 6} = \frac{20 - 440}{6} = -70 \text{ boxes per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

- indicate the rate of change
- the specific context of the number of boxes remaining
- the specific point in time, $t = 9$.

Responses do not need to include correct units or any units.

The word “decreasing” may be used in place of the word “changing”.

6-2

The response is not earned for:

- “the rate of boxes sold on day 9” as this does not indicate a change in the rate of boxes remaining.
- “decreasing at a rate of -70 boxes per day”

“the rate of change of S ” without specifying that S is the number of boxes remaining, does not earn the point.

Acceptable ways to provide the specific point in time:

- On day 9 or after 9 days.
- At $t = 9$ (missing the unit of time)
- On week 9 (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 9 days” or “from day 9 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(9)$ is the rate at which the number of boxes of snacks remaining is changing, in boxes per day, on day 9.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $S(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

- $y = 440 - 40(t - 6)$
- $y - 440 = -40(t - 6)$
- $y = -40t + 240 + 440$
- $y = -40t + 680$



0	1
---	---

The student response accurately includes the criteria below.

6-2

☐ Tangent line equation
Solution:

$$y = S'(6)(t - 6) + S(6)$$

$$y = -40(t - 6) + 440$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the point the response must include $S''(t) < 0$ and overestimate.

“ $S(t)$ is concave down” can be used instead of $S''(t) < 0$.

Information about $S'(t)$ by itself without reference to $S''(t) < 0$ is not sufficient and does not earn this point.

“Overestimate because it is concave down” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states, $S''(t) < 0$, and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $S'(t)$ is concave down, overestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason
Solution:

Because $S''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $8 \cdot Q'(5)$ earns this point.

Only expression $8 \cdot S'(5)$ also earns this point.

No justifications or explanations are required.

Any incorrect statement such as, $Q'(5) = 8 \cdot S'(5)$, does not earn this point.



0	1
---	---

6-2

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 5$ days is $8Q'(5)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is negative, not positive”. We will infer that negative is referring to the slopes shown in the slope field.

· “This would be correct for a positive derivative, but $\frac{dy}{dt}$ is negative.” We will read “This” as “the slope field” in this sentence.

· “All of the slopes in the slope field are positive, not negative.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are negative.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· “ $\frac{dy}{dt}$ is negative” lacks any reference to the slope field and so will not earn the point.

· “All the slopes in the graph are positive” lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as “The slope field has positive points” or “The field is increasing” will not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} < 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

6-2

50. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	2	5	10
$S(t)$ (boxes per day)	-60	-40	-30	-5

A student organization is selling boxes of snacks to raise money for a school trip. For time t days, $0 \leq t \leq 10$, the rate of change in the number of boxes of snacks remaining to sell is modeled by an increasing, twice-differentiable function S . Values of $S(t)$ at various times t are shown in the table above.

(a) Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{10} S(t) \, dt$. Show the calculations that lead to your answer.

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{10} S(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $S'(2)$ using the average rate of change of S over the interval $0 \leq t \leq 5$.

(d) Explain the meaning of $S'(2)$ in the context of the problem.

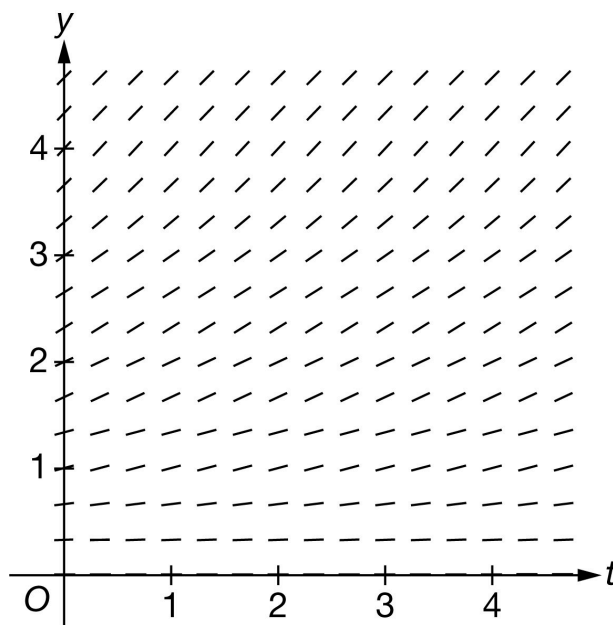
(e) It is known that $S'(5) = 6$. Write an equation for the line tangent to the graph of $y = S(t)$ at $t = 5$.

(f) Find the value of $\int_0^{10} S'\left(\frac{t}{2}\right) \, dt$. Show the calculations that lead to your answer.

(g) Another student organization is also selling boxes of snacks. For $0 \leq t \leq 10$, the total number of boxes of snacks remaining for the second student organization can be modeled by the decreasing and differentiable function M , where M is a function of t . Assume the weight of each box is 7.5 ounces. Write an expression for the rate of change, in ounces per day, of the total weight of the boxes of snacks remaining for the second student organization at $t = 6$ days.

6-2

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = M'(t)$ for the function M defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

-40	2	-80	or	$(2 - 0)(-40)$	$= -80$
-30	3	-90		$(5 - 2)(-30)$	$= -90$
-5	5	$+(-25)$		$(10 - 5)(-5)$	$= -25$
		-195			-195

Implicit responses such as those above are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (a).

The second point is for the correct numerical result.

In order to earn the second point, there must be supporting calculations.

The numerical values for $S(2)$, $S(5)$, and $S(10)$ must be pulled from the table

Any simplification shown must be correct.

Use the following as an illustration of how to score both points. (In most cases, you see scores for which the expression represents the entirety of relevant work shown for the response.)

6-2

- A response of $(-40) \cdot (2 - 0) + (-30) \cdot (5 - 2) + (-5) \cdot (10 - 5)$ earns both points.
- A response of $2 \cdot S(2) + 3 \cdot S(5) + 5 \cdot S(10)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $2 \cdot S(2) + 3 \cdot S(5) + 5 \cdot S(10) = -195$ earns both points.
- A response of $2 \cdot (-40) + 3 \cdot (-30) + 5 \cdot (-5)$ earns both points.
- A response of $2 \cdot (-40) + 3 \cdot (-30) + 4 \cdot (-5)$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $2 \cdot (-40) + 3 \cdot (-30) + 5 \cdot (-5) = -190$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $(-80) + (-90) + (-25)$ or $(-80) + (-90) + (-25) = -195$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$2 \cdot (-40) + 3 \cdot (-30) + 5 \cdot (-5)$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{10} S(t) \, dt &\approx S(2)(2 - 0) + S(5)(5 - 2) + S(10)(10 - 5) \\ &= (-40) \cdot 2 + (-30) \cdot 3 + (-5) \cdot 5 = -195 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because S is increasing.

The response does not need an explicit reference to part (a) or to a right Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following table contains some examples of what WILL or will NOT earn the point.

The following examples earn the point:

- Overestimate because S is increasing

6-2

- Overestimate because $S'(t) > 0$
- Overestimate because the graph of S is increasing (we know they are talking about the given function S)
- Overestimate because the function is increasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Overestimate because S is concave down/up
- Overestimate because the graph/curve is increasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Overestimate because the table [or table values] is/are increasing. The behavior of the values shown in the table is not the same as the behavior of the function S . The table does not tell us what is happening to S between the given points.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because S is an increasing function, and this is a right Riemann sum, this is an overestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $S(5)$ and $S(0)$ must be pulled from the table.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Boxes per day per day or Boxes/day/day
- Boxes per day² or Boxes/day²
- Boxes per day squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter “ B ” is not a standard abbreviation for “boxes.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

6-2

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points.

$$\frac{-30+60}{5} \text{ Boxes/day}^2 \text{ or } \frac{-30-(-60)}{5} \text{ Boxes/day}^2$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$S'(2) \approx \frac{S(5)-S(0)}{5-0} = \frac{-30-(-60)}{5} = 6 \text{ boxes per day per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of a rate
- Including the context of boxes [remaining to be sold]
- Explicitly referring to the specified point in time $t = 2$

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use 6 as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of 6 ” is OK [Units not required in part (d)].(the point is not yet earned with this phrase alone)
- “... decreasing at a rate of 6 ” is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)

Note: the context is complicated. $S(t)$ is describing the rate of change of the remaining inventory of boxes. The values in the table indicate that the inventory is decreasing more slowly as time passes. Common sense might tell us that this equates to the idea that the rate at which boxes are sold is decreasing. Explaining the meaning using “boxes sold” rather than “boxes remaining” will not prevent the response from earning this point.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

- The rate of change in the rate at which boxes are changing

6-2

- The rate of change in boxes is changing (or decreasing) at a rate of ...
- The rate at which the rate of change of boxes is changing

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “boxes per day” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

“Rate of change of S ” is not enough without describing what S represents in this context.

Context

A key component of the context for this explanation is “boxes.”

Any misstatement about boxes sold vs. boxes remaining to be sold will not prevent the response from earning this point.

Explicit reference to $t = 2$

Acceptable references to a specific point in time include (but are not limited to):

- On day 2 (where day could be replaced with any other unit of time)
- At $t = 2$ (with or without units of time)
- After 2 days (where days could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from day 2 on” or “during the first 2 days”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(2)$ is the rate at which the rate of change in the number of boxes of snacks remaining is changing, in boxes per day per day, on day 2.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

$$y = 6(t - 5) - 30$$

$$y + 30 = 6(t - 5)$$

6-2

$$y = 6t - 60$$

The key calculus concept being assessed in this part of the problem is that the value of $S'(5)$ is the slope of the tangent line. In that spirit...

- The response will not be penalized for using x instead of t .

We are accepting $S(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Tangent line equation

Solution:

$$y = S'(5)(t - 5) + S(5)$$

$$y = 6(t - 5) - 30$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that we need to see:

- Uses $u = \frac{t}{2}$, which may be done implicitly
- Uses the correct constant of 2
- Uses the correct limits for the substitution (0 and 5). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $2S\left(\frac{t}{2}\right)\bigg|_0^{10}$ or $2S(u)\bigg|_0^5$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_5^0 S'(u) du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 10$. No explanation or additional supporting work is necessary.

- $kS\left(\frac{t}{2}\right)\bigg|_0^b$

6-2

$$\cdot k(S(b) - S(0))$$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 60.

Values from the table must be used. $2(S(5) - S(0))$ is not enough to earn the answer point.

The expression $2(-30 - (-60))$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown they do not need to be correct to earn this point.

The response cannot earn this point without also earning the first two points.

Minimalist response: The expression $2(-30 - (-60))$ represents the minimum work that must be seen in a response in order to earn all 3 points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{2}$. Then $du = \frac{dt}{2}$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 5$.

$$\begin{aligned} \int_0^{10} S' \left(\frac{t}{2} \right) dt &= 2 \int_0^5 S'(u) du \\ &= 2(S(5) - S(0)) \\ &= 2(-30 - (-60)) = 60 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 6$ earns the point. It may be written as $(7.5) \cdot M'(6)$ or $(7.5) \cdot \frac{dM}{dt} \Big|_{t=6}$.

The expression $(7.5) \cdot M'(t)$ is not sufficient because it does not use $t = 6$.

The response must use M (not S or any other function name) in order to earn the point.

6-2



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in ounces per day, of the total weight of the boxes of snacks remaining for the second student organization at $t = 6$ days is $7.5M'(6)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation $\frac{dy}{dt} = M'(t)$ and the given slope field.

In order to earn the point, the response must state that the values of the differential equation are negative while the slopes in the given slope field are positive or 0 (nonnegative).

Some ways to say that the values of differential equation are negative:

- M is decreasing
- $M'(t) < 0$
- M has negative slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has nonnegative slopes.
- The slopes shown are positive or 0.
- The slope field contains positive slopes.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is increasing/positive.
- The slope field looks like it represents an increasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ M is a decreasing function, so the signs of the slopes in the slope field should be negative” can earn this point because “should be” is indicating contrast with what is known.

6-2

The response will not be penalized for saying that the slopes in the slope field are “always” positive or “all” positive.



0	1
---	---

The student response accurately includes the criteria below.

☐

Explanation

Solution:

M is a decreasing function, so $\frac{dy}{dt} = M'(t) \leq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

6-2

51. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

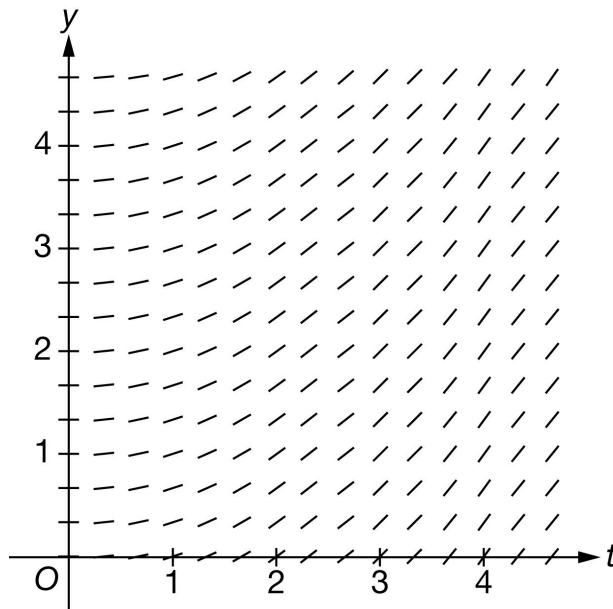
t (days)	0	5	9
$S(t)$ (boxes per day)	-70	-30	-16

A student organization is selling boxes of snacks to raise money for a school trip. For time t days, $0 \leq t \leq 9$, the rate of change in the number of boxes of snacks remaining to sell is modeled by an increasing, twice-differentiable function S . Values of $S(t)$ at various times t are shown in the table above.

- Using correct units, approximate $S'(5)$ using the average rate of change of S over the interval $0 \leq t \leq 9$.
- Explain the meaning of $S'(5)$ in the context of the problem.
- The function $B(t)$ models the number of boxes remaining to sell at time t days, where $B'(t) = S(t)$. It is known that $B(5) = 240$. Write an equation for the line tangent to the graph of $y = B(t)$ at $t = 5$.
- Find the value of $\int_0^{4.5} S'(2t) \, dt$. Show the calculations that lead to your answer.
- Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^9 S(t) \, dt$. Show the calculations that lead to your answer.
- Another student organization is also selling boxes of snacks. For $0 \leq t \leq 9$, the rate of change in the number of boxes of snacks remaining for the second student organization to sell is modeled by a decreasing, differentiable function M , where M is a function of t . Assume the weight of each box of snacks is 7 ounces. Write an expression in terms of M and S for the rate of change, in ounces per day, of the combined weight of the boxes of snacks remaining for the two organizations to sell at $t = 8$ days.

6-2

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = 0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. We need to see both a difference and a quotient for the supporting work.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is $\frac{-16+70}{9}$ boxes per day per day



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

6-2

$$S'(5) \approx \frac{S(9) - S(0)}{9 - 0} = \frac{-16 - (-70)}{9} = 6 \text{ boxes per day per day}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of boxes
- Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

- $S'(5)$ is the average (or an approximation of, or an estimate of) rate of change of . . .
- $S'(5)$ is the acceleration (or rate of change of velocity) of . . .

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $S'(5)$ is the rate of increase of the rate of change of the number of boxes at 5 days. (earns the point)
- Example: $S'(5)$ means the number of boxes per day is increasing by 5 boxes/day/day. (does not earn the point)

Acceptable references to time include (but are not limited to):

- On day 5
- At $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “from day 0 to day 5”) the response will not earn the point.

Examples that earn the point include:

- $S'(5)$ represents the rate at which the rate of change of the number of boxes is changing on day 5.
- $S'(5)$ is the rate of change in the rate of change of the number of boxes at 5 days.
- $S'(5)$ represents the rate at which the rate of change of the number of boxes is increasing at $t = 5$.
- “ $S'(5) = (\text{answer from part (a)})$ ” means the rate of change in the number of boxes per day is increasing at a rate of (answer from part (a)) at $t = 5$.
- $S'(5)$ represents the rate at which the rate of change of the number of boxes is decreasing on the fifth day.

Examples that do not earn the point include:

- $S'(5)$ represents the rate at which the number of boxes is changing on day 5.

6-2

- $S'(5)$ is the rate of change in the number of boxes per day at $t = 5$ days
- The rate of change in the number of boxes per day is increasing at $t = 5$.
- The rate of change in the rate of change of the number of boxes per day is increasing at $t = 5$.
- $S'(5) = -6$ means that the rate of change in the number of boxes is decreasing at a rate of -6 at $t = 5$ (double negative).



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(5)$ is the rate at which the rate of change in the number of boxes per day is changing, in boxes per day per day, on day 5.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. For example:

$$y = -30(t - 5) + 240$$

$$y - 240 = -30(t - 5)$$

$$y = -30t + 390$$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $B(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = B'(5)(t - 5) + B(5)$$

$$y = -30(t - 5) + 240$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-2

The first point is earned with a correct substitution.

Two things are required as evidence of correct substitution:

- Coefficient of $\frac{1}{2}$ correctly applied to an integral or an antiderivative
- Evidence that the correct limits of 0 to 9 will be applied

Examples that will earn the first point (these are just a few):

$$\cdot \frac{1}{2} \int_0^9 S'(u) \, du$$

$$\cdot \frac{1}{2} S(u) \Big|_0^9 \text{ or } \frac{1}{2} S(2t) \Big|_0^{4.5}$$

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\cdot \int_0^{4.5} S'(2t) \, dt = \frac{1}{2} \int_0^{4.5} S'(u) \, du = \frac{1}{2} S(2t) \Big|_0^{4.5} \text{ earns the first point, but will not earn the third point.}$$

$$\cdot \int_0^{4.5} S'(2t) \, dt = \frac{1}{2} \int_0^{4.5} S'(u) \, du = \frac{1}{2} [S(4.5) - S(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b S'(u) \, du = S(u) \Big|_a^b = S(b) - S(a)$$

This point can be earned with either $kS(2t) \Big|_a^b$, $kS(t) \Big|_a^b$, or $k(S(b) - S(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 4.5$ or $b = 9$.

The response can earn this point without earning the first point.

$$\frac{1}{2} [S(9) - S(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{2} [S(9) - S(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\cdot \int_0^{4.5} S'(2t) \, dt = \frac{1}{2} \int S'(u) \, du = \frac{1}{2} S(2t) \Big|_0^{4.5} = \dots \text{ is eligible for the third point.}$$

Eligibility for the third point: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{2}(-16 + 70)$.

6-2



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2 dt$.

When $t = 0$, $u = 0$.

When $t = 4.5$, $u = 9$.

$$\int_0^{4.5} S'(2t) dt = \frac{1}{2} \int_0^9 S'(u) du$$

$$= \frac{1}{2}(S(9) - S(0))$$

$$= \frac{1}{2}(-16 - (-70)) = \frac{1}{2}(54) = 27$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the form of a trapezoidal sum

Point is earned for the setup of the trapezoidal sum, with one numerical error allowed. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum, a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning the first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point (e.g. $\frac{1}{2}[5(S(0) - S(5)) + 4(S(5) - S(9))]$ earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one error in the numerical values.

$\cdot \frac{1}{2}(5)[S(0) + 2S(5) + S(9)]$ is equivalent to $\frac{1}{2}[5(S(0) + S(5)) + 5(S(5) + S(9))]$, has only one incorrect numerical value (the second interval width is 5 instead of 4), so can earn the point.

$\cdot \frac{1}{2}(4.5)[S(0) + 2S(5) + S(9)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

6-2

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn the first point but not the second:

$\cdot k \left[\left(\frac{-70-30}{2} \right) \cdot 5 + \left(\frac{-30-16}{2} \right) \cdot 4 \right]$ for $k > 0$ earns just the first point.

The minimum response needed to earn both points is: $\left(\frac{-100}{2} \right) \cdot 5 + \left(\frac{-46}{2} \right) \cdot 4$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^9 S(t) dt &\approx \left(\frac{S(0)+S(5)}{2} \right) (5-0) + \left(\frac{S(5)+S(9)}{2} \right) (9-5) \\ &= \left(\frac{-70-30}{2} \right) \cdot 5 + \left(\frac{-30-16}{2} \right) \cdot 4 = -342 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 7 as a factor of the expression, and/or the correct sum $M(8) + S(8)$.

If the sum is incorrect, then the constant factor of 7 must be applied either to a sum that involves only $M(t)$ and $S(t)$, $M'(t)$ and $S'(t)$, or integrals of $M(t)$ and $S(t)$, or to one of the correct terms of the sum $M(8)$ or $S(8)$, with no additional incorrect terms.

Examples that earn the point include:

$$\begin{aligned} &\cdot 7 [M(t) + S(t)] \\ &\cdot 7 \int [M(t) + S(t)] dt \\ &\cdot 7 [M' + S'] \\ &\cdot 7S(8) \end{aligned}$$

Examples that do not earn the point include:

6-2

$$\cdot 7[M(t) + S'(t)]$$

$$\cdot 7 \int M(t) dt$$

$$\cdot 7M' + S'$$

$$\cdot 7S(8) + t$$

Special case: $7M(8) + S(8)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[M(8) + S(8)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$7M(8) + 7S(8)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn a just the first point. For example, $7M(8) + 7S(8) + 7(2)$ earns only the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

The rate of change, in ounces per day, of the combined weight of the boxes of snacks remaining for the two organizations to sell at $t = 8$ days is $7M(8) + 7S(8)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

6-2

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(2, 2)$ the slope shown on the slope field is 1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-2

52. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	2	5	10
$S(t)$ (number of boxes of snacks remaining)	500	340	200	20

A student organization is selling boxes of snacks to raise money for a school trip. The student organization begins with 500 boxes of snacks and sells them over a ten-day period. For time t days, $0 \leq t \leq 10$, the total number of boxes of snacks remaining to sell is modeled by a decreasing, twice-differentiable function S . Values of $S(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $S'(5)$ using the average rate of change of S over the interval $2 \leq t \leq 10$.

(b) Explain the meaning of $S'(5)$ in the context of the problem.

(c) It is known that $S'(2) = -55$. Write an equation for the line tangent to the graph of $y = S(t)$ at $t = 2$.

(d) Use a left Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{10} S(t) \, dt$. Show the computations that lead to your answer.

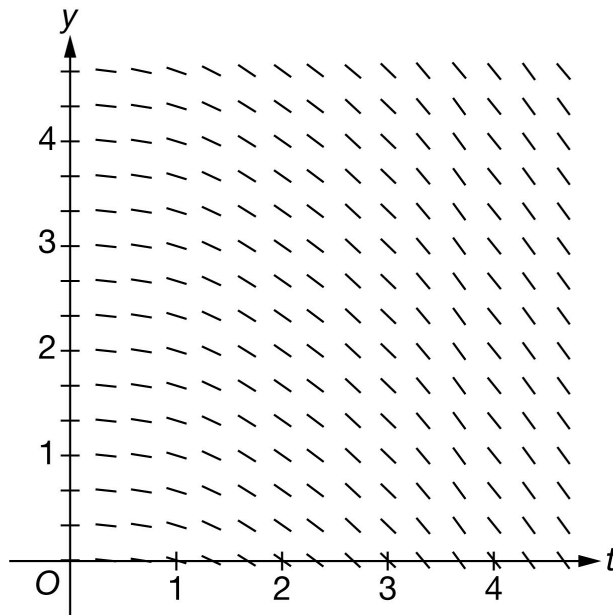
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{10} S(t) \, dt$? Give a reason for your answer.

(f) Find the value of $\int_0^5 S'(2t) \, dt$. Show the computations that lead to your answer.

(g) For $0 \leq t \leq 10$, the total number of boxes of snacks remaining can also be modeled by the differentiable function M , where M is a function of t . Assume the weight of each box is 7 ounces. Write an expression for the rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 8$ days.

6-2

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.2y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\cdot \frac{20-340}{10-2}$$

$$\cdot \frac{S(10)-S(2)}{8} = \frac{-320}{8}$$

$$\cdot \frac{S(10)-S(2)}{8} = -40$$

Examples that do not earn the first point include:

$$\cdot \frac{S(10)-S(2)}{8}$$

$$\cdot \frac{S(10)-S(2)}{10-2}$$

Minimal response that earns the point: $\frac{20-340}{10-2}$ or $\frac{20-340}{8}$

Neither $\frac{-320}{8}$ nor -40 , with no supporting work, will earn this point

The second point is for correct units

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

6-2

- Boxes per day
- Boxes of snacks per day
- Snacks per day
- Boxes/day

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$S'(5) \approx \frac{S(10) - S(2)}{10 - 2} = \frac{20 - 340}{8} = -40$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of change
- Including the context of boxes of snack collection
- Referring to the specific point in time, $t = 5$

Students are not penalized for incorrect units as a modifier in this part. (Whether correct or incorrect, units have no effect on the score in this part.) Example: “The number of coins is decreasing at a rate of 40 coins per day.”

Responses may use “changing” or “decreasing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “The number of boxes is decreasing at a rate of 40 boxes per day” is okay, but “The number of boxes is decreasing at a rate of -40 boxes per day” would not earn the point.

Acceptable references to a specific point in time include (but are not limited to):

- On day 5 and after 5 days
- At $t = 5$
- At $t = 5$ weeks or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from day 5 on,” “during the first 5 days,” or “for $t = 0$ to $t = 5$ ”), the point is not earned.

Examples that earn the point include:

6-2

- $S'(5)$ is the rate of change of the number of boxes of snacks per day at time $t = 5$
- The rate of change of number of boxes of snacks per day is -40 boxes of snacks per day on day 5
- $S'(5)$ is the rate of change in the number of boxes of snacks at time $t = 5$ in boxes of snacks per *month*
- -40 is the rate of decrease in the number of boxes of snacks per day on day 5

Examples that do not earn the point include:

- $S'(5)$ is the average rate of change of the total number of boxes of snacks per day on day 5
- $S'(5)$ is the total number of boxes of snacks per day on day 5
- The rate of change is -40 boxes of snacks per day (Missing “at $t = 5$ ”)
- $S'(5)$ is the slope of the tangent line at $t = 5$
- Rate of change of S at time $t = 5$
- Increase in number of boxes of snacks at $t = 5$
- Boxes per day increases on day 5



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(5)$ is the rate at which the number of boxes of snacks remaining is changing, in boxes per day, on day 5.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

Some examples include

- $y - 340 = -55(t - 2)$, $y - 340 = -55(x - 2)$
- $y = -55(t - 2) + 340$, $y = -55(x - 2) + 340$
- $y = -55t + 450$, $y = -55x + 450$

The key calculus concept being assessed in this part of the problem is that the value of $S'(2)$ is the slope of the tangent line. In that spirit, we are accepting $S(t) = \dots$ in lieu of $y = \dots$, but not $S(2) = \dots$



0	1
---	---

6-2

The student response accurately includes the criteria below.

- ☐ Tangent line equation

Solution:

$$y = S'(2)(t - 2) + S(2)$$

$$y = -55(t - 2) + 340$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of 3 products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

· Responses with the numerical error described above are not eligible for the second point.

NOTE: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

· In order to earn the second point, there must be supporting calculations.

· The numerical values for $S(2)$, $S(5)$, and $S(10)$ must be pulled from the table

· Any simplification shown must be correct.

Use the following as an illustration to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

· A response of $500 \cdot (2 - 0) + 340 \cdot (5 - 2) + 200 \cdot (10 - 5)$ earns both points.

· A response of $2 \cdot S(0) + 3 \cdot S(2) + 5 \cdot S(5)$ earns the first point. Eligible for second point if more work is on the page.

· A response of $2 \cdot S(0) + 3 \cdot S(2) + 5 \cdot S(5) = 3020$ earns both points.

· A response of $2 \cdot 500 + 3 \cdot 340 + 5 \cdot 200$ earns both points.

· A response of $2 \cdot 500 + 10 \cdot 340 + 5 \cdot 200$ contains one numerical error. Earns the first point. Not eligible for the second point.

· A response of $2 \cdot 500 + 3 \cdot 340 + 5 \cdot 200 = 3200$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.

· A response of $1000 + 1020 + 1000$ or $1000 + 1020 + 1000 = 3020$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

6-2

$$2 \cdot 500 + 10 \cdot 340 + 5 \cdot 200$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Left Riemann sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{10} S(t) dt &\approx S(0)(2-0) + S(2)(5-2) + S(5)(10-5) \\ &= 500 \cdot 2 + 340 \cdot 3 + 200 \cdot 5 = 3020 \end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because S is decreasing.

The response does not need an explicit reference to the approximation in part (d).

The following statements are sufficient for earning this point:

- Overestimate because S is decreasing
- Overestimate because the function is decreasing

NOTE: At this point, there is only one function being discussed in this problem, so we will assume “the function” is S .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function S , the point is not earned.

Examples that earn the point:

- The left Riemann sum overestimates because the function is decreasing.
- The sum is an overestimate because $S(t)$ is decreasing.
- The sum is an overestimate because $S'(t) < 0$.

Examples that do not earn the point:

- The sum is an overestimate because $S'(t)$ is decreasing. ($S'(t)$ decreasing is not known)
- The left Riemann sum overestimates because the graph is decreasing.
- The left Riemann sum overestimates because the function is decreasing and concave up.

6-2



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because S is a decreasing function, and this is a left Riemann sum, the answer from part (d) is an overestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that need to be seen:

- Uses $u = 2t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{2}$
- Uses the correct limits for the substitution (0 and 10). These limits may only be seen when evaluating the antiderivative. These don't need to see them as limits on a definite integral.
- Either of the expressions below will earn the substitution point without any additional supporting work: $\frac{1}{2} S(2t) \Big|_0^5$ or $\frac{1}{2} S(u) \Big|_0^{10}$

Special Handling: (Notation Errors)

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{10}^0 S'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 10$. No explanation or additional supporting work is necessary.

- $k S(2t) \Big|_0^b$
- $k (S(b) - S(0))$
- This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is -240 .

Values from the table must be used. $\frac{1}{2}[S(10) - S(0)]$ is not enough to earn the answer point

6-2

The expression $\frac{1}{2}(20 - 500)$ close parenthesis is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{2}(-480)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2 \, dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 10$.

$$\begin{aligned}\int_0^5 S'(2t) \, dt &= \frac{1}{2} \int_0^{10} S'(u) \, du \\ &= \frac{1}{2}(S(10) - S(0)) \\ &= \frac{1}{2}(20 - 500) = -240\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 8$. No supporting work or justification is required.

The correct expression of $7 \cdot S'(8)$ or $7 \cdot M'(8)$ earns this point.

No justifications or explanations are required.

Any incorrect statement such as, $S'(8) = 7 \cdot M'(8)$, does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

6-2

☐ Expression**Solution:**

The rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 8$ days is $7M'(8)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = -0.2y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field, but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point $(4, 0)$, the slope of the segment should be 0, but it is negative.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point. For example, merely stating either $\frac{dy}{dt}$ is negative, or that the slope field is negative does not earn the point.

- “The values for y are less than or equal to zero so $\frac{dy}{dt} \leq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are negative” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is decreasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation**Solution:**

6-2

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-2

53. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

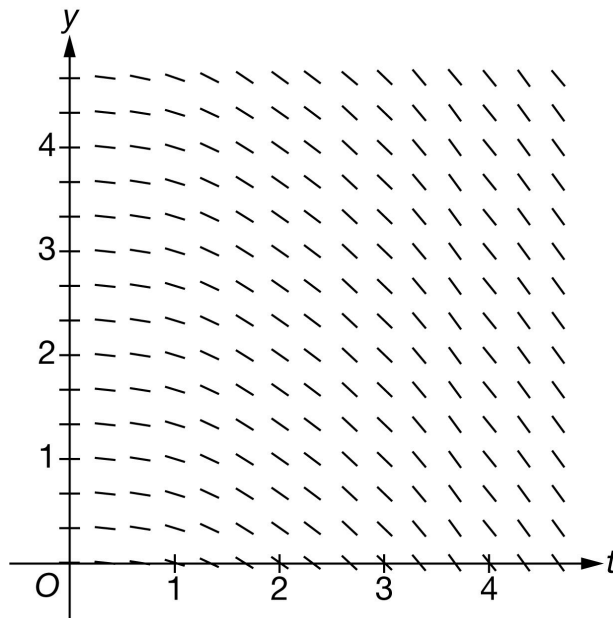
t (minutes)	0	8	24
$P(t)$ (postcards per minute)	-6	-10	-12

A student is delivering postcards to teachers from students on a school trip. For time t minutes, $0 \leq t \leq 24$, the rate of change in the number of postcards remaining for the student to deliver is modeled by a twice-differentiable function P , in postcards remaining per minute. Values of $P(t)$ at various times t are shown in the table above.

- Using correct units, approximate $P'(8)$ using the average rate of change of P over the interval $0 \leq t \leq 24$.
- Explain the meaning of $P'(8)$ in the context of the problem.
- The function $N(t)$ models the number of postcards remaining for the student to deliver at time t minutes, where $N'(t) = P(t)$. It is known that $N(8) = 185$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 8$.
- Find the value of $\int_0^8 P'(3t) \, dt$. Show the calculations that lead to your answer.
- Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{24} P(t) \, dt$. Show the calculations that lead to your answer.
- Another student is also delivering postcards. For $0 \leq t \leq 24$, the rate of change in the number of postcards remaining for the second student to deliver, in postcards per minute, can be modeled by the differentiable function R . Assume the weight of each postcard is 4 grams. Write an expression in terms of P and R for the rate of change, in grams per minute, of the combined weight of the postcards remaining for both students at $t = 13$ minutes.

6-2

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = -0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{-12+6}{24}$ postcards per minute per minute



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

6-2

$$P'(8) \approx \frac{P(24) - P(0)}{24 - 0} = \frac{-12 - (-6)}{24} = \frac{-6}{24}$$

$$= -\frac{1}{4} \text{ postcard per minute per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of postcards
- Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

- $P'(8)$ is the average (or an approximation, or an estimate of) rate of change of
- $P'(8)$ is the acceleration (or rate of change of velocity) of

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $P'(8)$ is the rate of decrease of the rate of change of the number of postcards at 8 minutes. (earns the point)
- Example: $P'(8)$ means the number of postcards per minute is decreasing by $\frac{1}{4}$ postcards/minute/minute. (does not earn the point)

Acceptable references to time include (but are not limited to):

- On (or at) minute 8
- At $t = 8$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 8$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “from minute 0 to minute 8”) the response will not earn the point.

Examples that earn the point include:

- $P'(8)$ represents the rate at which the rate of change of the number of postcards is changing on minute 8.
- $P'(8)$ is the rate of change in the rate of change of the number of postcards at 8 minutes.
- $P'(8)$ represents the rate at which the rate of change of the number of postcards is decreasing at $t = 8$.
- $P'(8)$ equals (answer from part (a)) means the rate of change in the number of postcards per minute is increasing at a rate of (answer from part (a)) at $t = 8$.
- $P'(8)$ represents the rate at which the rate of change of the number of postcards is increasing on the eighth minute.

6-2

Examples that do not earn the point include:

- $P'(8)$ represents the rate at which the number of postcards is changing on minute 8.
- $P'(8)$ is the rate of change in the number of postcards per minute at $t = 8$ minutes
- The rate of change in the number of postcards per minute is decreasing at $t = 8$.
- The rate of change in the rate of change of the number of postcards per day is decreasing at $t = 8$.
- $P'(8) = -\frac{1}{4}$ means that the rate of change in the number of postcards is decreasing at a rate of $-\frac{1}{4} t = 8$. (double negative)



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(8)$ is the rate at which the rate of change in the number of postcards remaining is changing, in postcards per minute per minute, at minute 8.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

- $y = -10(t - 8) + 185$
- $y - 185 = -10(t - 8)$
- $y = -10t + 265$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $N(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = N'(8)(t - 8) + N(8)$$

6-2

$$y = -10(t - 8) + 185$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required as evidence of correct substitution:

- Coefficient of $\frac{1}{3}$ correctly applied to an integral or an antiderivative
- Evidence that the correct limits of 0 to 24 will be applied

Examples that will earn the first point:

$$\cdot \frac{1}{3} \int_0^{124} P'(u) du$$

$$\cdot \frac{1}{3} P(u) \Big|_0^{24} \text{ or } \frac{1}{3} P(3t) \Big|_0^8$$

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\cdot \int_0^8 P'(3t) dt = \frac{1}{3} \int_0^8 P'(u) du = \frac{1}{3} P(3t) \Big|_0^8 \text{ earns the first point, but will not earn the third point.}$$

$$\cdot \int_0^8 P'(3t) dt = \frac{1}{3} \int_0^8 P'(u) du = \frac{1}{3} [P(8) - P(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b P'(u) du = P(u) \Big|_a^b = P(b) - P(a)$$

This point can be earned with either $kP(3t) \Big|_a^b$, $kP(t) \Big|_a^b$, or $k(P(b) - P(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 8$ or $b = 24$.

The response can earn this point without earning the first point.

$$\frac{1}{3} [P(24) - P(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{3} [P(24) - P(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\cdot \int_0^8 P'(3t) dt = \frac{1}{3} \int_0^8 P'(u) du = \frac{1}{3} P(3t) \Big|_0^8 = \dots \text{ is eligible for the third point.}$$

Eligibility: Must have earned the first two points.

6-2

The minimum response needed to earn all three points is: $\frac{1}{3}(-12 + 6)$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 dt$.

When $t = 0$, $u = 0$.

When $t = 8$, $u = 24$.

$$\int_0^8 P'(3t) dt = \frac{1}{3} \int_0^{24} P'(u) du$$

$$= \frac{1}{3}(P(24) - P(0))$$

$$= \frac{1}{3}(-12 - (-6)) = -\frac{6}{3} = -2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the form of a trapezoidal sum

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point (e.g., $\frac{1}{2}[8(P(0) - P(8)) + 16(P(8) - P(24))]$ earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one error in the numerical values.

$\frac{1}{2}(8)[P(0) + 2P(8) + P(24)]$ is equivalent to $\frac{1}{2}[8(P(0) + P(8)) + 8(P(8) + P(24))]$, thus has only one incorrect numerical value (the second interval width is 8 instead of 16), so can earn the point.

6-2

$\cdot \frac{1}{2}(12) [P(0) + 2P(8) + P(24)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation (+) of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will only the first point:

$\cdot k \left[\left(\frac{-6-10}{2} \right) \cdot 8 + \left(\frac{-10-12}{2} \right) \cdot 16 \right]$ for $k > 0$ earns just the first point.

The minimum response needed to earn both points: $\left(\frac{-16}{2} \right) \cdot 8 + \left(\frac{-22}{2} \right) \cdot 16$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\int_0^{24} P(t) dt \approx \left(\frac{P(0)+P(8)}{2} \right) (8-0) + \left(\frac{P(8)+P(24)}{2} \right) (24-8)$$

$$= \left(\frac{-6-10}{2} \right) \cdot 8 + \left(\frac{-10-12}{2} \right) \cdot 16 = -240$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 4 as a factor of the expression, and/or the correct sum $P(13) + R(13)$.

If the sum is incorrect, then the constant factor of 4 must be applied either to a sum that involves only $P(t)$ and $R(t)$, $P'(t)$ and $R'(t)$, or integrals of $P(t)$ and $R(t)$, or to one of the correct terms of the sum $P(13)$ or $R(13)$, with no additional incorrect terms.

Examples that earn the point include:

$$\cdot 4 [P(t) + R(t)]$$

$$\cdot 4 \int [P(t) + R(t)] dt$$

$$\cdot 4 [P' + R']$$

6-2

$$\cdot 4P(13)$$

Examples that do not earn the point include:

$$\cdot 4[P(t) + R'(t)]$$

$$\cdot 4 \int P(t) dt$$

$$\cdot 4P' + R'$$

$$\cdot 4R(13) + t$$

Special case: $4P(13) + R(13)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[P(13) + R(13)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$4P(13) + 4R(13)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn only the first point. For example, $4P(13) + 4R(13) + 4R(2)$ earns the first point only.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

The rate of change, in grams per minute, of the combined weight of the postcards remaining for both students at $t = 13$ minutes is $4P(13) + 4R(13)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y as t changes.”)

6-2

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(2, 2)$ the slope shown on the slope field is -1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-2

54. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	24	35
$P(t)$ (number of postcards remaining)	316	260	100	0

A principal is delivering postcards to teachers from students on a school trip. The principal begins with 316 postcards and delivers them over 35 minutes. For time t minutes, $0 \leq t \leq 35$, the total number of postcards remaining to deliver is modeled by a decreasing, twice-differentiable function P . Values of $P(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $P'(8)$ using the average rate of change of P over the interval $0 \leq t \leq 24$.

(b) Explain the meaning of $P'(8)$ in the context of the problem.

(c) It is known that $P'(24) = -11$. Write an equation for the line tangent to the graph of $y = P(t)$ at $t = 24$.

(d) Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{35} P(t) \, dt$. Show the computations that lead to your answer.

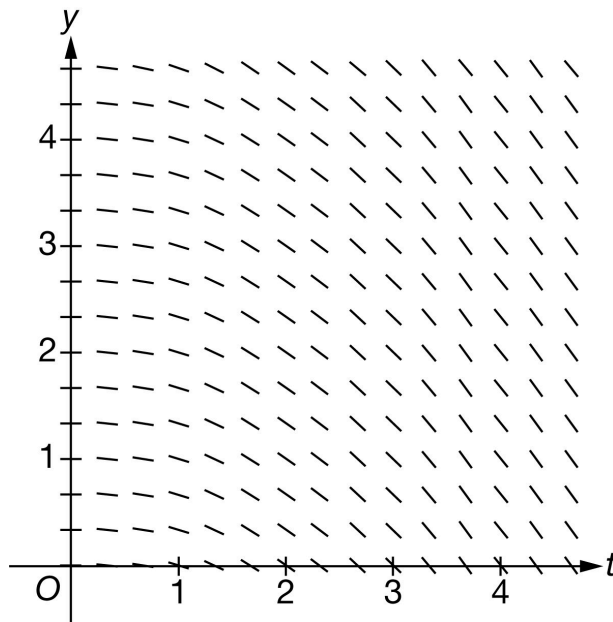
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{35} P(t) \, dt$? Give a reason for your answer.

(f) Find the value of $\int_0^8 P'(3t) \, dt$. Show the computations that lead to your answer.

(g) For $0 \leq t \leq 35$, the total number of postcards remaining can also be modeled by the differentiable function R , where R is a function of t . Assume the weight of each postcard is 4 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 13$ minutes.

6-2

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\frac{100-316}{24-0}$$

$$\frac{P(24)-P(0)}{24} = \frac{-216}{24}$$

$$\frac{P(24)-P(0)}{24} = -9$$

An example that does not earn the first point is $\frac{P(24)-P(0)}{24}$.

Minimal response that earns the first point: $\frac{100-316}{24-0}$ or $\frac{100-316}{24}$

Neither $\frac{-216}{24}$ nor -9 , with no supporting work, will earn this point

The second point is for correct units

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

· Postcards per minute

· Cards per min

6-2

· Postcards/min

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$P'(8) \approx \frac{P(24) - P(0)}{24 - 0} = \frac{100 - 316}{24} = -9 \text{ postcards per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of change
- Including the context of postcards
- Referring to the specific point in time, $t = 8$

The response is not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “decreasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “the number of postcards is decreasing at 9 postcards per minute.” is okay, but “the number of postcards is decreasing at a rate of -9 postcards per day...” would not earn the point.

Acceptable reference to a specific point in time includes (but is not limited to):

- On minute 8 and after 8 minutes
- At $t = 8$
- At $t = 8$ hours or any other incorrect unit of time
- If the response clearly indicates an interval of time (e.g., “from minute 8 on,” “during the first 8 minutes,” or “for $t = 0$ to $t = 8$ ”), the point is not earned.

Examples that earn the point include:

- $P'(8)$ is the rate of change of the number of postcards per minute at time $t = 8$
- The rate of change of number of postcards is -9 postcards per minute at minute 8
- $P'(8)$ is the rate of change in the number of postcards at time $t = 8$ in postcards per hour

6-2

- -9 is the rate of decrease in the number of postcards per minute at minute 8

Examples that do not earn the point include:

- $P'(8)$ is the average rate of change of the total number of postcards /minute at minute 8
- $P'(8)$ is the number of postcards per minute at minute 8
- The rate of change is -9 postcards per minute (Missing “at $t = 8$ ”)
- $P'(8)$ is the slope of the tangent line at $t = 8$
- Rate of change of P at time $t = 8$
- Postcards per minute decreases on minute 8



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(8)$ is the rate at which the number of postcards remaining is changing, in postcards per minute, at time $t = 8$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

Some examples include:

- $y - 100 = -11(t - 24)$, $y - 100 = -11(x - 24)$
- $y = -11(t - 24) + 100$, $y = -11(x - 24) + 100$
- $y = -11t + 364$, $y = -11x + 364$

The key calculus concept being assessed in this part of the problem is that the value of $P'(24)$ is the slope of the tangent line. In that spirit, we are accepting $P'(t) = \dots$ in lieu of $y = \dots$, but we are not accepting $P'(24) = \dots$.

Note that $P(t) = -11(t - 24) + 100$ earns the point, but $P(24) = -11(t - 24) + 100$ does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

6-2

Solution:

$$y = P'(24)(t - 24) + P(24)$$

$$y = -11(t - 24) + 100$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

- A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

- Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (d).

The second point is for the correct numerical result.

- In order to earn the second point, there must be supporting calculations.

- The numerical values for $P(8)$, $P(24)$, and $P(35)$ must be pulled from the table

- Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $260 \cdot (8 - 0) + 100 \cdot (24 - 8) + 0 \cdot (35 - 24)$ earns both points.

- A response of $8 \cdot P(8) + 16 \cdot P(24) + 11 \cdot P(35)$ earns the first point. Eligible for second point if more work is on the page.

- A response of $8 \cdot P(8) + 16 \cdot P(24) + 11 \cdot P(35) = 3680$ earns both points.

- A response of $8 \cdot 260 + 16 \cdot 100 + 11 \cdot 0$ earns both points.

- A response of $8 \cdot 260 + 12 \cdot 100 + 11 \cdot 0$ contains one numerical error. Earns the first point. Not eligible for the second point.

- A response of $8 \cdot 260 + 16 \cdot 100 + 11 \cdot 0 = 3860$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.

- A response of $2080 + 1600 + 0 = 3680$ or $2080 + 1600 = 3680$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$8 \cdot 260 + 16 \cdot 100 + 11 \cdot 0$$

6-2



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{35} P(t) dt \approx P(8)(8-0) + P(24)(24-8) + P(35)(35-24)$$

$$= 260 \cdot 8 + 100 \cdot 16 + 0 \cdot 11 = 3680$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because P is decreasing.

The response does not need an explicit reference to the approximation in part (d), nor does there need to be an answer given in part (d). The following statements are sufficient for earning this point:

- Underestimate because P is decreasing
- Underestimate because the function is decreasing

Note: At this point, there is only one function being discussed in this problem, so we will assume “the function” is P .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function P , the point is not earned.

Examples that earn the point:

- The right Riemann sum underestimates because the function is decreasing.
- The sum is an underestimate because $P(t)$ is decreasing.
- The sum is an underestimate because $P'(t) < 0$.

Examples that do not earn the point:

- The sum is an underestimate because $P'(t)$ is decreasing. ($P'(t)$ decreasing is not known)
- The right Riemann sum underestimates because the graph is decreasing.
- The right Riemann sum underestimates because the function is decreasing and concave up.



0	1
---	---

6-2

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because P is a decreasing function, and this is a right Riemann sum, the answer from part (d) is an underestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = 3t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{3}$
- Uses the correct limits for the substitution (0 and 24). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.
- Either of the expressions below will earn the substitution point without any additional supporting work: $\frac{1}{3} P(3t) \Big|_0^8$ or $\frac{1}{3} P(u) \Big|_0^{24}$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{24}^0 P'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 8$ or $b = 24$. No explanation or additional supporting work is necessary.

- $kP(3t) \Big|_0^b$
- $k(P(b) - P(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is -72 .

Values from the table must be used. $\frac{1}{3}[P(24) - P(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{3}(100 - 316)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning the FTC point.

6-2

Minimalist response: The expression $\frac{1}{3}(-216)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 \, dt$.

When $t = 0$, $u = 0$.

When $t = 8$, $u = 24$.

$$\begin{aligned}\int_0^8 P'(3t) dt &= \frac{1}{3} \int_0^{24} P'(u) du \\ &= \frac{1}{3}(P(24) - P(0)) \\ &= \frac{1}{3}(100 - 316) = -\frac{216}{3} = -72\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 13$. No supporting work is required.

The correct expression, $4R'(13)$ or $4P'(13)$, earns this point.

No justification or explanations are required.

Any incorrect statement such as, $P'(13) = 4R'(13)$, does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 13$ minutes is $4R'(13)$.

6-2

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = -0.4y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field., but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point (4, 0), the slope of the segment should be 0, but it is negative.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point. For example, merely stating either $\frac{dy}{dt}$ is negative, or that the slope field is negative discretely does not earn the point.

- “The values for y are less than or equal to zero so $\frac{dy}{dt} \leq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are negative” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is decreasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-2

55. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

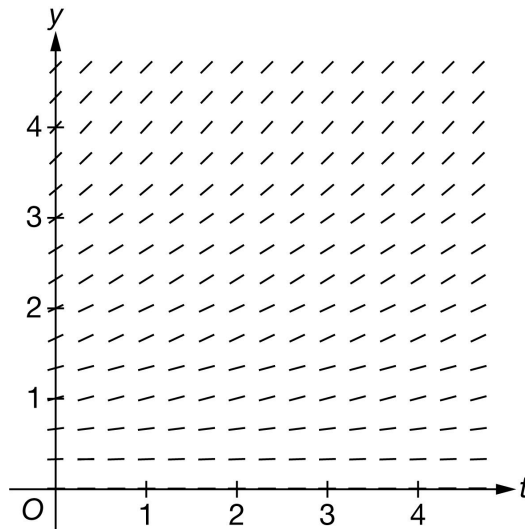
t (minutes)	0	9	18	27	36
$P(t)$ (number of postcards remaining)	320	256	100	40	0

A principal is delivering postcards to teachers from students on a school trip. The principal begins with 320 postcards and delivers them over 36 minutes. For time t minutes, $0 \leq t \leq 36$, the total number of postcards remaining to deliver is modeled by a twice-differentiable function P . Values of $P(t)$ at various times t are shown in the table above.

- (a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate $\int_0^{36} P(t) \, dt$. Show the computations that lead to your answer.
- (b) Find the value of $\int_0^9 P'(4t) \, dt$. Show the computations that lead to your answer.
- (c) Using correct units, approximate $P'(18)$ using the average rate of change of P over the interval $9 \leq t \leq 27$.
- (d) Explain the meaning of $P'(18)$ in the context of the problem.
- (e) It is known that $P'(9) = -12$. Write an equation for the line tangent to the graph of $y = P(t)$ at $t = 9$.
- (f) It is known that $P''(t) < 0$ for $9 \leq t \leq 18$. If $P(15)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $P(15)$? Give a reason for your answer.
- (g) For $0 \leq t \leq 36$, the total number of postcards remaining can be modeled by the differentiable function Z , where Z is a function of t . Assume the weight of each postcard is 4.5 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 19$ minutes.

6-2

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.3y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 18 \cdot (P(9) + P(27)) \text{ or } 18 \cdot P(9) + 18 \cdot P(27)$$

$$\cdot (18 - 0) \cdot 256 + (36 - 18) \cdot 40$$

$$\cdot 18 \cdot (256 + 40) \text{ or } 18 \cdot 256 + 18 \cdot 40$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{18}{b} \cdot (P(9) + P(27))$, for any nonzero value of b would earn the point. Note: The value $\frac{18}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (P(9) + P(27))$ for $a \neq 18$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 18 \cdot (256 + 40)$$

$$\cdot 18 \cdot 256 + 18 \cdot 40$$

6-2



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint sum
- ☐ Answer

Solution:

$$\int_0^{36} P(t) dt \approx P(9)(18 - 0) + P(27)(36 - 18)$$

$$= 256 \cdot 18 + 40 \cdot 18 = 5328$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 4t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{4}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int P'(4t) dt$ or $\int P'(u) du$.

The point is earned for any of the following:

· The equation $\int P'(4t) dt = P(4t)$ or $\int P'(u) du = P(u)$

· $\frac{1}{4}P(4t)\big|_0^9$ or $\frac{1}{4}P(u)\big|_0^{36}$

· $P(36) - P(0)$, $P(36)$, or $P(0)$

This point can be earned if the substitution point is not.

A response that applies the FTC to $P'(t)$ does not earn this point.

Note: $\int_a^b P'(u) du = P(b) - P(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of P' is P .

6-2

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$P(36)$ and $P(0)$ must be evaluated; the expression $\frac{1}{4}(P(36) - P(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{4}(0 - 320)$ or $-\frac{320}{4}$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

$$\cdot \frac{1}{4}P(4t) \Big|_0^9$$

$$\cdot \frac{1}{4}(P(36) - P(0))$$

Responses of only the following would earn only the second point:

$$\cdot P(36) - P(0)$$

$$\cdot P(9) - P(0)$$

$$\cdot P(36)$$

$$\cdot \int_0^9 P'(u) du = P(9)$$

$$\cdot P(4t) \Big|_0^9 \text{ or } P(u) \Big|_0^{36}$$

The following responses would earn no points:

$$\cdot -80 \text{ in the absence of one of the first two points.}$$

$$\cdot \int_0^9 P'(4t) dt = P(9)$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

6-2

Solution:

Let $u = 4t$. Then $du = 4 \, dt$.

When $t = 0$, $u = 0$.

When $t = 9$, $u = 36$.

$$\begin{aligned}\int_0^9 P'(4t) \, dt &= \frac{1}{4} \int_0^{36} P'(u) \, du \\ &= \frac{1}{4}(P(36) - P(0)) \\ &= \frac{1}{4}(0 - 320) = -80\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{40-256}{18}$ or $\frac{-216}{27-9}$.

Second point (Units)

The units must be attached to a value (either correct or incorrect).

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$P'(18) \approx \frac{P(27) - P(9)}{27 - 9} = \frac{40 - 256}{18} = -12 \text{ postcards per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

6-2

- indicate the rate of change
- the specific context of the number of postcards remaining
- the specific point in time, $t = 18$.

Responses do not need to include correct units or any units.

The word “decreasing” may be used in place of the word “changing”.

The response is not earned for:

- “the rate of postcards delivered in minute 18” as this does not indicate a change in the rate of delivery.
 - “decreasing at a rate of -12 postcards per minute”
- “the rate of change of P ” without specifying that P is the number of postcards delivered, does not earn the point.

Acceptable ways to provide the specific point in time:

- In minute 18 or after 18 minutes.
- At $t = 18$ (missing the unit of time)
- After 18 days (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 18 minutes” or “from minute 18 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(18)$ is the rate at which the number of postcards remaining is changing, in postcards per minute, at time $t = 18$ minutes

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $P(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

o $y = 256 - 12(t - 9)$

o $y - 256 = 12(t - 9)$

6-2

$$\circ y = -12t + 108 + 256$$

$$\circ y = -12t + 364$$



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = P'(9)(t - 9) + P(9)$$

$$y = -12(t - 9) + 256$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point the response must include $P''(t) < 0$ and overestimate.

“ $P(t)$ is concave down” can be used instead of $P''(t) < 0$.

Information about $P'(t)$ by itself without reference to $P''(t) < 0$ is not sufficient and does not earn this point.

“Overestimate because it is concave down” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states $P''(t) < 0$ and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $P'(t)$ is concave down, overestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because $P''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $4.5 \cdot Z'(19)$ earns this point.

6-2

Only expression $4.5 \cdot P'(19)$ also earns this point.

No justifications or explanations are required.

Any incorrect statement such as, $Z'(19) = 4.5 \cdot P'(19)$, does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 19$ minutes is $4.5Z'(19)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

- “ $\frac{dy}{dt}$ is negative, not positive”. We will infer that negative is referring to the slopes shown in the slope field.
- “This would be correct for a positive derivative, but $\frac{dy}{dt}$ is negative.” We will read “This” as “the slope field” in this sentence.
- “All of the slopes in the slope field are positive, not negative.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are negative.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

- “ $\frac{dy}{dt}$ is negative” lacks any reference to the slope field and so will not earn the point.
- “All the slopes in the graph are positive” lacks any reference to the differential equation, and so will not earn the point.
- Vague answers such as “The slope field has positive points” or “The field is increasing” will not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

6-2☐

Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} < 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

6-2

56. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

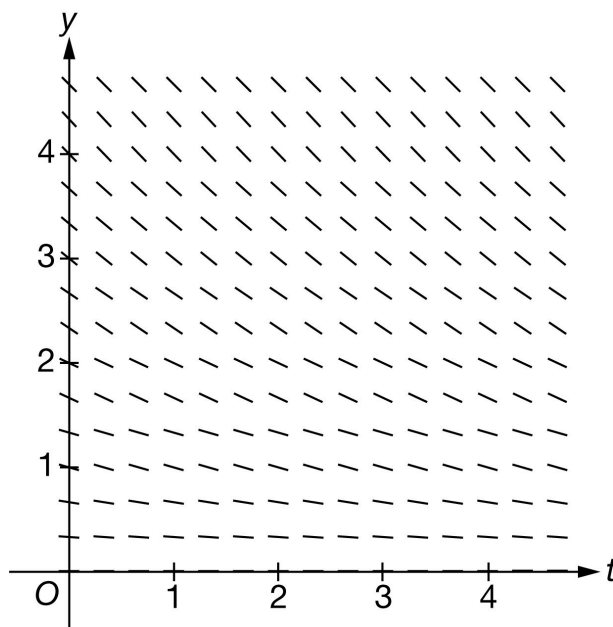
t (minutes)	0	8	24	35
$P(t)$ (number of postcards remaining per minute)	-10	-8	-5	-4

A student is delivering postcards to teachers from students on a school trip. For time t minutes, $0 \leq t \leq 35$, the rate at which the total number of postcards remaining to deliver is changing, in number of postcards remaining per minute, is modeled by an increasing, twice-differentiable function P . Values of $P(t)$ at various times t are shown in the table above.

- (a) Use a left Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{35} P(t) \, dt$. Show the calculations that lead to your answer.
- (b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{35} P(t) \, dt$? Give a reason for your answer.
- (c) Using correct units, approximate $P'(24)$ using the average rate of change of P over the interval $8 \leq t \leq 35$.
- (d) Explain the meaning of $P'(24)$ in the context of the problem.
- (e) It is known that $P'(8) = 0.2$. Write an equation for the line tangent to the graph of $y = P(t)$ at $t = 8$.
- (f) Find the value of $\int_0^{24} P'\left(\frac{t}{3}\right) \, dt$. Show the calculations that lead to your answer.
- (g) Another student brings a bag full of postcards to deliver as well. For $0 \leq t \leq 35$, the total number of postcards delivered to teachers by this student can be modeled by the increasing, differentiable function R , where R is a function of t . Assume the weight of each postcard is 4 grams. Write an expression for the rate of change, in grams per minute, of the total weight of postcards delivered by the second student at $t = 11$ minutes.

6-2

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = R'(t)$ for the function R defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only 1 numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

8	(-10)	-80	or	$(8 - 0)(-10)$	$= -80$
16	(-8)	-128		$(24 - 8)(-8)$	$= -128$
11	(-5)	$+(-55)$		$(35 - 24)(-5)$	$= -55$
		-263			-263

Implicit responses such as those above are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

In order to earn the second point, there must be supporting calculations.

The numerical values for $P(0)$, $P(8)$, and $P(24)$ must be pulled from the table

Any simplification shown must be correct.

Use the following as an illustration how to score of both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

6-2

- A response of $(-10) \cdot (8 - 0) + (-8) \cdot (24 - 8) + (-5) \cdot (35 - 24)$ earns both points.
- A response of $8 \cdot P(0) + 16 \cdot P(8) + 11 \cdot P(24)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $8 \cdot P(0) + 16 \cdot P(8) + 11 \cdot P(24) = -263$ earns both points.
- A response of $8 \cdot (-10) + 16 \cdot (-8) + 11 \cdot (-5)$ earns both points.
- A response of $8 \cdot (-10) + 18 \cdot (-8) + 11 \cdot (-5)$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $8 \cdot (-10) + 16 \cdot (-8) + 11 \cdot (-5) = -200$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $(-80) + (-128) + (-55)$ or $(-80) + (-128) + (-55) = -263$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points. $8 \cdot (-10) + 16 \cdot (-8) + 11 \cdot (-5)$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{35} P(t) dt \approx P(0)(8 - 0) + P(8)(24 - 8) + P(24)(35 - 24)$$

$$= (-10) \cdot 8 + (-8) \cdot 16 + (-5) \cdot 11 = -263$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because P is increasing.

The response does not need an explicit reference to part (a) or to a left Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following examples earn the point:

- Underestimate because P is increasing
- Underestimate because $P'(t) > 0$
- Underestimate because the graph of P is increasing (we know they are talking about the given function P)

6-2

· Underestimate because the function is increasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

· Underestimate because P is concave down/up

· Underestimate because the graph/curve is increasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.

· Underestimate because the table [or table values] is/are increasing. The behavior of the values shown in the table is not the same as the behavior of the function P . The table does not tell us what is happening to P between the given points.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because P is an increasing function, and this is a left Riemann sum, this is an underestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $P(35)$ and $P(8)$ must be pulled from the table.

Decimal approximation is not expected, but if presented, then the decimal value must be correct to three decimal places, either rounded or truncated. We do not read anything past the third decimal place.

· If the first three places match 0.148 then the point is earned

· It does not matter whether any subsequent decimal places match the correct value or not. (For example, the value 0.1487 will earn the point because the first three decimal places match the correct value.)

· Responses with fewer than three decimal places (0.14, 0.15, 0.1, etc.) will not earn the point.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

· Postcards per minute per minute or Postcards/min/min

· Postcards per min² or Postcards/min²

· Postcards per minute squared

6-2

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter “ P ” is not a standard abbreviation for “Postcards.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points. $\frac{-4+8}{27}$ postcards/min² or $\frac{-4-(-8)}{27}$ postcards/min²



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$P'(24) \approx \frac{P(35)-P(8)}{35-8} = \frac{-4-(-8)}{27} = \frac{4}{27} \text{ postcard per minute per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of a rate
- Including the context of postcard delivery (postcards remaining)
- Explicitly referring to the specified point in time $t = 24$

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use $\frac{4}{27}$ as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of $\frac{4}{27}$ ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)
- “... decreasing at a rate of $\frac{4}{27}$ ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)

Note: the context is complicated. $P(t)$ is describing the rate of change of the remaining undelivered postcards. The values in the table indicate that the number of undelivered postcards is decreasing more slowly as time passes. Common sense might tell us that this equates to the idea that the rate at which postcards are delivered is decreasing. Explaining the meaning using “postcards delivered” rather than “postcards remaining” will not prevent the response from earning this point.

6-2

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

- The rate of change in the rate at which postcards are changing
- The rate of change in postcards is changing/increasing/decreasing at a rate of . . .

The rate at which the rate of change of postcards is changing.

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “postcards per minute” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

“Rate of change of P ” is not enough without describing what P represents in this context.

Context

A key component of the context for this explanation is “postcards.”

Any misstatement about postcards delivered vs. postcards remaining to be delivered will not prevent the response from earning this point.

Explicit reference to $t = 24$

Acceptable references to a specific point in time include (but are not limited to):

- At minute 24 (where minute could be replaced with any other unit of time)
- At $t = 24$ (with or without units of time)
- After 24 minutes (where minutes could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from minute 24 on” or “during the first 24 minutes”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(24)$ is the rate at which the rate of change in the number of postcards remaining is changing, in postcards per minute per minute, at minute 24.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-2

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

$$\cdot y = 0.2(t - 8) - 8$$

$$\cdot y + 8 = 0.2(t - 8)$$

$$\cdot y = 0.2t - 9.6$$

The key calculus concept being assessed in this part of the problem is that the value of $P'(8)$ is the slope of the tangent line. In that spirit:

- The response will not be penalized for using x instead of t .
- We are accepting $P(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = P'(8)(t - 8) + P(8)$$

$$y = 0.2(t - 8) - 8$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = \frac{t}{3}$, which may be done implicitly
- Uses the correct constant of 3
- Uses the correct limits for the substitution (0 and 8). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $3P\left(\frac{t}{3}\right)\Big|_0^{24}$ or $3P(u)\Big|_0^8$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral

6-2

· Reversed limits of integration, for example: $\int_8^0 P'(u) \, du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 8$ or $b = 24$. No explanation or additional supporting work is necessary.

· $kP\left(\frac{t}{3}\right)\Big|_0^b$

· $k(P(b) - P(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 6.

Values from the table must be used. $3(P(8) - P(0))$ is not enough to earn the answer point.

The expression $3(-8 - (-10))$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they do not need to be correct to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $3(-8 - (-10))$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{3}$. Then $du = \frac{dt}{3}$.

When $t = 0$, $u = 0$.

When $t = 24$, $u = 8$.

$$\int_0^{24} P'\left(\frac{t}{3}\right) dt = 3 \int_0^8 P'(u) du$$

$$= 3(P(8) - P(0))$$

$$= 3(-8 - (-10)) = 6$$

Part G

6-2

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 11$ earns the point. It may be written as $4 \cdot R'(11)$ or $4 \cdot \frac{dR}{dt} \Big|_{t=11}$

The expression $4 \cdot R'(t)$ is not sufficient because it does not use $t = 11$.

The response must use R (not M or G or any other function name) in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of postcards delivered by the second student at $t = 11$ minutes is $4R'(11)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation $\frac{dy}{dt} = R'(t)$ and the given slope field.

In order to earn the point, the response must state that the values of the differential equation are positive while the slopes in the given slope field are negative or 0 (non-positive).

Some ways to say that the values of differential equation are positive:

- R is increasing
- $R'(t) > 0$
- R has positive slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has nonpositive slopes.
- The slopes shown are negative or 0.
- The slope field contains negative slopes.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is decreasing/negative.

6-2

- The slope field looks like it represents a decreasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ R is an increasing function, so the signs of the slopes in the slope field should be positive” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” negative or “all” negative.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

R is an increasing function, so $\frac{dy}{dt} = R'(t) \geq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

6-2

57. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	20	30
$D(t)$ (meters per minute)	0	100	120	150

A student on the school track and field team is running laps around a track. For time t minutes, $0 \leq t \leq 30$, the rate at which the student runs, in meters per minute, is modeled by an increasing, twice-differentiable function D . Values of $D(t)$ at various times t are shown in the table above.

(a) Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{30} D(t) \, dt$. Show the calculations that lead to your answer.

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{30} D(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $D'(5)$ using the average rate of change of D over the interval $0 \leq t \leq 20$.

(d) Explain the meaning of $D'(5)$ in the context of the problem.

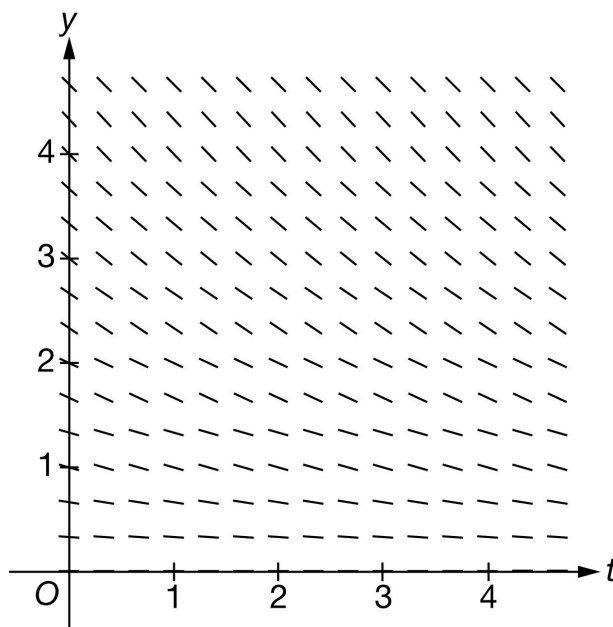
(e) It is known that $D'(20) = 2.5$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 20$.

(f) Find the value of $\int_0^{20} D'\left(\frac{t}{4}\right) \, dt$. Show the calculations that lead to your answer.

(g) A friend on the track and field team is using her running practice as a fundraiser. For $0 \leq t \leq 30$, the total number of laps run by the friend can be modeled by the increasing, differentiable function L , where L is a function of t . For every lap run, the friend raises 4 dollars. Write an expression for the rate of change of the money, in dollars per minute, raised by the friend at $t = 19$ minutes.

6-2

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = L'(t)$ for the function L defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

100	5	500
120	15	1800
150	10	+1500
		3800

or

$(5 - 0)(100)$	$= 500$
$(20 - 5)(120)$	$= 1800$
$(30 - 20)(150)$	$= 1500$
	3800

Implicit responses such as those above are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (a).

The second point is for the correct numerical result.

In order to earn the second point, there must be supporting calculations.

The numerical values for $D(5)$, $D(20)$, and $D(30)$ must be pulled from the table.

6-2

Any simplification shown must be correct.

Use the following as an illustration of how to score both points. (In most cases, you see scores for which the expression represents the entirety of relevant work shown for the response.)

- A response of $100 \cdot (5 - 0) + 120 \cdot (20 - 5) + 150 \cdot (30 - 20)$ earns both points.
- A response of $5 \cdot D(5) + 15 \cdot D(20) + 10 \cdot D(30)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $5 \cdot D(5) + 15 \cdot D(20) + 10 \cdot D(30) = 3800$ earns both points.
- A response of $5 \cdot 100 + 15 \cdot 120 + 10 \cdot 150$ earns both points.
- A response of $5 \cdot 100 + 25 \cdot 120 + 10 \cdot 150$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $5 \cdot 100 + 15 \cdot 120 + 10 \cdot 150 = 3300$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $500 + 1800 + 1500$ or $500 + 1800 + 1500 = 3800$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points. $5 \cdot 100 + 15 \cdot 120 + 10 \cdot 150$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{30} D(t) dt \approx D(5)(5 - 0) + D(20)(20 - 5) + D(30)(30 - 20)$$

$$= 100 \cdot 5 + 120 \cdot 15 + 150 \cdot 10 = 3800$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because D is increasing.

The response does not need an explicit reference to part (a) or to a right Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following table contains some examples of what WILL or will NOT earn the point.

The following examples earn the point:

6-2

- Overestimate because D is increasing
- Overestimate because $D'(t) > 0$
- Overestimate because the graph of D is increasing (we know they are talking about the given function D)
- Overestimate because the function is increasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Overestimate because D is concave down/up
- Overestimate because the graph/curve is increasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Overestimate because the table [or table values] is/are increasing. The behavior of the values shown in the table is not the same as the behavior of the function D . The table does not tell us what is happening to D between the given points.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

Solution:

Because D is an increasing function, and this is a right Riemann sum, this is an overestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $D(20)$ and $D(0)$ must be pulled from the table.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Meters per minute per minute or Meters/min/min
- Meters per min^2 or Meters/ min^2
- Meters per minute squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

Use caution when interpreting abbreviations: m should be meters and min should be minutes, but responses may vary. For example,

6-2

- if a response includes units “Meters/m/m ” we read “ m ” as minutes; or
- if a response includes units “m/min/min ” we read “ m ” as meters; or
- if a response includes units “m/m/m ” we cannot interpret the units and the response will not earn the second point.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points.

$$\frac{120-0}{20} \text{ meters/min}^2 \text{ or } \frac{120}{20-0} \text{ meters/min}^2$$

In the minimalist response, we still need to see subtraction (a difference), so subtracting 0 must be explicit either in the numerator or in the denominator.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$D'(5) \approx \frac{D(20)-D(0)}{20-0} = \frac{120-0}{20} = 6 \text{ meters per minute per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

- Three components of a correct answer:
- Indicating a rate of a rate
- Including the context of running
- Explicitly referring to the specified point in time ($t = 5$)

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use 6 as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of 6 ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)
- “... decreasing at a rate of 6 ” is not OK and the point is not earned.

6-2

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change in the context of motion [running]. For example:

- The rate of change in the rate at which students are running
- The rate of change in speed is changing (or increasing) at a rate of . . .

Since the runners are unlikely to be running backwards, we interpret speed as being the same as velocity.

- The rate at which the velocity of the runner is changing
- Acceleration of the runner (this is correct for the context)

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates

Units are not interpreted as rates. (The phrase “meters per minute” is not enough to indicate a rate.)

Slope of tangent line is not an appropriate interpretation for this context.

“Rate of change of D ” is not enough without describing what D represents in this context.

Context

The context is running, which could be indicated by words such as (but not limited to): meters, minutes, laps, speed, velocity, runner

Explicit reference to $t = 5$

Acceptable references to a specific point in time include (but are not limited to):

- At minute 5 (where minute could be replaced with any other unit of time)
- At $t = 5$ (with or without units of time)
- After 5 minutes (where minutes could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from minute 5 on” or “during the first 5 minutes”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$D'(5)$ is the rate at which the rate at which the student runs is changing, in meters per minute per minute, at minute 5.

Part E

6-2

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

· $y = 2.5(t - 20) + 120$

· $y - 120 = 2.5(t - 20)$

· $y = 2.5t + 70$

The key calculus concept being assessed in this part of the problem is that the value of $D'(20)$ is the slope of the tangent line. In that spirit:

- The response will not be penalized for using x instead of t .
- We are accepting $D(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = D'(20)(t - 20) + D(20)$$

$$y = 2.5(t - 20) + 120$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that we need to see:

- Uses $u = \frac{t}{4}$, which may be done implicitly
- Uses the correct constant of 4
- Uses the correct limits for the substitution (0 and 5). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $4D\left(\frac{t}{4}\right)\bigg|_0^{20}$ or $4D(u)\bigg|_0^5$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),

6-2

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\int_5^0 D'(u) \, du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 20$. No explanation or additional supporting work is necessary.

· $kD\left(\frac{t}{4}\right)\Big|_0^b$

· $k(D(b) - D(0))$ or $kD(b)$ (because $D(0) = 0$)

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 400.

Values from the table must be used. $4(D(5) - D(0))$ is not enough to earn the answer point.

The expressions $4(100 - 0)$ or $4(100)$ are sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they do not need to be correct to earn this point.

The response cannot earn this point without also earning the first two points.

Minimalist response: The expression $4(100)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{4}$. Then $du = \frac{dt}{4}$.

When $t = 0$, $u = 0$.

When $t = 20$, $u = 5$.

$$\int_0^{20} D'\left(\frac{t}{4}\right) dt = 4 \int_0^5 D'(u) du$$

$$= 4(D(5) - D(0))$$

$$= 4(100 - 0) = 400$$

6-2

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 19$ earns the point. It may be written as $4 \cdot L'(19)$ or $4 \cdot \frac{dL}{dt} \Big|_{t=19}$.

The expression $4 \cdot L'(t)$ is not sufficient because it does not use $t = 19$.

The response must use L (not M or D or any other function name) in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change of the money, in dollars per minute, raised by the friend at $t = 19$ minutes is $4L'(19)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation $\frac{dy}{dt} = L'(t)$ and the given slope field.

In order to earn the point, the response must:

State that the values of the differential equation are positive while the slopes in the given slope field are negative or 0.

Some ways to say that the values of differential equation are positive:

- L is increasing
- $L'(t) > 0$
- L has positive slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has negative (non-positive) slopes.
- The slopes shown are negative or 0.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is decreasing/negative.

6-2

- The slope field looks like it represents a decreasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ L is an increasing function, so the signs of the slopes in the slope field should be positive” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” negative or “all” negative.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

L is an increasing function, so $\frac{dy}{dt} = L'(t) \geq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

6-2

58. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	20	30
$D(t)$ (meters)	0	1000	4500	6500

A student on the school track and field team is running laps around a track. For time t minutes, $0 \leq t \leq 30$, the total distance run, in meters, is modeled by an increasing, twice-differentiable function D . Values of $D(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $D'(20)$ using the average rate of change of D over the interval $5 \leq t \leq 30$.

(b) Explain the meaning of $D'(20)$ in the context of the problem.

(c) It is known that $D'(5) = 200$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 5$.

(d) Use a left Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{30} D(t) \, dt$. Show the computations that lead to your answer.

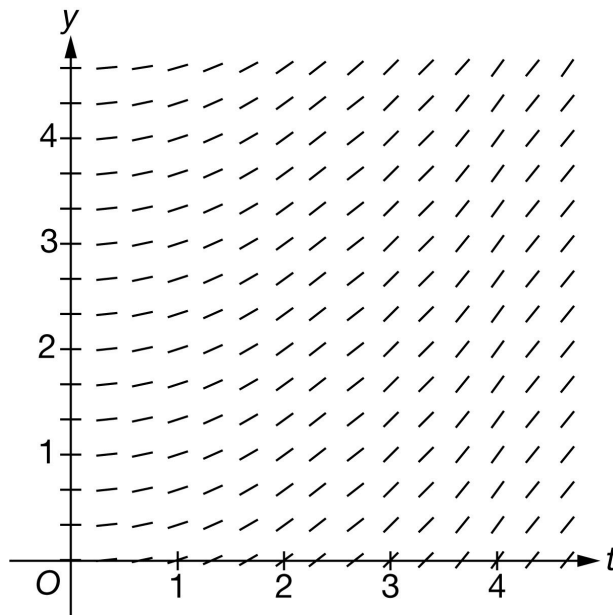
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{30} D(t) \, dt$? Give a reason for your answer.

(f) Find the value of $\int_0^5 D'(4t) \, dt$. Show the computations that lead to your answer.

(g) The student is using their running practice as a fundraiser. For $0 \leq t \leq 30$, the total number of laps run can be modeled by the differentiable function L , where L is a function of t . For every lap run, the student raises 5 dollars. Write an expression for the rate of change, in dollars per minute, of the total money raised at $t = 22$ minutes.

6-2

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.5y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\cdot \frac{6500 - 1000}{30 - 5}$$

$$\cdot \frac{D(30) - D(5)}{25} = \frac{5500}{25}$$

$$\cdot \frac{D(30) - D(5)}{25} = 220$$

An example that does not earn the first point is $\frac{D(30) - D(5)}{30 - 5}$.

Minimal response that earns the point: $\frac{5500 - 1000}{30 - 5}$ or $\frac{6500 - 1000}{25}$

Neither $\frac{5500}{25}$ nor 220, with no supporting work, will earn this point

The second point is for correct units

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

· Meters per minute

· Meters/minute

6-2

· m/min

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$D'(20) \approx \frac{D(30) - D(5)}{30 - 5} = \frac{6500 - 1000}{25} = 220 \text{ meters per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of change
- Including the context of student running
- Referring to the specific point in time $t = 20$

The response is not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “increasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “The number of meters is increasing at a rate of 220 meters per minute” is okay.

Acceptable reference to a specific point in time includes (but is not limited to):

- On minute 20 and after 20 minutes
- At $t = 20$
- At $t = 20$ hours or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from minute 20 on,” “during the first 20 minutes,” or “for $t = 0$ to $t = 20$ ”), the point is not earned.

Examples that earn the point include:

- $D'(20)$ is the rate of change of the distance of meters per minute at time $t = 20$
- The rate of change (or speed/velocity/ how fast) is 220 meters per minute at 20 minutes
- $D'(20)$ is the rate of change in the distance ran at time $t = 20$ in meters per *second*

Examples that do not earn the point include:

6-2

- 220 is the rate of increase in the distance of meters per minute at 20 minutes
- $D'(20)$ is the average rate of change of the total meters ran per minute at minute 20
- $D'(20)$ is the number of meters per minute at minute 20
- The rate of change is 220 meters per minute (Missing “at $t = 20$ ”)
- $D'(20)$ is the slope of the tangent line at $t = 20$
- Rate of change of D at time $t = 20$
- Increase in number of meters at $t = 20$



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$D'(20)$ is the rate at which the total distance run is changing, in meters per minute, at time $t = 20$ minutes.

– OR –

$D'(20)$ is the speed at which the student is running, in meters per minute, at time $t = 20$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

Some examples include

- $y = 200(t - 5) + 1000$, $y = 200(x - 5) + 1000$
- $y - 1000 = 200(t - 5)$,
- $y - 1000 = 200(x - 5)$
- $y = 200t$, $y = 200x$

The key calculus concept being assessed in this part of the problem is that the value of $D'(5)$ is the slope of the tangent line. In that spirit, we are accepting $D'(t) = \dots$ in lieu of $y = \dots$, but we are not accepting $D'(5) = \dots$

Note that $D(t) = 200(t - 5) + 1000$ earns the point, but $D(5) = 200(t - 5) + 1000$ does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

6-2

☐ Tangent line equation

Solution:

$$y = D'(5)(t - 5) + D(5)$$

$$y = 200(t - 5) + 1000$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of three products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

· Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

· In order to earn the second point, there must be supporting calculations.

· The numerical values for $D(0)$, $D(5)$, and $D(20)$ must be pulled from the table

· Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

· A response of $0 \cdot (5 - 0) + 1000 \cdot (20 - 5) + 4500 \cdot (30 - 20)$ earns both points.

· A response of $5 \cdot D(0) + 15 \cdot D(5) + 10 \cdot D(20)$ earns the first point. Eligible for second point if more work is on the page.

· A response of $5 \cdot D(0) + 15 \cdot D(5) + 10 \cdot D(20) = 6000$ earns both points.

· A response of $5 \cdot 0 + 15 \cdot 1000 + 10 \cdot 4500$ earns both points.

· A response of $5 \cdot 0 + 10 \cdot 1000 + 10 \cdot 4500$ contains one numerical error. Earns the first point. Not eligible for the second point.

· A response of $5 \cdot 0 + 15 \cdot 1000 + 10 \cdot 4500 = 50,000$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.

· A response of $0 + 15,000 + 45,000$ or $0 + 15,000 + 45,000 = 60,000$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$5 \cdot 0 + 15 \cdot 1000 + 10 \cdot 4500$$

6-2



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{30} D(t) dt \approx D(0)(5-0) + D(5)(20-5) + D(20)(30-20)$$

$$= 0 \cdot 5 + 1000 \cdot 15 + 4500 \cdot 10 = 60,000$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because D is increasing.

The response does not need an explicit reference in part (d). The following statements are sufficient for earning this point:

- Underestimate because D is increasing
- Underestimate because the function is increasing

Note: At this point, there is only one function being discussed in this problem, so we will assume “the function” is D .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function D , the point is not earned.

Examples that earn the point:

- The left Riemann sum underestimates because the function is increasing.
- The sum is an underestimate because $D(t)$ is increasing.
- The sum is an underestimate $D'(t) > 0$.

Examples that do not earn the point:

- The sum is an underestimate because $D'(t)$ is increasing. ($D'(t)$ increasing is not known)
- The left Riemann sum underestimates because the graph is increasing.
- The left Riemann sum underestimates because the function is increasing and concave up.



0	1
---	---

The student response accurately includes the criteria below.

6-2

☐ Answer with reason

Solution:

Because D is an increasing function, and this is a left Riemann sum, the answer from part (d) is an underestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = 4t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{4}$
- Uses the correct limits for the substitution (0 and 20). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

· Either of the expressions below will earn the substitution point without any additional supporting work: $\frac{1}{4} D(4t) \Big|_0^5$ or $\frac{1}{4} D(u) \Big|_0^{20}$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{20}^0 D'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 20$. No explanation or additional supporting work is necessary.

- $kD(4t) \Big|_0^b$
- $k(D(b) - D(0))$

The response cannot earn this point without also earning the FTC point.

The third point is for the correct numerical value, which is 1125.

Values from the table must be used. $\frac{1}{4}[D(20) - D(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{4}(4500 - 0)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning first two points.

6-2

Minimalist response: The expression $\frac{1}{4}(4500)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4 \, dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 20$.

$$\begin{aligned}\int_0^5 D'(4t) \, dt &= \frac{1}{4} \int_0^{20} D'(u) \, du \\ &= \frac{1}{4}(D(20) - D(0)) \\ &= \frac{1}{4}(4500 - 0) = 1125\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 22$. No supporting work is required.

The correct expression of $5 \cdot L'(22)$ earns this point.

No justifications or explanations are required.

Any incorrect statement such as, $D'(22) = 5 \cdot L'(22)$, does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Expression

Solution:

The rate of change, in dollars per minute, of the total money raised at $t = 22$ minutes is $5L'(22)$.

6-2

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = 0.5y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field, but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point (4, 0), the slope of the segment should be 0, but it is positive.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point. For example, merely stating either $\frac{dy}{dt}$ is positive, or that the slope field is positive does not earn the point.

- “The values for are greater than or equal to zero so $\frac{dy}{dt} \geq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are positive” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is increasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-2

59. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

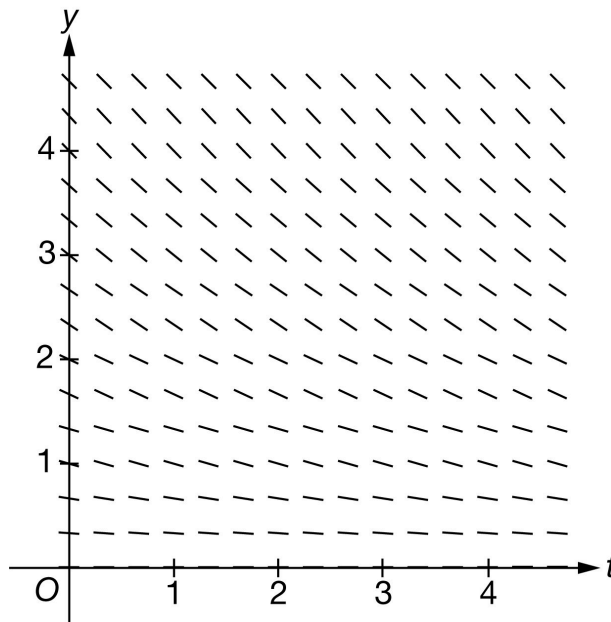
t (minutes)	0	8	16	24	32
$D(t)$ (meters)	0	1500	2800	3900	5000

A student on the school track and field team is running laps around a track. For time t minutes, $0 \leq t \leq 32$, the total distance run, in meters, is modeled by a twice-differentiable function D . Values of $D(t)$ at various times t are shown in the table above.

- (a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate $\int_0^{32} D(t) \, dt$. Show the computations that lead to your answer.
- (b) Find the value of $\int_0^{16} D'(2t) \, dt$. Show the computations that lead to your answer.
- (c) Using correct units, approximate $D'(16)$ using the average rate of change of D over the interval $8 \leq t \leq 24$.
- (d) Explain the meaning of $D'(16)$ in the context of the problem.
- (e) It is known that $D'(24) = 140$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 24$.
- (f) It is known that $D''(t) < 0$ for $24 \leq t \leq 31$. If $D(30)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $D(30)$? Give a reason for your answer.
- (g) A second student is raising funds by running laps for 60 minutes. For $0 \leq t \leq 60$, the total number of laps run by this student is modeled by $L(t)$, a differentiable function of t . For each lap run, the student raises 4 dollars. Write an expression for the rate of change, in dollars per minute, of the total money raised at time $t = 45$ minutes.

6-2

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 16 \cdot (D(8) + D(24)) \text{ or } 16 \cdot D(8) + 16 \cdot D(24)$$

$$\cdot (16 - 0) \cdot 1500 + (32 - 16) \cdot 3900$$

$$\cdot 16 \cdot (1500 + 3900) \text{ or } 16 \cdot 1500 + 16 \cdot 3900$$

To earn the point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{16}{b} \cdot (D(8) + D(24))$, for any nonzero value of b would earn the point. Note: The value $\frac{16}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (D(8) + D(24))$, for $a \neq 16$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 16 \cdot (1500 + 3900)$$

$$\cdot 16 \cdot 1500 + 16 \cdot 3900$$

6-2



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint sum
- ☐ Answer

Solution:

$$\int_0^{32} D(t) dt \approx D(8)(16 - 0) + D(24)(32 - 16)$$

$$= 1500 \cdot 16 + 3900 \cdot 16 = 86,400$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all 3 components of the substitution done correctly.

Uses $u = 2t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{2}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int D'(2t) dt$ or $\int D'(u) du$.

The point is earned for any of the following:

· The equation $\int D'(2t) dt = D(2t)$ or $\int D'(u) du = D(u)$

· $\frac{1}{2} D(2t) \Big|_0^{16}$ or $\frac{1}{2} D(u) \Big|_0^{32}$

· $D(32) - D(0)$, $D(32)$, or $D(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b D'(u) du = D'(b) - D'(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of D' is D .

Third Point (Answer)

Must be the correct numerical answer.

6-2

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$D(32)$ and $D(0)$ must be evaluated; the expression $\frac{1}{2}(D(32) - D(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{2}(5000 - 0)$ or $\frac{1}{2}(5000)$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

· $\frac{1}{2}D(2t)\big|_0^{16}$

· $\frac{1}{2}(D(32) - D(0))$

Responses of only the following would earn only the second point:

· $D(32) - D(0)$

· $D(16) - D(0)$

· $D(32)$

· $\int_0^{16} D'(u) \, du = D(16)$

· $D(2t)\big|_0^{16}$ or $D(u)\big|_0^{32}$

The following responses would earn no points:

· 2500 in the absence of one of the first two points.

· $\int_0^{16} D'(2t) \, dt = D(16)$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2 \, dt$.

When $t = 0$, $u = 0$.

6-2

When $t = 16$, $u = 32$.

$$\begin{aligned}\int_0^{16} D'(2t) \, dt &= \frac{1}{2} \int_0^{32} D'(u) \, du \\ &= \frac{1}{2}(D(32) - D(0)) \\ &= \frac{1}{2}(5000 - 0) = 2500\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{3900-1500}{16}$ or $\frac{2400}{24-8}$.

Second point (Units)

The units must be attached to a value (either correct or incorrect).



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$\begin{aligned}D'(16) &\approx \frac{D(24) - D(8)}{24 - 8} = \frac{3900 - 1500}{16} \\ &= 150 \text{ meters per minute}\end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components

- indicate the rate of change, the velocity, or the speed
- the specific context of the total distance run
- the specific point in time, $t = 16$.

Responses do not need to include correct units or any units.

6-2

The phrase “the student is running” may be used in place of “the total distance run is changing”.

The phrase “the student is running” is not enough. The response must attach “running” to the word “rate”, as in “the student is running at a rate of”.

The response is not earned for “the rate of total distance run” as this does not indicate a change in the rate of distance.

“the rate of change of D ” without specifying that D is the total distance run, does not earn the point.

Acceptable ways to provide the specific point in time:

- At minute 16 or after 16 minutes.
- At $t = 16$ (missing the unit of time)
- At 16 seconds (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 16 minutes” or “from minute 16 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$D'(16)$ is the rate at which the total distance run is changing, in meters per minute, at time $t = 16$ minutes.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $D(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

- $y = 3900 + 140(t - 24)$
- $y - 3900 = 140(t - 24)$
- $y = 140t - 3360 + 3900$
- $y = 140t + 540$



0	1
---	---

6-2

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = D'(24)(t - 24) + D(24)$$

$$y = 140(t - 24) + 3900$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point the response must include $D''(t) < 0$ and overestimate.

“ $D(t)$ is concave down” can be used instead of $D''(t) < 0$.

Information about $D'(t)$ by itself without reference to $D''(t) < 0$ is not sufficient and does not earn this point.

“Overestimate because it is concave down” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states $D''(t) < 0$ and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $D'(t)$ is concave down, overestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because $D''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $4 \cdot L'(45)$ earns this point.

No justifications or explanations are required.

The expression $4 \cdot D'(45)$ does not earn this point.

Any incorrect statement such as, $L'(45) = 4 \cdot D'(45)$, does not earn this point.

6-2



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in dollars per minute, of the total money raised at time $t = 45$ minutes is $4L'(45)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

- “ $\frac{dy}{dt}$ is positive, not negative”. We will infer that negative is referring to the slopes shown in the slope field.
- “This would be correct for a negative derivative, but $\frac{dy}{dt}$ is positive.” We will read “This” as “the slope field” in this sentence.
- “All of the slopes in the slope field are negative, not positive.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are positive.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

- “ $\frac{dy}{dt}$ is positive” lacks any reference to the slope field and so will not earn the point.
- “All the slopes in the graph are negative” lacks any reference to the differential equation, and so will not earn the point.
- Vague answers such as “The slope field has negative points” or “The field is decreasing” will not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

Solution:

When $y > 0$, $\frac{dy}{dt} > 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

6-2

60. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

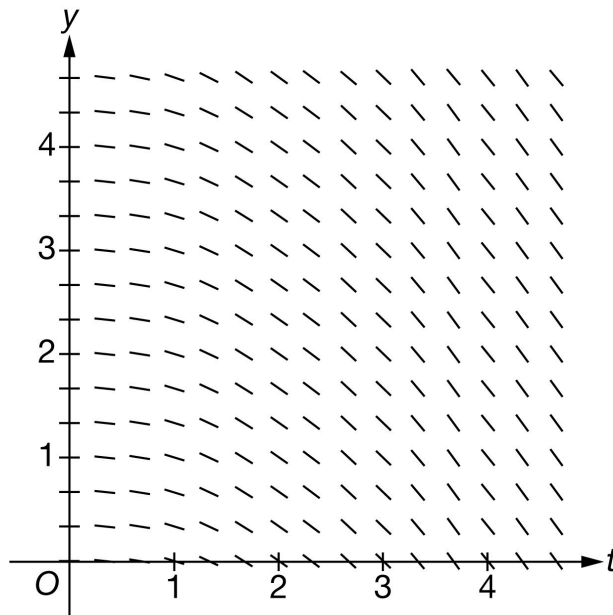
t (minutes)	0	5	20
$R(t)$ (meters per minute)	40	200	160

A student on the school track and field team is running on the track. For time t minutes, $0 \leq t \leq 20$, the rate at which the student runs, in meters per minute, is modeled by a twice-differentiable function, $R(t)$. Values of $R(t)$ at various times t are shown in the table above.

- (a) Using correct units, approximate $R'(5)$ using the average rate of change of R over the interval $0 \leq t \leq 20$.
- (b) Explain the meaning of $R'(5)$ in the context of the problem.
- (c) The function $D(t)$ models the number of meters run by the student at time t minutes, where $D'(t) = R(t)$. It is known that $D(5) = 800$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 5$.
- (d) Find the value of $\int_0^5 R'(4t) \, dt$. Show the calculations that lead to your answer.
- (e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{20} R(t) \, dt$. Show the calculations that lead to your answer.
- (f) The student and a friend team up to run on the school track for a fundraiser for 20 minutes. For $0 \leq t \leq 20$, the total number of laps run by the student is modeled by $M(t)$ and the total number of laps run by the friend is modeled by $L(t)$. Both M and L are differentiable functions of t . Each person raises 5 dollars for each lap run. Write an expression for the rate of change, in dollars per minute, of the total money raised at time $t = 18$ minutes.

6-2

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = -0.35y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{160-40}{20}$ meters per minute per minute.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

6-2

$$R'(5) \approx \frac{R(20) - R(0)}{20 - 0} = \frac{160 - 40}{20} = 6 \text{ meters per minute per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of the student running
- Referring to the specific point in time

Because the context is one of motion, a description of $R'(5)$ as the acceleration of the runner at time $t = 5$ earns the point.

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statement makes the response ineligible for the point. For example:

$R'(5)$ is the average (or an approximation of, or an estimate of) rate of change of

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $R'(5)$ is the rate of increase of the rate at which the student runs at 5 minutes. (earns the point)
- Example: $R'(5)$ means the rate at which the student runs is increasing by 6 meters / min / min. at time $t = 5$ (does not earn the point)

Acceptable references to time include (but are not limited to):

- At the fifth minute
- At $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “during the first 5 minutes”) the response will not earn the point.

Examples that earn the point include:

- $R'(5)$ represents the rate at which the rate at which the student runs is changing on day 5.
- $R'(5)$ is the rate of change in the velocity of the runner at 5 days.
- $R'(5)$ represents the rate at which the velocity of the runner is increasing at $t = 5$.
- $R'(5)$ equals (answer from part (a)) means the velocity of the runner is increasing at a rate of (answer from part (a)) at $t = 5$.

Examples that do not earn the point include:

- $R'(5)$ represents the rate at which the meters per minute is changing on day 5.
- $R'(5)$ is the rate of change in the meters run at $t = 5$ days.

6-2

- The rate at which the velocity changes is increasing at $t = 5$.
- The rate of change in the rate of change of the runner is increasing at $t = 5$.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$R'(5)$ is the rate of change in the rate at which the student runs, in meters per minute per minute, at time $t = 5$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

o $y = 200(t - 5) + 800$

o $y - 800 = 200(t - 5)$

o $y = 200t - 200$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $D(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = D'(5)(t - 5) + D(5)$$

$$y = 200(t - 5) + 800$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required to award this point:

6-2

- Coefficient of $\frac{1}{4}$ correctly applied to an integral or an antiderivative
- Evidence that the correct limits of 0 to 20 will be or have been applied

Examples that will earn the first point (these are just a few):

$$\cdot \frac{1}{4} \int_0^{20} R'(u) \, du$$

$$\cdot \frac{1}{4} R(u) \Big|_0^{20} \text{ or } \frac{1}{4} R(4t) \Big|_0^5$$

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\cdot \int_0^5 R'(4t) \, dt = \frac{1}{4} \int_0^5 R'(u) \, du = \frac{1}{4} R(4t) \Big|_0^5 \text{ earns the first point, but will not earn the third point.}$$

$$\cdot \int_0^5 R'(4t) \, dt = \frac{1}{4} \int_0^5 R'(u) \, du = \frac{1}{4} [R(5) - R(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b R'(u) \, du = R(u) \Big|_a^b = R(b) - R(a)$$

This point can be earned with either $kR(4t) \Big|_a^b$, $kR(t) \Big|_a^b$, or $k(R(b) - R(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 5$ or $b = 20$.

A response can earn this point without earning the first point.

$$\frac{1}{4} [R(20) - M(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{4} [R(20) - M(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\int_0^5 R'(4t) \, dt = \frac{1}{4} \int R'(u) \, du = \frac{1}{4} R(4t) \Big|_0^5 = \dots \text{ is eligible for the third point.}$$

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{4}(160 - 40)$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

6-2

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4 dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 20$.

$$\begin{aligned}\int_0^5 R'(4t) dt &= \frac{1}{4} \int_0^{20} R'(u) du \\ &= \frac{1}{4}(R(20) - R(0)) \\ &= \frac{1}{4}(160 - 40) = 30\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The **first point** is for the form of a trapezoidal sum

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point

(e.g. $\frac{1}{2}[5(R(0) - R(5)) + 15(R(5) - R(20))]$ earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one incorrect numerical value.

$\frac{1}{2}(5)[R(0) + 2R(5) + R(20)]$ is equivalent to $\frac{1}{2}[5(R(0) + R(5)) + 5(M(5) + R(20))]$, thus has only one incorrect numerical value (the second interval width is 5 instead of 15), so can earn the point.

$\frac{1}{2}(10)[R(0) + 2R(5) + R(20)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

6-2

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first point if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation (+) of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn the first but not the second point:

$$\cdot k \left[\left(\frac{40+200}{2} \right) \cdot 5 + \left(\frac{200+160}{2} \cdot 15 \right) \right] \text{ for } k > 0 \text{ earns only the first point.}$$

The minimum response needed to earn both points: $\left(\frac{240}{2} \right) \cdot 5 + \left(\frac{360}{2} \right) \cdot 15$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{20} R(t) \, dt &\approx \left(\frac{R(0)+R(5)}{2} \right) (5-0) + \left(\frac{R(5)+R(20)}{2} \right) (20-5) \\ &= \left(\frac{40+200}{2} \right) \cdot 5 + \left(\frac{200+160}{2} \right) \cdot 15 = 3300 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 5 as a factor of the expression, and/or the correct sum $L'(18) + M'(18)$.

If the sum is incorrect, then the constant factor of 5 must be applied either to a sum that involves only $L(t)$ and $M(t)$, $L'(t)$ and $M'(t)$, or integrals of $L(t)$ and $M(t)$, or to one of the correct terms of the sum $L'(18)$ or $M'(18)$, with no additional incorrect terms.

Examples that earn the point include:

$$\begin{aligned} &\cdot 5 [M'(t) + L'(t)] \\ &\cdot 5 \int [M(t) + L(t)] \, dt \\ &\cdot 5 [M + L] \\ &\cdot 5M'(18) \end{aligned}$$

Examples that do not earn the point include:

$$\cdot 5 [M(t) + L'(t)]$$

6-2

$$\cdot 5 \int M(t) dt$$

$$\cdot 5M' + L$$

$$\cdot 5L'(18) + t$$

Special case: $5L'(18) + M'(18)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[L'(18) + M'(18)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$5L'(18) + 5M'(18)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn a just the first point. For example, $5L'(18) + 5M'(18) + 5(2)$ earns only the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

The rate of change, in dollars per minute, of the total money raised at time $t = 18$ minutes is $5M'(18) + 5L'(18)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

6-2

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(2, 2)$ the slope shown on the slope field is -1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

6-2

61. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	6	9	12
$p(t)$ (feet)	3	5.5	7	13	17

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

- (a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{12} p(t) \, dt$. Show the computations that lead to your answer.
- (b) Use the data in the table to approximate $p'(10)$ using the average rate of change of $p(t)$ over the interval $9 \leq t \leq 12$. Show the computations that lead to your answer. Indicate units of measure.
- (c) Interpret the meaning of $p'(10)$ in the context of the problem.
- (d) Must there exist a value of c , for $0 < c < 12$, such that $p'(c)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 12$? Justify your answer.
- (e) Find $\int_0^3 p'(2t+3) \, dt$. Show the computations that lead to your answer.
- (f) Let $h(x) = \int_2^{3x} p(3t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.
- (g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(2) = -6$, $g'(2) = 4$, and $g''(2) = -3$. Is the speed of particle Q increasing or decreasing at time $t = 2$? Give a reason for your answer.
- (h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 4x^2y^3$ with initial condition $f\left(\frac{1}{2}\right) = -2$. Write an equation for the line tangent to the graph of f at the point $\left(\frac{1}{2}, -2\right)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(3) \cdot (6 - 0) + p(9) \cdot (12 - 6)$ or $p(3) \cdot (6) + p(9) \cdot (6)$ earns first point.

6-2

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

111 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products ($33 + 78$) or a graph with areas labeled and summed to 111 or equivalent, can earn this point.

Examples:

- $111 = 33 + 78$ by itself earns second point, not first point
- 111 declared, with $5.5 \cdot 6$ and $13 \cdot 6$ identified (perhaps graphically) earns second point, not first point
- $k(5.5 \cdot 6 + 13 \cdot 6)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(3) \cdot (6) + p(9) \cdot (12 - 6)$ by itself earns first point, not second point



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{12} p(t) dt \approx p(3)(6 - 0) + p(9)(12 - 6)$$

$$= 5.5 \cdot 6 + 13 \cdot 6 = 111$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(12)-p(9)}{12-9}$ by itself doesn't earn first point; however $\frac{17-13}{12-9}$ or $\frac{17-13}{3}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

6-2

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(10) \approx \frac{17-13}{12-9} = \frac{4}{3} \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(10)$ interpretation.

Need two components in response:

- “rate of change of position” or “velocity” of the particle
- at $t = 10$

Correct ways to describe rate at time at $t = 10$ include:

- Rate at which position is changing/increasing
- Rate at which particle is moving right
- Instantaneous velocity or velocity of particle

Ambiguous/incorrect ways to describe rate include:

- Rate at which particle is changing/increasing
- Rate at which particle is moving

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Interpretation

Solution:

$p'(10)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 10$ seconds.

6-2

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(c) = \frac{7}{6}$ must be correct to earn second point.

Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

6-2

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 12$ to guarantee that there exists a value of c , for $0 < c < 12$, such that $p'(c)$ equals the average rate of change of p over the interval $0 \leq x \leq 12$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

- $u = 2t + 3$ (can be explicit or implicit)
- $du = 2 dt$ (yielding integral coefficient of $\frac{1}{2}$)
- Correct integral limits throughout, with t from 0 to 3 and u from 3 to 9

Omitting the coefficient of $\frac{1}{2}$ or using some other value will not earn first point; response could earn second point, but not third point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 3, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 3$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^3 p'(2t + 3) dt = \frac{1}{2} p(2t + 3) \Big|_0^3$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

- $k(p(9) - p(3))$ or $k(p(2 \cdot 3 + 3) - p(2 \cdot 0 + 3))$ for k a nonzero constant
- $k p(2t + 3) \Big|_0^3$ or $k p(u) \Big|_3^9$ for k a nonzero constant
- $\int p'(2t + 3) dt = k p(2t + 3)$ or $\int p'(u) du = k p(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^3 p'(2t + 3) dt = \frac{1}{2} p(2t + 3) \Big|_0^3$

Third Point

Earned only for an answer of 3.75 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

6-2

Scoring Examples:

$$\cdot \frac{1}{2} \int p'(u) \, du = \frac{1}{2} p(u) \text{ earns } 0 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) \, du = \frac{1}{2} p(u) \Big|_0^3 = \frac{1}{2} (p(3) - p(0)) \text{ earns } 0 - 1 - 0$$

$$\cdot \int p'(u) \, du = p(u) \Big|_3^9 = (p(9) - p(3)) = 7.5 \text{ earns } 0 - 1 - 0$$

$$\cdot 2 \int p'(u) \, du = 2p(u) \Big|_0^3 = 2[13 - 5.5] \text{ earns } 0 - 1 - 0$$

$$\cdot \int_0^3 p'(2t+3) \, dt = p(t^2+3t) \Big|_0^3 = p(18) - p(0) \text{ earns } 0 - 0 - 0$$

$$\cdot \frac{1}{2} \int_0^3 p'(u) \, du = \frac{1}{2} (p(9) - p(3)) = 3.75 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^3 p'(2t+3) \, dt = \frac{1}{2} (p(9) - p(3)) = 3.75 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^3 p'(2t+3) \, dt = \frac{1}{2} p(u) \Big|_3^9 = \frac{1}{2} (p(9) - p(3)) = 3.75 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^3 p'(2t+3) \, dt = \frac{1}{2} p(2t+3) \Big|_0^3 = \frac{1}{2} (p(9) - p(3)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) = \frac{1}{2} p(u); \frac{1}{2} p(2t+3) \Big|_0^3 = \frac{1}{2} (13 - 5.5) \text{ earns } 1 - 1 - 1$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:Let $u = 2t + 3$. Then $du = 2 \, dt$.When $t = 0$, $u = 3$.When $t = 3$, $u = 9$.

$$\int_0^3 p'(2t+3) \, dt = \frac{1}{2} \int_3^9 p'(u) \, du$$

$$= \frac{1}{2} (p(9) - p(3))$$

$$= \frac{1}{2} (13 - 5.5) = 3.75$$

6-2

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(9x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(3 \cdot 3x)$ or $p(9x)$ (Note $p(9t)$ would also earn this point.)

The expression $p(3 \cdot 3x) - p(3 \cdot 2)$ or $(p(3 \cdot 3x) - p(3 \cdot 2)) \cdot 3$ earns this point. The $p(3 \cdot 2)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 3t$ earns first point.

Part of this response may include $\int_2^{3x} p(3t) dt = \frac{1}{3} \int_6^{9x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(1)$. That is accomplished by multiplying the term involving $p(x)$ by 3.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(9x)$ for any $a \neq 3$.

The point is not earned for responses of $3p(3t) \big|_2^{3x}$ or $3(p(9x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of 39 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$\cdot p(3 \cdot 3x) - p(6); p(9) - p(6) = 13 - 7$ earns $1 - 0 - 0$

$\cdot p(3 \cdot 3x) \cdot 3 - p(6) \cdot 0; 39$ earns $1 - 1 - 1$

$\cdot 3p(3t) \big|_2^{3x} = 3(p(9) - p(6))$ earns $1 - 0 - 0$

$\cdot p(3x) \cdot 3; p(3) \cdot 3 = 5.5 \cdot 3$ earns $0 - 0 - 0$

$\cdot p(9x) \cdot 3; p(9) \cdot 3$ earns $1 - 1 - 0$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

6-2

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_2^{3x} p(3t) dt = p(3 \cdot 3x) \cdot 3$$

$$h'(1) = p(9) \cdot 3 = 13 \cdot 3 = 39$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- “decreasing since $g < 0$ and $g' > 0$ ”
- “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

- “decreasing since velocity and acceleration have opposite signs” earns neither point
- “decreasing since velocity < 0 and acceleration > 0 ” earns neither point
- “decreasing since $(-6) \cdot 4 = -24 < 0$ ” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

6-2

Solution:

Because $g(2) = -6 < 0$, the velocity of the particle is negative. Because $g'(2) = 4 > 0$, the acceleration of the particle is positive. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 2$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

$$\cdot y = -8\left(x - \frac{1}{2}\right) - 2 \text{ or } y + 2 = -8\left(x - \frac{1}{2}\right)$$

$$\cdot y = -8x + 2$$

$$\cdot y = \left(4 \cdot \left(\frac{1}{2}\right)^2 \cdot (-2)^3\right)\left(x - \frac{1}{2}\right) - 2$$

Incorrect responses include:

$$\cdot y = (4x^2y^3)\left(x - \frac{1}{2}\right) - 2$$

$$\cdot y = 8x + 2$$

$$\cdot f(x) = -8x + 2$$



0	1
---	---

The student response accurately includes the criteria below.

☐

Tangent line equation

Solution:

$$\frac{dy}{dx} \Big|_{\left(\frac{1}{2}, -2\right)} = 4 \cdot \left(\frac{1}{2}\right)^2 \cdot (-2)^3 = -8$$

$$y = -8\left(x - \frac{1}{2}\right) - 2$$

6-2

62. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	8	12	16
$p(t)$ (feet)	3	9	16.5	26	36

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

- (a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{16} p(t) \, dt$. Show the computations that lead to your answer.
- (b) Use the data in the table to approximate $p'(11)$ using the average rate of change of $p(t)$ over the interval $8 \leq t \leq 12$. Show the computations that lead to your answer. Indicate units of measure.
- (c) Interpret the meaning of $p'(11)$ in the context of the problem.
- (d) Must there exist a value of k , for $0 < k < 16$, such that $p'(k)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 16$? Justify your answer.
- (e) Find $\int_0^4 p'(2t + 4) \, dt$. Show the computations that lead to your answer.
- (f) Let $h(x) = \int_1^{4x} p(2t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.
- (g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(3) = 2$, $g'(3) = -5$, and $g''(3) = 11$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.
- (h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -xy^2 + 3$ with initial condition $f(5) = 2$. Write an equation for the line tangent to the graph of f at the point $(5, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(4) \cdot (8 - 0) + p(12) \cdot (16 - 8)$ or $p(3) \cdot (6) + p(9) \cdot (6)$ earns first point.

6-2

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

280 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products ($72 + 208$) or a graph with areas labeled and summed to 280 or equivalent, can earn this point.

Examples:

- $280 = 72 + 208$ by itself earns second point, not first point
- 280 declared, with $9 \cdot 8$ and $26 \cdot 8$ identified (perhaps graphically) earns second point, not first point
- $k(9 \cdot 8 + 26 \cdot 8)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(4) \cdot (8) + p(12) \cdot (16 - 8)$ by itself earns first point, not second point



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{16} p(t) dt \approx p(4)(8 - 0) + p(12)(16 - 8)$$

$$= 9 \cdot 8 + 26 \cdot 8 = 280$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(12)-p(8)}{12-8}$ by itself doesn't earn first point; however $\frac{26-16.5}{12-8}$ or $\frac{26-16.5}{4}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

6-2

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(11) \approx \frac{26-16.5}{12-8} = \frac{9.5}{4} = 2.375 \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(11)$ interpretation.

Need two components in response:

- “rate of change of position” or “velocity” of the particle
- at $t = 11$

Correct ways to describe rate at time at $t = 11$ include:

- Rate at which position is changing/increasing
- Rate at which particle is moving right
- Instantaneous velocity or velocity of particle

Ambiguous/incorrect ways to describe rate include:

- Rate at which particle is changing/increasing
- Rate at which particle is moving

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Interpretation

Solution:

$p'(11)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 11$ seconds.

6-2

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(k) = \frac{33.5}{16}$ must be correct to earn second point.

Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

6-2

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 16$ to guarantee that there exists a value of k , for $0 < k < 16$, such that $p'(k)$ equals the average rate of change of p over the interval $0 \leq x \leq 16$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

- $u = 2t + 4$ (can be explicit or implicit)
- $du = 2 dt$ (yielding integral coefficient of $\frac{1}{2}$)
- Correct integral limits throughout, with t from 0 to 4 and u from 4 to 12

Omitting the coefficient of $\frac{1}{2}$ or using some other value will not earn first point; response could earn second point, but not third point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 4, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 4$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^4 p'(2t + 4) dt = \frac{1}{2}p(2t + 4) \Big|_0^4$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

- $k(p(12) - p(4))$ or $k(p(2 \cdot 4 + 4) - p(2 \cdot 0 + 4))$ for k a nonzero constant
- $k p(2t + 4) \Big|_0^3$ or $k p(u) \Big|_4^{12}$ for k a nonzero constant
- $\int p'(2t + 4) dt = k p(2t + 4)$ or $\int p'(u) du = k p(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^4 p'(2t + 4) dt = \frac{1}{2}p(2t + 4) \Big|_0^4$

Third Point

Earned only for an answer of 8.5 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

6-2

Scoring Examples:

$$\cdot \frac{1}{2} \int p'(u) \, du = \frac{1}{2} p(u) \text{ earns } 0 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) \, du = \frac{1}{2} p(u) \Big|_0^4 = \frac{1}{2} (p(4) - p(0)) \text{ earns } 0 - 1 - 0$$

$$\cdot \int p'(u) \, du = p(u) \Big|_4^{12} = (p(12) - p(4)) = 17 \text{ earns } 0 - 1 - 0$$

$$\cdot 2 \int p'(u) \, du = 2p(u) \Big|_0^4 = 2[26 - 9] \text{ earns } 0 - 1 - 0$$

$$\cdot \int_0^4 p'(2t + 4) \, dt = p(t^2 + 4t) \Big|_0^4 = p(32) - p(0) \text{ earns } 0 - 0 - 0$$

$$\cdot \frac{1}{2} \int_0^4 p'(u) \, du = \frac{1}{2} (p(12) - p(4)) = 8.5 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^4 p'(2t + 4) \, dt = \frac{1}{2} (p(12) - p(4)) = 8.5 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^4 p'(2t + 4) \, dt = \frac{1}{2} p(u) \Big|_4^{12} = \frac{1}{2} (p(12) - p(4)) = 8.5 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^4 p'(2t + 4) \, dt = \frac{1}{2} p(2t + 4) \Big|_0^4 = \frac{1}{2} (p(12) - p(4)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) \, du = \frac{1}{2} p(u); \text{ and } \frac{1}{2} p(2t + 4) \Big|_0^4 = \frac{1}{2} (26 - 9) \text{ earns } 1 - 1 - 1$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:Let $u = 2t + 4$. Then $du = 2 \, dt$.When $t = 0$, $u = 4$.When $t = 4$, $u = 12$.

$$\begin{aligned} \int_0^4 p'(2t + 4) \, dt &= \frac{1}{2} \int_4^{12} p'(u) \, du \\ &= \frac{1}{2} (p(12) - p(4)) \\ &= \frac{1}{2} (26 - 9) = 8.5 \end{aligned}$$

6-2

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(8x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(2 \cdot 4x)$ or $p(8x)$ (Note $p(8t)$ would also earn this point.)

The expression $p(2 \cdot 4x) - p(2 \cdot 1)$ or $(p(2 \cdot 4x) - p(2 \cdot 1)) \cdot 4$ earns this point. The $p(2 \cdot 1)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 2t$ earns first point.

Part of this response may include $\int_1^{4x} p(2t) dt = \frac{1}{3} \int_2^{8x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(1)$. That is accomplished by multiplying the term involving $p(x)$ by 4.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(8x)$ for any $a \neq 4$.

The point is not earned for responses of $4p(2t)|_1^{4x}$ or $4(p(8x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of 66 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$\cdot p(2 \cdot 4x) - p(2)$; $p(8) - p(2)$ earns $1 - 0 - 0$

$\cdot p(2 \cdot 4x) \cdot 4 - p(2) \cdot 0$; 66 earns $1 - 1 - 1$

$\cdot 4p(2t)|_1^{4x} = 4(p(8) - p(2))$ earns $1 - 0 - 0$

$\cdot p(2x) \cdot 2$; $p(2) \cdot 2$ earns $0 - 0 - 0$

$\cdot p(2 \cdot 4x) \cdot 4$; $p(8) \cdot 4$ earns $1 - 1 - 0$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

6-2

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_1^{4x} p(2t) dt = p(2 \cdot 4x) \cdot 4$$

$$h'(1) = p(8) \cdot 4 = 16.5 \cdot 4 = 66$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- “decreasing since $g > 0$ and ” $g' < 0$ ”
- “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

- “decreasing since velocity and acceleration have opposite signs” earns neither point
- “decreasing since velocity > 0 and acceleration < 0 ” earns neither point
- “decreasing since $(2) \cdot (-5) = -10 < 0$ ” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

6-2

Solution:

Because $g(3) = 2 > 0$, the velocity of the particle is positive. Because $g'(3) = -5 < 0$, the acceleration of the particle is negative. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

$$\cdot y = -17(x - 5) + 2 \text{ or } y - 2 = -17(x - 5)$$

$$\cdot y = -17x + 87$$

$$\cdot y = (-5 \cdot 2^2 + 3)(x - 5) + 2$$

Incorrect responses include:

$$\cdot y = (-xy^2 + 3)(x - 5) + 2$$

$$\cdot y = 17x + 87$$

$$\cdot f(x) = -17x + 87$$



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(5,2)} = -5 \cdot 2^2 + 3 = -17$$

$$y = -17(x - 5) + 2$$

6-2

63. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	6	9	12
$p(t)$ (feet)	-5	-8.5	-12.5	-15	-20

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{12} p(t) \, dt$. Show the computations that lead to your answer.

(b) Use the data in the table to approximate $p'(5)$ using the average rate of change of $p(t)$ over the interval $3 \leq t \leq 6$. Show the computations that lead to your answer. Indicate units of measure.

(c) Interpret the meaning of $p'(5)$ in the context of the problem.

(d) Must there exist a value of k , for $0 < k < 12$, such that $p'(k)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 12$? Justify your answer.

(e) Find $\int_0^3 p'(2t + 6) \, dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_1^{2x} p(3t) \, dt$. Find $h'(2)$. Show the computations that lead to your answer.

(g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(3) = -1$, $g'(3) = -5$, and $g''(3) = 7$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 12x^2 - y^3$ with initial condition $f\left(\frac{1}{2}\right) = -2$. Write an equation for the line tangent to the graph of f at the point $\left(\frac{1}{2}, -2\right)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(3) \cdot (6 - 0) + p(9) \cdot (12 - 6)$ or $p(3) \cdot (6) + p(9) \cdot (6)$

6-2

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point. However, if one of these values is incorrect, the response does not earn the second point.

Second Point

Earned for an answer of -141 or equivalent.

Can be earned without the first point for the Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made, but not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products $(-51 - 90)$, or a graph with areas labeled and summed to -141 , can earn this point.

Examples:

- $-141 = -51 - 90$ by itself earns second point, not first point
- -141 declared, with $-8.5 \cdot 6$ and $-15 \cdot 6$ identified earns second point, not first point (perhaps graphically)
- $k(-8.5 \cdot 6 - 15 \cdot 6)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(3) \cdot (6 - 0) + p(9) \cdot (12 - 6)$ by itself earns first point, not second point



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{12} p(t) dt \approx p(3)(6 - 0) + p(9)(12 - 6)$$

$$= -8.5 \cdot 6 + (-15) \cdot 6 = -141$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for a correct difference quotient, with appropriate values from the table.

The response must show both a difference and a quotient.

$\frac{p(6)-p(3)}{6-3}$ by itself doesn't earn first point; however $\frac{-12.5-(-8.5)}{6-3}$ or $A \frac{-12.5-(-8.5)}{3}$ by itself earns the first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

6-2

Correct units must appear in part (b); correct units in part (c) only do not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(5) \approx \frac{-12.5 - (-8.5)}{6 - 3} = \frac{-4}{3} \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(5)$ interpretation.

Need two components in the response:

- “rate of change of position” or “velocity” of the particle
- at $t = 5$

Correct ways to describe the rate of change include:

- rate at which position (of the particle) is changing/decreasing
- rate at which particle is moving left
- instantaneous velocity or velocity of the particle

Ambiguous/incorrect ways to describe rate include:

- rate at which particle is changing/decreasing
- rate at which particle is moving

The response does not need to include the value of the approximation found in part (b), but will not be penalized for doing so (e.g. “the rate at which the particle's position is decreasing is $\frac{4}{3}$ at time $t = 5$ ” earns the point).

If the response includes the approximation value from part (b) and makes an incorrect statement such as “the rate at which the position of the particle is decreasing is $-\frac{4}{3}$ at time $t = 5$ ”, the point is not earned.

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
---	---

The student response accurately includes the criteria below.

6-2

☐ Interpretation
Solution:

$p'(5)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 5$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(c) = -\frac{5}{4}$ must be correct to earn second point.

Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

6-2

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 12$ to guarantee that there exists a value of k , for $0 < k < 12$, such that $p'(k)$ equals the average rate of change of p over the interval $0 \leq x \leq 12$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution.

A correct substitution requires three elements (with consistent use of u or another substitution variable):

- $u = 2t + 6$ (can be explicit or implicit)
- $du = 2 dt$ (yielding integral coefficient of $\frac{1}{2}$)
- Correct integral limits throughout, with t from 0 to 3 and u from 6 to 12

The substitution could be implicit: $\int_0^3 p'(2t + 6) dt = \frac{1}{2} p(2t + 6) \Big|_0^3$

Omitting the coefficient of $\frac{1}{2}$ or using some other constant will not earn the substitution or answer points; the response could still earn the second point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 3, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 6$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point, if handled correctly.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

- $k(p(12) - p(6))$ or $k(p(2 \cdot 3 + 6) - p(2 \cdot 0 + 6))$ for k a nonzero constant
- $k p(2t + 6) \Big|_0^3$ or $k p(u) \Big|_6^{12}$ for k a nonzero constant
- $\int p'(2t + 6) dt = k p(2t + 6)$ or $\int p'(u) du = k p(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

6-2

The first two points can be earned simultaneously by responses such as $\int_0^3 p'(2t+6) dt = \frac{1}{2}p(2t+6) \Big|_0^3$

Third Point

Earned only for an answer of -3.75 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

· $\frac{1}{2} \int p'(u) du = \frac{1}{2}p(u)$ earns 0 – 1 – 0

· $\frac{1}{2} \int p'(u) du = \frac{1}{2}p(u) \Big|_0^3 = \frac{1}{2}(p(3) - p(0))$ earns 0 – 1 – 0

· $\int p'(u) du = p(u) \Big|_6^{12} = (p(12) - p(6)) = -7.5$ earns 0 – 1 – 0

· $2 \int p'(u) du = 2p(u) \Big|_0^3 = 2[-8.5 + 5]$ earns 0 – 1 – 0

· $\int_0^3 p'(2t+6) dt = p(t^2+6t) \Big|_0^3 = p(27) - p(0)$ earns 0 – 0 – 0

· $\frac{1}{2} \int_0^3 p'(u) du = \frac{1}{2}(p(12) - p(6)) = -3.75$ earns 0 – 1 – 1

· $\int_0^3 p'(2t+6) dt = \frac{1}{2}(p(12) - p(6)) = -3.75$ earns 1 – 1 – 1

· $\int_0^3 p'(2t+6) dt = \frac{1}{2}p(u) \Big|_6^{12} = \frac{1}{2}(p(12) - p(6)) = -3.75$ earns 1 – 1 – 1

· $\int_0^3 p'(2t+6) dt = \frac{1}{2}p(2t+6) \Big|_0^3 = \frac{1}{2}(p(12) - p(6))$ earns 1 – 1 – 0

· $\frac{1}{2} \int p'(u) = \frac{1}{2}p(u)$; and $\frac{1}{2}p(2t+6) \Big|_0^3 = \frac{1}{2}(-20 - (12.5))$ earns 1 – 1 – 1



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t + 6$. Then $du = 2 dt$.

When $t = 0$, $u = 6$.

6-2

When $t = 3$, $u = 12$.

$$\begin{aligned}\int_0^3 p'(2t + 6) dt &= \frac{1}{2} \int_6^{12} p'(u) du \\ &= \frac{1}{2}(p(12) - p(6)) \\ &= \frac{1}{2}(-20 - (-12.5)) = -3.75\end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(6x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(3 \cdot 2x)$ or $p(6x)$ (Note $p(6t)$ would also earn this point.)

The expression $p(3 \cdot 2x) - p(3 \cdot 1)$ or $(p(3 \cdot 2x) - p(3 \cdot 1)) \cdot 3$ earns this point. The $p(3 \cdot 1)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 3t$ earns first point.

Part of this response may include $\int_1^{2x} p(3t) dt = \frac{1}{3} \int_3^{6x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(2)$. That is accomplished by multiplying the term involving $p(x)$ by 2.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(6x)$ for any $a \neq 2$.

The point is not earned for responses of $2p(3t) \big|_1^{2x}$ or $2(p(6x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of -40 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$$\cdot p(3 \cdot 2x) - p(3); p(12) - p(3) = -20 + 8.5 \text{ earns } 1 - 0 - 0$$

$$\cdot p(3 \cdot 2x) \cdot 2 - p(3) \cdot 0; -40 \text{ earns } 1 - 1 - 1$$

$$\cdot 2p(3t) \big|_1^{2x} = 2(p(12) - p(3)) \text{ earns } 1 - 0 - 0$$

$$\cdot p(3x) \cdot 2; p(6) \cdot 2 = -12.5 \cdot 2 \text{ earns } 0 - 0 - 0$$

$$\cdot p(6x) \cdot 2; p(12) \cdot 2 \text{ earns } 1 - 1 - 0$$

6-2



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_1^{2x} p(3t) dt = p(3 \cdot 2x) \cdot 2$$

$$h'(2) = p(12) \cdot 2 = -20 \cdot 2 = -40$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “increasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- “increasing since $g < 0$ and $g' < 0$ ”
- “particle speeding up since g and g' have the same signs”

Can only be earned if response earned first point.

A response that calls out all three given values ($g(3)$, $g'(3)$, and $g''(3)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g and g' only, both points can be earned.

Examples:

- “increasing since velocity and acceleration have the same signs” earns neither point
- “increasing since velocity < 0 and acceleration < 0 ” earns neither point
- “increasing since $(-1) \cdot (-5) = 5 > 0$ ” earns both points

6-2



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(3) = -1 < 0$, the velocity of the particle is negative. Because $g'(3) = -5 < 0$, the acceleration of the particle is negative. Because the velocity and acceleration have the same signs, the speed of the particle is increasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation, written in any form.

Correct responses include:

$$y = 11\left(x - \frac{1}{2}\right) - 2 \text{ or } y + 2 = 11\left(x - \frac{1}{2}\right)$$

$$y = 11x - \frac{15}{2}$$

$$y = \left(12 \cdot \left(\frac{1}{2}\right)^2 - (-2)^3\right)\left(x - \frac{1}{2}\right) - 2$$

Incorrect responses include:

$$y = (12x^2 - y^3)\left(x - \frac{1}{2}\right) - 2$$

$$y = 11x - 2$$

$f(x) = 11x - \frac{15}{2}$. ($f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.)



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Tangent line equation

Solution:

$$\frac{dy}{dx} \Big|_{\left(\frac{1}{2}, -2\right)} = 12 \cdot \left(\frac{1}{2}\right)^2 - (-2)^3 = 11$$

$$y = 11\left(x - \frac{1}{2}\right) - 2$$

6-2

64. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	5	10	15	20
$p(t)$ (feet)	-25	-20	-14	-8	-2.5

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

(a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{20} p(t) \, dt$. Show the computations that lead to your answer.

(b) Use the data in the table to approximate $p'(7)$ using the average rate of change of $p(t)$ over the interval $5 \leq t \leq 10$. Show the computations that lead to your answer. Indicate units of measure.

(c) Interpret the meaning of $p'(7)$ in the context of the problem.

(d) Must there exist a value of b , for $0 < b < 20$, such that $p'(b)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 20$? Justify your answer.

(e) Find $\int_0^5 p'(2t + 5) \, dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_3^{5x} p(4t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(1) = -2$, $g'(1) = 4$, and $g''(1) = 7$. Is the speed of particle Q increasing or decreasing at time $t = 1$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -2x^3y^2 - 5$ with initial condition $f(-1) = 2$. Write an equation for the line tangent to the graph of f at the point $(-1, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(5) \cdot (10 - 0) + p(15) \cdot (20 - 10)$ or $p(5) \cdot (10) + p(15) \cdot (10)$ earns first point.

6-2

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

−280 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products (−200 − 80) or a graph with areas labeled and summed to −280 or equivalent, can earn this point.

Examples:

- −280 = −200 − 80 by itself earns second point, not first point
- −280 declared, with $-20 \cdot 10$ and $-8 \cdot 10$ identified (perhaps graphically) earns second point, not first point
- $k(-20 \cdot 10 - 8 \cdot 10)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(3) \cdot (10) + p(15) \cdot (20 - 10)$ by itself earns first point, not second point



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{20} p(t) dt \approx p(5)(10 - 0) + p(15)(20 - 10)$$

$$= -20 \cdot 10 + (-8) \cdot 10 = -280$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(10)-p(5)}{10-5}$ by itself doesn't earn first point; however $\frac{-14-(-20)}{10-5}$ or $\frac{-14-(-20)}{5}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

6-2

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(7) \approx \frac{-14 - (-20)}{10 - 5} = \frac{6}{5} \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(7)$ interpretation.

Need two components in response:

- “rate of change of position” or “velocity” of the particle
- at $t = 7$

Correct ways to describe rate at time at $t = 7$ include:

- Rate at which position is changing/increasing
- Rate at which particle is moving right
- Instantaneous velocity or velocity of particle

Ambiguous/incorrect ways to describe rate include:

- Rate at which particle is changing/increasing
- Rate at which particle is moving

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Interpretation

Solution:

$p'(7)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 7$ seconds.

6-2

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(b) = \frac{22.5}{20}$ must be correct to earn second point.

Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

6-2

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 20$ to guarantee that there exists a value of b , for $0 < b < 20$, such that $p'(b)$ equals the average rate of change of p over the interval $0 \leq x \leq 20$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

- $u = 2t + 5$ (can be explicit or implicit)
- $du = 2dt$ (yielding integral coefficient of $\frac{1}{2}$)
- Correct integral limits throughout, with t from 0 to 5 and u from 5 to 15

Omitting the coefficient of $\frac{1}{2}$ or using some other value will not earn first point; response could earn second point, but not third point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 5, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 5$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^5 p'(2t + 5)dt = \frac{1}{2}p(2t + 5)\big|_0^5$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

- $k(p(15) - p(5))$ or $k(p(2 \cdot 5 + 5) - p(2 \cdot 0 + 5))$ for k a nonzero constant
- $kp(2t + 5)\big|_0^5$ or $kp(u)\big|_5^{15}$ for k a nonzero constant
- $\int p'(2t + 5)dt = kp(2t + 5)$ or $\int p'(u)du = kp(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^5 p'(2t + 5)dt = \frac{1}{2}p(2t + 5)\big|_0^5$

Third Point

Earned only for an answer of 6 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

6-2

Scoring Examples:

$$\cdot \frac{1}{2} \int p'(u) du = \frac{1}{2} p(u) \text{ earns } 0 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) du = \frac{1}{2} p(u) \Big|_0^5 = \frac{1}{2} (p(5) - p(0)) \text{ earns } 0 - 1 - 0$$

$$\cdot \int p'(u) du = p(u) \Big|_5^{15} = (p(15) - p(5)) = 12 \text{ earns } 0 - 1 - 0$$

$$\cdot 2 \int p'(u) du = 2p(u) \Big|_0^5 = 2[-8 - (-20)] \text{ earns } 0 - 1 - 0$$

$$\cdot \int_0^5 p'(2t + 5) dt = p(t^2 + 5t) \Big|_0^5 = p(50) - p(0) \text{ earns } 0 - 0 - 0$$

$$\cdot \frac{1}{2} \int_0^5 p'(u) du = \frac{1}{2} (p(15) - p(5)) = 6 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^5 p'(2t + 5) dt = \frac{1}{2} (p(15) - p(5)) = 6 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^5 p'(2t + 5) dt = \frac{1}{2} p(u) \Big|_5^{15} = \frac{1}{2} (p(15) - p(5)) = 6 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^5 p'(2t + 5) dt = \frac{1}{2} p(2t + 5) \Big|_0^5 = \frac{1}{2} (p(15) - p(5)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) = \frac{1}{2} p(u); \text{ and } \frac{1}{2} p(2t + 5) \Big|_0^5 = \frac{1}{2} (-8 - (-20)) \text{ earns } 1 - 1 - 1$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:Let $u = 2t + 5$. Then $du = 2 dt$.When $t = 0$, $u = 5$.When $t = 5$, $u = 15$.

$$\begin{aligned} \int_0^5 p'(2t + 5) dt &= \frac{1}{2} \int_5^{15} p'(u) du \\ &= \frac{1}{2} (p(15) - p(5)) \\ &= \frac{1}{2} (-8 - (-20)) = 6 \end{aligned}$$

6-2

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(20x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(4 \cdot 5x)$ or $p(20x)$ (Note $p(20t)$ would also earn this point.)

The expression $p(4 \cdot 5x) - p(4 \cdot 3)$ or $(p(4 \cdot 5x) - p(4 \cdot 3)) \cdot 5$ earns this point. The $p(4 \cdot 3)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 4t$ earns first point.

Part of this response may include $\int_3^{5x} p(4t) dt = \frac{1}{4} \int_{12}^{20x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(1)$. That is accomplished by multiplying the term involving $p(x)$ by 5.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(20x)$ for any $a \neq 5$.

The point is not earned for responses of $5p(4t) \big|_3^{5x}$ or $5(p(20x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of -12.5 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$\cdot p(4 \cdot 5x) - p(12)$; $p(20) - p(12)$ earns $1 - 0 - 0$

$\cdot p(4 \cdot 5x) \cdot 5 - p(12) \cdot 0$; -12.5 earns $1 - 1 - 1$

$\cdot 5p(4t) \big|_3^{5x} = 5(p(20) - p(12))$ earns $1 - 0 - 0$

$\cdot p(4x) \cdot 5$; $p(4) \cdot 5$ earns $0 - 0 - 0$

$\cdot p(4 \cdot 5x) \cdot 5$; $p(20) \cdot 5$ earns $1 - 1 - 0$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

6-2

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_3^{5x} p(4t) dt = p(4 \cdot 5x) \cdot 5$$

$$h'(1) = p(20) \cdot 5 = -2.5 \cdot 5 = -12.5$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- “decreasing since $g < 0$ and $g' > 0$ ”
- “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

- “decreasing since velocity and acceleration have opposite signs” earns neither point
- “decreasing since velocity < 0 and acceleration > 0 ” earns neither point

“decreasing since $(-2) \cdot 4 = -8 < 0$ ” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

6-2

Solution:

Because $g(1) = -2 < 0$, the velocity of the particle is negative. Because $g'(1) = 4 > 0$, the acceleration of the particle is positive. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

· $y = 3(x + 1) + 2$ or $y - 2 = 3(x + 1)$

· $y = 3x + 5$

· $y = (-2 \cdot (-1)^3 \cdot 2^2 - 5)(x + 1) + 2$

Incorrect responses include:

· $y = (-2x^3y^2 - 5)(x + 1) + 2$

· $y = -3x + 5$

· $f(x) = 3x + 5$



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(-1, 2)} = -2(-1)^3 \cdot 2^2 - 5 = 3$$

$$y = 3(x + 1) + 2$$

6-2

65. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	8	12	16
$p(t)$ (feet)	25	21	16	12	7.5

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

- (a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{16} p(t) \, dt$. Show the computations that lead to your answer.
- (b) Use the data in the table to approximate $p'(2)$ using the average rate of change of $p(t)$ over the interval $0 \leq t \leq 4$. Show the computations that lead to your answer. Indicate units of measure.
- (c) Interpret the meaning of $p'(2)$ in the context of the problem.
- (d) Must there exist a value of d , for $0 < d < 16$, such that $p'(d)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 16$? Justify your answer.
- (e) Find $\int_0^4 p'(3t+4) \, dt$. Show the computations that lead to your answer.
- (f) Let $h(x) = \int_1^{2x} p(3t) \, dt$. Find $h'(2)$. Show the computations that lead to your answer.
- (g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(3) = 2$, $g'(3) = -4$, and $g''(3) = -8$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.
- (h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -2x^2 + y^2$ with initial condition $f(3) = 5$. Write an equation for the line tangent to the graph of f at the point $(3, 5)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(4) \cdot (8 - 0) + p(12) \cdot (16 - 8)$ or $p(4) \cdot (8) + p(12) \cdot (8)$ earns first point.

6-2

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

264 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products ($168 + 96$) or a graph with areas labeled and summed to 264 or equivalent, can earn this point.

Examples:

- $264 = 168 + 96$ by itself earns second point, not first point
- 264 declared, with $21 \cdot 8$ and $12 \cdot 8$ identified (perhaps graphically) earns second point, not first point
- $k(21 \cdot 8 + 21 \cdot 8)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(4) \cdot (8) + p(12) \cdot (16 - 8)$ by itself earns first point, not second point



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{16} p(t) dt \approx p(4)(8 - 0) + p(12)(16 - 8)$$

$$= 21 \cdot 8 + 12 \cdot 8 = 264$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(4)-p(0)}{4-0}$ by itself doesn't earn first point; however $\frac{21-25}{4-0}$ or $\frac{21-25}{4}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

6-2

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(2) \approx \frac{21-25}{4-0} = \frac{-4}{4} = -1 \text{ foot per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(2)$ interpretation.

Need two components in response:

- “rate of change of position” or “velocity” of the particle
- at $t = 2$

Correct ways to describe rate at time at $t = 2$ include:

- Rate at which position is changing/decreasing
- Rate at which particle is moving left
- Instantaneous velocity or velocity of particle

Ambiguous/incorrect ways to describe rate include:

- Rate at which particle is changing/decreasing
- Rate at which particle is moving

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Interpretation

Solution:

$p'(2)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 2$ seconds.

6-2

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(d) = -\frac{17.5}{16}$ must be correct to earn second point.

Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

6-2

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 16$ to guarantee that there exists a value of d , for $0 < d < 16$, such that $p'(d)$ equals the average rate of change of p over the interval $0 \leq x \leq 16$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

- $u = 3t + 4$ (can be explicit or implicit)
- $du = 3 dt$ (yielding integral coefficient of $\frac{1}{3}$)
- Correct integral limits throughout, with t from 0 to 4 and u from 4 to 16

Omitting the coefficient of $\frac{1}{3}$ or using some other value will not earn first point; response could earn second point, but not third point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 4, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $3t + 4$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^4 p'(3t + 4) dt = \frac{1}{3}p(3t + 4) \Big|_0^4$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

- $k(p(16) - p(4))$ or $k(p(3 \cdot 4 + 4) - p(3 \cdot 0 + 4))$ for k a nonzero constant
- $k p(3t + 4) \Big|_0^4$ or $k p(u) \Big|_4^{16}$ for k a nonzero constant
- $\int p'(3t + 4) dt = k p(3t + 4)$ or $\int p'(u) du = k p(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^4 p'(3t + 4) dt = \frac{1}{2}p(3t + 4) \Big|_0^4$

Third Point

Earned only for an answer of -4.5 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

6-2

Scoring Examples:

$$\cdot \frac{1}{3} \int p'(u) \, du = \frac{1}{3} p(u) \text{ earns } 0 - 1 - 0$$

$$\cdot \frac{1}{3} \int p'(u) \, du = \frac{1}{3} p(u) \Big|_0^4 = \frac{1}{3} (p(4) - p(0)) \text{ earns } 0 - 1 - 0$$

$$\cdot \int p'(u) \, du = p(u) \Big|_4^{16} = (p(16) - p(4)) = -13.5 \text{ earns } 0 - 1 - 0$$

$$\cdot 3 \int p'(u) \, du = 3p(u) \Big|_4^{16} = 3[7.5 - 21] \text{ earns } 0 - 1 - 0$$

$$\cdot \int_0^4 p'(3t+4) \, dt = p(1.5t^2 + 4t) \Big|_0^4 = p(40) - p(0) \text{ earns } 0 - 0 - 0$$

$$\cdot \frac{1}{3} \int_0^4 p'(u) \, du = \frac{1}{3} (p(16) - p(4)) = -4.5 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^4 p'(3t+4) \, dt = \frac{1}{3} (p(16) - p(4)) = -4.5 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^4 p'(3t+4) \, dt = \frac{1}{3} p(u) \Big|_4^{16} = \frac{1}{3} (p(16) - p(4)) = -4.5 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^4 p'(3t+4) \, dt = \frac{1}{3} p(3t+4) \Big|_0^4 = \frac{1}{3} (p(16) - p(4)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{3} \int p'(u) = \frac{1}{3} p(u); \text{ and } \frac{1}{3} p(3t+4) \Big|_0^4 = \frac{1}{3} (7.5 - 21) \text{ earns } 1 - 1 - 1$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:Let $u = 3t + 4$. Then $du = 3 \, dt$.When $t = 0$, $u = 4$.When $t = 4$, $u = 16$.

$$\begin{aligned} \int_0^4 p'(3t+4) \, dt &= \frac{1}{3} \int_4^{16} p'(u) \, du \\ &= \frac{1}{3} (p(16) - p(4)) \\ &= \frac{1}{3} (7.5 - 21) = -4.5 \end{aligned}$$

6-2

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(6x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(3 \cdot 2x)$ or $p(6x)$ (Note $p(6t)$ would also earn this point.)

The expression $p(3 \cdot 2x) - p(3 \cdot 1)$ or $(p(3 \cdot 2x) - p(3 \cdot 1)) \cdot 2$ earns this point. The $p(3 \cdot 1)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 3t$ earns first point.

Part of this response may include $\int_1^{2x} p(3t) dt = \frac{1}{3} \int_3^{6x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(2)$. That is accomplished by multiplying the term involving $p(x)$ by 2.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(6x)$ for any $a \neq 2$.

The point is not earned for responses of $2p(3t) \big|_1^{2x}$ or $2(p(6x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of 24 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

• $p(3 \cdot 2x) - p(3)$; $p(12) - p(3)$ earns 1 – 0 – 0

• $p(3 \cdot 2x) \cdot 2 - p(3) \cdot 0$; 24 earns 1 – 1 – 1

• $2p(3t) \big|_1^{2x} = 2(p(12) - p(3))$ earns 1 – 0 – 0

• $p(3x) \cdot 2$; $p(6) \cdot 2$ earns 0 – 0 – 0

• $p(6x) \cdot 2$; $p(12) \cdot 2$ earns 1 – 1 – 0



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

6-2

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_1^{2x} p(3t) dt = p(3 \cdot 2x) \cdot 2$$

$$h'(2) = p(12) \cdot 2 = 12 \cdot 2 = 24$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- “decreasing since $g < 0$ and $g' > 0$ ”
- “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

- “decreasing since velocity and acceleration have opposite signs” earns neither point
- “decreasing since velocity < 0 and acceleration > 0 ” earns neither point

“decreasing since $(2) \cdot (-4) = -8 < 0$ ” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

6-2

Solution:

Because $g(3) = 2 > 0$, the velocity of the particle is positive. Because $g'(3) = -4 < 0$, the acceleration of the particle is negative. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

$$y = 7(x - 3) + 5 \text{ or } y - 5 = 7(x - 3)$$

$$y = 7x - 16$$

$$y = (-2 \cdot 3^2 + 5^2)(x - 3) + 5$$

Incorrect responses include:

$$y = (-2x^2 + y^2)(x - 3) + 5$$

$$y = -7x - 16$$

$$f(x) = 7x - 16$$



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(3,5)} = -2 \cdot 3^2 + 5^2 = 7$$

$$y = 7(x - 3) + 5$$

6-2

66. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	8	11
$v(t)$ (meters per second)	-15	-12	-8	-3

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

(a) Use the data in the table to approximate $v'(10)$ using the average rate of change of $v(t)$ over the interval $8 \leq t \leq 11$. Show the computations that lead to your answer. Indicate units of measure.

(b) Interpret the meaning of $v'(10)$ in the context of the problem.

(c) Justify why there must be a time $t = k$, for $0 \leq k \leq 3$, when the velocity of the particle is -13 meters per second.

(d) Use a right Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{11} v(t) \, dt$. Show the computations that lead to your answer.

(e) Find $\int_4^{14} v'\left(\frac{t}{2} + 1\right) \, dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_3^{\frac{11}{2}x} v(2t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(2) = -5.5$, $g'(2) = 4$, and $g''(2) = 3$. Is the speed of particle Q increasing or decreasing at time $t = 2$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = y^2 - xy$ with initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of f at the point $(3, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation.

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{-3+8}{11-8}$, $\frac{-3-(-8)}{3}$, and $\frac{5}{11-8}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

6-2

Numerical values must be pulled from the table. $\frac{v(11)-v(8)}{11-8}$ alone will not earn the point as it does not pull values from the table (yet).

Second point: Units

$\frac{m}{s^2}$ and fraction or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$v'(10) \approx \frac{-3 - (-8)}{11 - 8} = \frac{5}{3} \text{ meters per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

· “rate of change of velocity” or “acceleration” of the particle

· at $t = 10$

“The rate at which the velocity is increasing at $t = 10$ ” is acceptable. If the answer to part (a) was reported to be negative, “the rate at which the velocity is decreasing at $t = 10$ ” is acceptable. We will not penalize for the same error twice.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

6-2

Solution:

$v'(10)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 10$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

“ v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values -15 , and -12 pulled from the table and correctly compared to -13 , in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always increasing on the interval $[0, 3]$. This point will not be earned if the response contains such errors in reasoning.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding -13 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(0) = -15 < -13 < -12 = v(3)$$

By the Intermediate Value Theorem, there must be a time $t = k$, for $0 \leq k \leq 3$, such that the velocity of the particle is -13 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of right Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$.

6-2

If exactly 5 out of 6 of these numbers ($b_1, b_2, b_3, h_1, h_2, h_3$) are correctly placed in the above expression, we are convinced that the response carries the form of a right Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(3) \cdot (3 - 0) + v(8) \cdot (8 - 3) + v(11) \cdot (11 - 8)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

second point: Answer

−85 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a right Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products ($-36 + (-40) + (-9)$), or a graph with areas labeled and summed to −85 or equivalent, can earn this point.

The response of only $(-12) \cdot 3 + (-8) \cdot 5 + (-3) \cdot 3$ earns both points.

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{11} v(t) dt \approx v(3)(3 - 0) + v(8)(8 - 3) + v(11)(11 - 8)$$

$$= -12 \cdot 3 + (-8) \cdot 5 + (-3) \cdot 3 = -85$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

· $u = \frac{t}{2} + 1$

· $du = \frac{dt}{2}$ or equivalent

· Appropriate limits of integration throughout (3 and 8 for u , 4 and 14 for t).

Substitution can be implicit. A coefficient of 2 accompanying a correct antiderivative convinces us of this:

$$\int_4^{14} v' \left(\frac{t}{2} + 1 \right) dt = 2v \left(\frac{t}{2} + 1 \right) \Big|_4^{14}$$

6-2

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (4 and 14), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{2} + 1$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Expressions/equations that earn this point are:

· $k(v(8) - v(3))$ or $k(v(\frac{14}{2} + 1) - v(\frac{4}{2} + 1))$, where k is a nonzero constant.

· $kv(\frac{t}{2} + 1)|_4^{14}$ or $k v(u)|_3^8$, where k is a nonzero constant.

· $\int v'(\frac{t}{2} + 1)dt = kv(\frac{t}{2} + 1)$ or $k \int v'(u)du = kv(u)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

· $\int_4^{14} v'(\frac{t}{2} + 1)dt = kv(\frac{t^2}{4} + t)$ where k is a nonzero constant is a common response. This does not earn the point. Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

8 or equivalent, with credible supporting work, earns this point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{2} + 1$. Then $du = \frac{dt}{2}$.

When $t = 4$, $u = 3$.

When $t = 14$, $u = 8$.

$$\begin{aligned} \int_4^{14} v' \left(\frac{t}{2} + 1 \right) dt &= 2 \int_3^8 v'(u) du \\ &= 2(v(8) - v(3)) \\ &= 2(-8 - (-12)) = 8 \end{aligned}$$

6-2

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Fundamental Theorem of Calculus (FTC)

Substitutes $\frac{11}{2}x$ into $v(2t)$. We must see either $v\left(2 \cdot \frac{11}{2}x\right)$ or $v(11x)$.

The expression $v\left(2 \cdot \frac{11}{2}x\right) - v(2(3))$ or $v\left(2 \cdot \frac{11}{2}x\right) \cdot \frac{11}{2} - v(2(3))$ will earn this point. The error of subtracting $v(2(3))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2}du = dt$ and rewrites: $\int_3^{\frac{11}{2}x} v(2t)dt = \frac{1}{2} \int_6^{11x} v(u)du$.

· Or $u = \frac{11}{2}x$, $\frac{du}{dx} = \frac{11}{2}$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by $\frac{11}{2}$

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 11 multiplied separately.

Third point: Answer

$-\frac{33}{2}$ or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_3^{\frac{11}{2}x} v(2t) dt = v\left(2 \cdot \frac{11}{2}x\right) \cdot \frac{11}{2}$$

$$h'(1) = v(11) \cdot \frac{11}{2} = -3 \cdot \frac{11}{2} = -\frac{33}{2}$$

6-2

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “increasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “increasing because velocity and acceleration are both negative” earns neither of the two points. It is not clear that the response is considering g' and g'' . A response that calls out all three given values ($g(2)$, $g'(2)$, and $g''(2)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “speeding up” will be interpreted as “increasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(2) = 4 > 0$, the velocity of the particle is positive. Because $g''(2) = 3 > 0$, the acceleration of the particle is positive. Therefore, because both the velocity and acceleration are positive, the speed of the particle is increasing at time $t = 2$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

· $y - 2 = ((2)^2 - 3 \cdot 2)(x - 3)$

· $y - 2 = -2(x - 3)$

· $y = -2x + 8$

6-2

$y - 2 = (y^2 - xy)(x - 3)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = -2(x - 3) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(3,2)} = 2^2 - 3 \cdot 2 = -2$$

$$y = -2(x - 3) + 2$$

6-2

67. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	2	5	11
$v(t)$ (meters per second)	3	5.5	7	13

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

(a) Use the data in the table to approximate $v'(8)$ using the average rate of change of $v(t)$ over the interval $5 \leq t \leq 11$. Show the computations that lead to your answer. Indicate units of measure.

(b) Interpret the meaning of $v'(8)$ in the context of the problem.

(c) Justify why there must be a time $t = b$, for $0 \leq b \leq 2$, when the velocity of the particle is 5 meters per second.

(d) Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{11} v(t) \, dt$. Show the computations that lead to your answer.

(e) Find $\int_0^9 v'\left(\frac{t}{3} + 2\right) \, dt$. Show the computations that lead to your answer.

(f) Let $h(x) = \int_1^{\frac{5}{2}x} v(2t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.

(g) The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(2) = 5.5$, $g'(2) = -4$, and $g''(2) = -3$. Is the speed of particle Q increasing or decreasing at time $t = 2$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -3x^2y^3$ with initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of f at the point $(3, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{13-7}{11-5}$, $\frac{13-7}{6}$, and $\frac{6}{11-5}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

6-2

Numerical values must be pulled from the table. $\frac{v(11)-v(5)}{11-5}$ alone will not earn the point as it does not pull values from the table (yet).

Note: the reciprocal of this quotient gives the same numerical value. This does not earn the point.

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$v'(8) \approx \frac{13-7}{11-5} = 1 \text{ meter per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

· “rate of change of velocity” or “acceleration” of the particle

· at $t = 8$

“The rate at which the velocity is increasing at $t = 8$ ” is acceptable. If the answer to part (a) was reported to be negative, “the rate at which the velocity is decreasing at $t = 8$ ” is acceptable. We will not penalize for the same error twice.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

Interpretation

6-2

**Solution:**

$v'(8)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 8$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

“ v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values 3 and 5.5 pulled from the table and correctly compared to 5, in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always increasing on the interval $[0, 2]$. This point will not be earned if the response contains such errors in reasoning.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding 5 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(0) = 3 < 5 < 5.5 = v(2)$$

By the Intermediate Value Theorem, there must be a time $t = b$, for $0 \leq b \leq 2$, such that the velocity of the particle is 5 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of left Riemann sum

6-2

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$.

If exactly 5 out of 6 of these numbers $(b_1, b_2, b_3, h_1, h_2, h_3)$ are correctly placed in the above expression, we are convinced that the response carries the form of a left Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(0) \cdot (2 - 0) + v(2) \cdot (5 - 2) + v(5) \cdot (11 - 5)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

64.5 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a left Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products $(6 + 16.5 + 42)$, or a graph with areas labeled and summed to 64.5 or equivalent, can earn this point.

The response of only $3 \cdot 2 + 5.5 \cdot 3 + 7 \cdot 6$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{11} v(t) dt \approx v(0)(2 - 0) + v(2)(5 - 2) + v(5)(11 - 5)$$

$$= 3 \cdot 2 + 5.5 \cdot 3 + 7 \cdot 6 = 64.5$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

· $u = \frac{t}{3} + 2$

· $du = \frac{dt}{3}$ or equivalent

· Appropriate limits of integration throughout (2 and 5 for u , 0 and 9 for t).

6-2

Substitution can be implicit. A coefficient of 3 accompanying a correct antiderivative convinces us of this:

$$\int_0^9 v' \left(\frac{t}{3} + 2 \right) dt = 3v \left(\frac{t}{3} + 2 \right) \Big|_0^9$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (0 and 9), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{3} + 2$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Expressions/equations that earn this point are:

• $k(v(5) - v(2))$ or $k(v(\frac{9}{3} + 2) - v(\frac{0}{3} + 2))$, where k is a nonzero constant.

• $kv(\frac{t}{3} + 2) \Big|_0^9$ or $k v(u) \Big|_2^5$, where k is a nonzero constant.

• $\int v' \left(\frac{t}{3} + 2 \right) dt = kv \left(\frac{t}{3} + 2 \right)$ or $k \int v'(u) du = kv(u)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

• $\int_0^9 v' \left(\frac{t}{3} + 2 \right) dt = k v \left(\frac{t^2}{6} + 2t \right)$ where k is a nonzero constant is a common response. This does not earn the point.

Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

4.5 or equivalent, with credible supporting work, earns this point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{3} + 2$. Then $du = \frac{dt}{3}$.

When $t = 0$, $u = 2$.

When $t = 9$, $u = 5$.

$$\int_0^9 v' \left(\frac{t}{3} + 2 \right) dt = 3 \int_2^5 v'(u) du$$

6-2

$$= 3(v(5) - v(2))$$

$$= 3(7 - 5.5) = 4.5$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Fundamental Theorem of Calculus (FTC)

Substitutes $\frac{5}{2}x$ into $v(2t)$. We must see either $v(2 \cdot \frac{5}{2}x)$ or $v(5x)$.

The expression $v(2 \cdot \frac{5}{2}x) - v(2(1))$ or $v(2 \cdot \frac{5}{2}x) \cdot \frac{5}{2} - v(2(1))$ will earn this point. The error of subtracting $v(2(1))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2} du = dt$ and rewrite $\int_1^{\frac{5}{2}x} v(2t) dt = \frac{1}{2} \int_2^{5x} v(u) du$.

· Or $u = \frac{5}{2}x$, $\frac{du}{dx} = \frac{5}{2}$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by $\frac{5}{2}$

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 5 multiplied separately.

Third point: Answer

$\frac{35}{2}$ or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_1^{\frac{5}{2}x} v(2t) dt = v\left(2 \cdot \frac{5}{2}x\right) \cdot \frac{5}{2}$$

6-2

$$h'(1) = v(5) \cdot \frac{5}{2} = 7 \cdot \frac{5}{2} = \frac{35}{2}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “increasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “increasing because velocity and acceleration are both negative” earns neither of the two points. It is not clear that the response is considering g' and g'' . A response that calls out all three given values ($g(2)$, $g'(2)$, and $g''(2)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “speeding up” will be interpreted as “increasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(2) = -4 < 0$, the velocity of the particle is negative. Because $g''(2) = -3 < 0$, the acceleration of the particle is negative. Therefore, because both the velocity and acceleration are negative, the speed of the particle is increasing at time $t = 2$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

$$y - 2 = -3(3)^2(2)^3(x - 3)$$

$$y - 2 = -216(x - 3)$$

6-2

$$y = -216x + 650$$

$y - 2 = -3x^2y^3(x - 3)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = -216(x - 3) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(3,2)} = -3 \cdot 3^2 \cdot 2^3 = -216$$

$$y = -216(x - 3) + 2$$

6-2

68. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	2	8	14
$v(t)$ (meters per second)	15	12	6	0

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

(a) Use the data in the table to approximate $v'(6)$ using the average rate of change of $v(t)$ over the interval $2 \leq t \leq 8$. Show the computations that lead to your answer. Indicate units of measure.

(b) Interpret the meaning of $v'(6)$ in the context of the problem.

(c) Justify why there must be a time $t = k$, for $8 \leq k \leq 14$, when the velocity of the particle is 3 meters per second.

(d) Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{14} v(t) \, dt$. Show the computations that lead to your answer.

(e) Find $\int_3^{21} v'\left(\frac{t}{3} + 7\right) \, dt$. Show the computations that lead to your answer.

(f) Let $p(x) = \int_0^{4x} v(2t) \, dt$. Find $p'(1)$. Show the computations that lead to your answer.

(g) The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(4) = 5$, $g'(4) = 2$, and $g''(4) = -6$. Is the speed of particle Q increasing or decreasing at time $t = 4$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -2x + 3y^2$ with initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of f at the point $(3, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation.

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{6-12}{8-2}$, $\frac{6-12}{6}$, and $-\frac{6}{8-2}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

6-2

Numerical values must be pulled from the table. $\frac{v(8)-v(2)}{8-2}$ alone will not earn the point as it does not pull values from the table (yet).

Note: the reciprocal of the quotient gives the same numerical answer. This does not earn the point.

Also: $\frac{v(14)-v(8)}{14-8}$ happens to equal -1 as well. A response using an incorrect pair of points does not earn the point.

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$v'(6) \approx \frac{6-12}{8-2} = -1 \text{ meter per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

- “rate of change of velocity” or “acceleration” of the particle
- at $t = 6$

“The rate at which the velocity is decreasing at $t = 6$ ” is acceptable. If the answer to part (a) was reported to be positive, “the rate at which the velocity is increasing at $t = 6$ is acceptable. We will not penalize for the same error twice.

Responses that say “velocity is decreasing at a rate of $-1 \frac{m}{s^2}$ ” or similar do not earn the point. This is interpreted to mean that the velocity is increasing.

Units are not required in part (b). If units are presented for t , they must be s (seconds).

6-2



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$v'(6)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 6$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values 0 and 6 pulled from the table and correctly compared with 3, in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always decreasing on the interval $[8, 14]$. This point will not be earned if the response contains such errors in reasoning.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ v continuous

☐ Bounding 3 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(14) = 0 < 3 < 6 = v(8)$$

By the Intermediate Value Theorem, there must be a time $t = k$, for $8 < k < 14$, such that the velocity of the particle is 3 meters per second.

6-2

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of left Riemann sum

Must see a sum of three terms, each of which is a product of two factors:

$$(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$$

If exactly 5 out of 6 of these numbers (b_1 , b_2 , b_3 , h_1 , h_2 , h_3) are correctly placed in the above expression, we are convinced that the response carries the form of a left Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(0) \cdot (2 - 0) + v(2) \cdot (8 - 2) + v(8) \cdot (14 - 8)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

138 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a left Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products ($30 + 72 + 36$), or a graph with areas labeled and summed to 138 or equivalent, can earn this point.

The response of only $15 \cdot 2 + 12 \cdot 6 + 6 \cdot 6$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Form of left Riemann sum

☐ Answer

Solution:

$$\int_0^{14} v(t) dt \approx v(0)(2 - 0) + v(2)(8 - 2) + v(8)(14 - 8)$$

$$= 15 \cdot 2 + 12 \cdot 6 + 6 \cdot 6 = 138$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

1st point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

6-2

$$\cdot u = \frac{t}{3} + 7$$

$$\cdot du = \frac{dt}{3} \text{ or equivalent}$$

· Appropriate limits of integration throughout (8 and 14 for u , 3 and 21 for t).

Substitution can be implicit. A coefficient of 3 accompanying a correct antiderivative convinces us of this:

$$\int_3^{21} v' \left(\frac{t}{3} + 7 \right) dt = 3v \left(\frac{t}{3} + 7 \right) \Big|_3^{21}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (3 and 21), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{3} + 7$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Expressions/equations that earn this point are:

$$\cdot k(v(14) - v(8)) \text{ or } k\left(v\left(\frac{21}{3} + 7\right) - v\left(\frac{3}{3} + 7\right)\right), \text{ where } k \text{ is a nonzero constant}$$

$$\cdot kv\left(\frac{t}{3} + 7\right) \Big|_3^{21} \text{ or } kv(u) \Big|_8^{14}, \text{ where } k \text{ is a nonzero constant}$$

$$\cdot \int v' \left(\frac{t}{3} + 7 \right) dt = kv \left(\frac{t}{3} + 7 \right) \text{ or } k \int v' (u) du = kv(u), \text{ where } k \text{ is a nonzero constant, with or without appropriate limits of integration displayed on either integral}$$

$$\cdot \int_3^{21} v' \left(\frac{t}{3} + 7 \right) dt = k v \left(\frac{t^2}{6} + 7t \right) \text{ where } k \text{ is a nonzero constant is a common response. This does not earn the point.}$$

Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

−18 or equivalent, with credible supporting work, earns this point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\text{Let } u = \frac{t}{3} + 7.$$

6-2

Then $du = \frac{dt}{3}$.

When $t = 3$, $u = 8$.

When $t = 21$, $u = 14$.

$$\begin{aligned}\int_3^{21} v' \left(\frac{t}{3} + 7 \right) dt &= 3 \int_8^{14} v' (u) du \\ &= 3 (v (14) - v (8)) \\ &= 3(0 - 6) = -18\end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Fundamental Theorem of Calculus (FTC)

Substitutes $4x$ into $v(2t)$. We must see either $v(2 \cdot 4x)$ or $v(8x)$.

The expression $v(2 \cdot 4x) - v(2(0))$ or $v(2 \cdot 4x) \cdot 4 - v(2(0))$ will earn this point. The error of subtracting $v(2(0))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2}du = dt$ and rewrites: $\int_0^{4x} v(2t) dt = \frac{1}{2} \int_0^{8x} v(u) du$.

· Or Alt text: $u = 4x$, $\frac{du}{dx} = 4$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by 4

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 8 multiplied separately.

Third point: Answer

24 or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule

6-2

☐ Answer
Solution:

$$p'(x) = \frac{d}{dx} \int_0^{4x} v(2t) dt = v(2 \cdot 4x) \cdot 4$$

$$p'(1) = v(8) \cdot 4 = 6 \cdot 4 = 24$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “decreasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “decreasing because velocity and acceleration have different signs” earns neither of the two points. It is not clear that the response is considering g' and g'' .

A response that calls out all three given values ($g(4)$, $g'(4)$, and $g''(4)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “slowing down” will be interpreted as “decreasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(4) = 2 > 0$, the velocity of the particle is positive. Because $g''(4) = -6 < 0$, the acceleration of the particle is negative. Therefore, because the signs of the velocity and acceleration are different, the speed of the particle is decreasing at time $t = 4$.

Part H

6-2

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

$$y - 2 = (-2(3) + 3(2)^2)(x - 3)$$

$$y - 2 = 6(x - 3)$$

$$y = 6x - 16$$

$y - 2 = (-2x + 3y^2)(x - 3)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = 6(x - 3) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(3,2)} = -2 \cdot 3 + 3 \cdot 2^2 = 6$$

$$y = 6(x - 3) + 2$$

6-2

69. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	8	11
$v(t)$ (meters per second)	-2	-5	-11	-12

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

(a) Use the data in the table to approximate $v'(5)$ using the average rate of change of $v(t)$ over the interval $3 \leq t \leq 8$. Show the computations that lead to your answer. Indicate units of measure.

(b) Interpret the meaning of $v'(5)$ in the context of the problem.

(c) Justify why there must be a time $t = d$, for $0 \leq d \leq 3$, when the velocity of the particle is -4 meters per second.

(d) Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{11} v(t) \, dt$. Show the computations that lead to your answer.

(e) Find $\int_0^{10} v' \left(\frac{t}{2} + 3 \right) \, dt$. Show the computations that lead to your answer.

(f) Let $r(x) = \int_0^{\frac{3}{2}x} v(2t) \, dt$. Find $r'(1)$. Show the computations that lead to your answer.

(g) The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(3) = -4$, $g'(3) = -2$, and $g''(3) = 6$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.

(h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 3x^2 - y^3$ with initial condition $f(-1) = 2$. Write an equation for the line tangent to the graph of f at the point $(-1, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation.

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{-11 - (-5)}{8 - 3}$, $\frac{-11 - (-5)}{5}$, and $\frac{-6}{8 - 3}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

6-2

Numerical values must be pulled from the table. $\frac{v(8)-v(3)}{8-3}$ alone will not earn the point as it does not pull values from the table (yet).

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$v'(5) \approx \frac{-11 - (-5)}{8 - 3} = -\frac{6}{5} \text{ meters per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

· “rate of change of velocity” or “acceleration” of the particle

· at $t = 5$

“The rate at which the velocity is decreasing at $t = 5$ ” is acceptable. If the answer to part (a) was reported to be positive, “the rate at which the velocity is increasing at $t = 5$ ” is acceptable. We will not penalize for the same error twice.

Responses that say “velocity is decreasing at a rate of $-\frac{6}{5} \frac{m}{s^2}$ ” or similar do not earn the point. This is interpreted to mean that the velocity is increasing.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

6-2

☐ Interpretation
Solution:

$v'(5)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 5$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

“ v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values -5 and -2 pulled from the table and correctly compared with -4 , in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always decreasing on the interval $[0, 3]$. This point will not be earned if the response contains such errors in reasoning.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding -4 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(3) = -5 < -4 < -2 = v(0)$$

By the Intermediate Value Theorem, there must be a time $t = d$, $0 < d < 3$, such that the velocity of the particle is -4 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of left Riemann sum

6-2

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$

If exactly 5 out of 6 of these numbers ($b_1, b_2, b_3, h_1, h_2, h_3$) are correctly placed in the above expression, we are convinced that the response carries the form of a left Riemann sum. In this case, award the first point. Such a response cannot earn the second point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(0) \cdot (3 - 0) + v(3) \cdot (8 - 3) + v(8) \cdot (11 - 8)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

−64 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a left Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products ($-6 + -25 + -33$) or a graph with areas labeled and summed to −64 or equivalent, can earn this point.

The response of only $(-2 \cdot 3) + (-5 \cdot 5) + (-11 \cdot 3)$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{11} v(t) \, dt \approx v(0)(3 - 0) + v(3)(8 - 3) + v(8)(11 - 8)$$

$$= -2 \cdot 3 + (-5) \cdot 5 + (-11) \cdot 3 = -64$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

· $u = \frac{t}{2} + 3$

· $du = \frac{dt}{2}$ or equivalent

· Appropriate limits of integration throughout (3 and 8 for u , 0 and 10 for t).

6-2

Substitution can be implicit. A coefficient of 2 accompanying a correct antiderivative convinces us of this:

$$\int_0^{10} v' \left(\frac{t}{2} + 3 \right) dt = 2v \left(\frac{t}{2} + 3 \right) \Big|_0^{10}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (0 and 10), this point is not earned.

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{2} + 3$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Expressions/equations that earn this point are:

$$k(v(8) - v(3)) \text{ or } k\left(v\left(\frac{10}{2} + 3\right) - v\left(\frac{0}{2} + 3\right)\right),$$

· where k is a nonzero constant.

$$\cdot kv\left(\frac{t}{2} + 3\right) \Big|_0^{10} \text{ or } kv(u) \Big|_3^8, \text{ where } k \text{ is a nonzero constant.}$$

$$\cdot \int v' \left(\frac{t}{2} + 3 \right) dt = kv \left(\frac{t}{2} + 3 \right) \text{ or } k \int v'(u) du = kv(u), \text{ where } k \text{ is a nonzero constant, with or without appropriate limits of integration displayed on either integral.}$$

$$\cdot \int_0^{10} v' \left(\frac{t}{2} + 3 \right) dt = kv \left(\frac{t^2}{4} + 3t \right) \text{ where } k \text{ is a nonzero constant is a common response. This does not earn the point.}$$

Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

−12 or equivalent, with credible supporting work, earns this point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\text{Let } u = \frac{t}{2} + 3. \text{ Then } du = \frac{dt}{2}.$$

$$\text{When } t = 0, u = 3.$$

$$\text{When } t = 10, u = 8.$$

6-2

$$\int_0^{10} v' \left(\frac{t}{2} + 3 \right) dt = 2 \int_3^8 v' (u) du$$

$$= 2 (v (8) - v (3))$$

$$= 2 (-11 - (-5)) = -12$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Fundamental Theorem of Calculus

Substitutes $\frac{3}{2}x$ into $v(2t)$. We must see either $v\left(2 \cdot \frac{3}{2}x\right)$ or $v(3x)$.

The expression $v\left(2 \cdot \frac{3}{2}x\right) - v(2(0))$ or $v\left(2 \cdot \frac{3}{2}x\right) \cdot \frac{3}{2} - v(2(0))$ will earn this point. The error of subtracting $v(2(0))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 2t$, $\frac{1}{2}du = dt$ and rewrites: $\int_0^{\frac{3}{2}x} v(2t) dt = \frac{1}{2} \int_0^{3x} v(u) du$.

· Or $u = \frac{3}{2}x$, $\frac{du}{dx} = \frac{3}{2}$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by $\frac{3}{2}$.

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{2}$ and 3 multiplied separately.

Third point: Answer

$-\frac{15}{2}$ or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

6-2

$$r'(x) = \frac{d}{dx} \int_0^{\frac{3}{2}x} v(2t) dt = v\left(2 \cdot \frac{3}{2}x\right) \cdot \frac{3}{2}$$

$$r'(1) = v(3) \cdot \frac{3}{2} = -5 \cdot \frac{3}{2} = -\frac{15}{2}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

Second point: Answer with reason

A correct response must:

- Conclude “decreasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “decreasing because velocity and acceleration have different signs” earns neither of the two points. It is not clear that the response is considering g' and g''

A response that calls out all three given values ($g(3)$, $g'(3)$, and $g''(3)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “slowing down” will be interpreted as “decreasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(3) = -2 < 0$, the velocity of the particle is negative. Because $g''(3) = 6 > 0$, the acceleration of the particle is positive. Because the signs of the velocity and acceleration are different, the speed of the particle is decreasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

6-2

Some common forms:

$$y - 2 = (3(-1)^2 - 2^3)(x + 1)$$

$$y - 2 = -5(x + 1)$$

$$y = -5x - 3$$

$y - 2 = (3x^2 - y^3)(x + 1)$ does not earn the point. This is not the equation of a line. The response can go on to present one of the correct linear forms, and then would earn the point.

$f(x) = -5(x + 1) + 2$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(-1, 2)} = 3 \cdot (-1)^2 - 2^3 = -5$$

$$y = 2 - 5(x + 1)$$

6-2

70. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	10	12
$v(t)$ (meters per second)	2	6	15	20

The velocity of a particle, P , moving along the x -axis is modeled by a differentiable function v , where time t is measured in seconds and $v(t)$ is measured in meters per second. Selected values of $v(t)$ are shown in the table above.

- Use the data in the table to approximate $v'(11)$ using the average rate of change of $v(t)$ over the interval $10 \leq t \leq 12$. Show the computations that lead to your answer. Indicate units of measure.
- Interpret the meaning of $v'(11)$ in the context of the problem.
- Justify why there must be a time $t = c$, for $4 \leq c \leq 10$, when the velocity of the particle is 8 meters per second.
- Use a right Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^{12} v(t) \, dt$. Show the computations that lead to your answer.
- Find $\int_4^{12} v' \left(\frac{t}{2} - 2 \right) \, dt$. Show the computations that lead to your answer.
- Let $h(x) = \int_2^{4x} v(3t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.
- The position of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(4) = 7$, $g'(4) = -3$, and $g''(4) = -2$. Is the speed of particle Q increasing or decreasing at time $t = 4$? Give a reason for your answer.
- Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 3x^2 - y^3$ with initial condition $f(2) = 1$. Write an equation for the line tangent to the graph of f at the point $(2, 1)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Approximation

Need to see both a difference and a quotient as the supporting work. Minimal answers that earn the point include $\frac{20-15}{12-10}$, $\frac{20-15}{2}$, and $\frac{5}{12-10}$.

The answer does not need to be simplified, but any attempt at simplification must be correct.

6-2

Numerical values must be pulled from the table. $\frac{v(12)-v(10)}{12-10}$ alone is not sufficient as it does not pull values from the table (yet).

Second point: Units

$\frac{m}{s^2}$ or meters per second per second will earn the point.

The units must be tied to a number. Units alone do not earn this point.

Beware of $\left(\frac{m}{s}\right)^2$ or $\frac{m^2}{s}$ (which do not earn the point).

This point must be earned in part (a). If no units are present in part (a) and correct units are shown in part (b), this point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$v'(11) \approx \frac{20-15}{12-10} = \frac{5}{2} \text{ meters per second per second}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Interpretation

There are two components to look for:

· “rate of change of velocity” or “acceleration” of the particle

· at $t = 11$

“The rate at which the velocity is increasing at $t = 11$ ” is acceptable. If the answer to part (a) was reported to be negative, “the rate at which the velocity is decreasing at $t = 11$ ” is acceptable. We will not penalize for the same error twice.

Units are not required in part (b). If units are presented for t , they must be s (seconds).



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

6-2

$v'(11)$ is the rate at which the velocity of the particle is changing, in meters per second per second, at time $t = 11$ seconds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: v is continuous

It must be clear that the reason given as to why v is continuous is that v is differentiable.

Neither “ v is differentiable and continuous” nor “ v differentiable, v continuous” are sufficient to earn this point.

“ v is differentiable hence continuous,” by contrast, earns the point.

“ v differentiable $\therefore v$ continuous” earns the point.

Second point: Bounding with table values

Must see the values 6 and 15 pulled from the table and correctly compared with 8, in words or using inequality symbols.

The Intermediate Value Theorem (IVT) does not need to be referenced by name.

That said, this point will not be earned if other theorems (MVT, etc.) are invoked.

Responses often “say too much”; especially common are responses that indicate that v is always increasing on the interval $[6, 15]$. This point will not be earned if the response contains such errors in reasoning.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ v continuous
- ☐ Bounding 8 with table values

Solution:

v is differentiable. $\Rightarrow v$ is continuous.

$$v(4) = 6 < 8 < 15 = v(10)$$

By the Intermediate Value Theorem, there must be a time $t = c$, for $4 < c < 10$, such that the velocity of the particle is 8 meters per second.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Form of right Riemann sum

Must see a sum of three terms, each of which is a product of two factors: $(b_1 \cdot h_1) + (b_2 \cdot h_2) + (b_3 \cdot h_3)$.

If exactly 5 out of 6 of these numbers ($b_1, b_2, b_3, h_1, h_2, h_3$) are correctly placed in the above expression, we are convinced that the response carries the form of a right Riemann sum. In this case, award the first point. Such a response cannot earn the second

6-2

point. If fewer than 5 out of 6 values are correct, no points are earned in this problem.

To earn this point, values do not need yet to be imported from the table: the expression $v(4) \cdot (4 - 0) + v(10) \cdot (10 - 4) + v(12) \cdot (12 - 10)$ is sufficient.

Once the first point is earned, it cannot be lost. Subsequent errors result in the second point not being earned.

Second point: Answer

154 or equivalent.

It is possible to earn this point without earning the first point. If there is credible evidence that an attempt was made to find a right Riemann sum but not enough work is shown, this point can be earned in the presence of the correct answer. For example, a sum of three terms with no evidence of products ($24 + 90 + 40$), or a graph with areas labeled and summed to 154 or equivalent, can earn this point.

The response of only $6 \cdot 4 + 15 \cdot 6 + 20 \cdot 2$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{12} v(t) dt \approx v(4)(4 - 0) + v(10)(10 - 4) + v(12)(12 - 10)$$

$$= 6 \cdot 4 + 15 \cdot 6 + 20 \cdot 2 = 154$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Substitution

There are three elements to look for in a correct substitution. (another variable may be chosen instead of u .)

- $u = \frac{t}{2} - 2$
- $du = \frac{dt}{2}$ or equivalent
- Appropriate limits of integration throughout (0 and 4 for u , 4 and 12 for t).

Substitution can be implicit. A coefficient of 2 accompanying a correct antiderivative convinces us of this:

$$\int_4^{12} v' \left(\frac{t}{2} - 2 \right) dt = 2v \left(\frac{t}{2} - 2 \right) \Big|_4^{12}$$

To elaborate on the third element above: if a response shows an otherwise correct substitution but the limits of integration on a definite integral are unchanged (4 and 12), this point is not earned.

6-2

Alternately, the integral can be worked with as an indefinite integral with the variable returned to $\frac{t}{2} - 2$ before evaluating as a means of eliminating the need to change limits. Such a response can earn the first point, if handled correctly.

Second point: Fundamental Theorem of Calculus

We must be convinced that the response has indicated an appropriate connection between $\int v'$ and v .

Expressions/equations that earn this point are:

· $k(v(4) - v(0))$ or $k(v(\frac{12}{2} - 2) - v(\frac{4}{2} - 2))$, where k is a nonzero constant.

· $kv(\frac{t}{2} - 2)|_4^{12}$ or $kv(u)|_0^4$, where k is a nonzero constant.

· $\int v'(\frac{t}{2} - 2) dt = kv(\frac{t}{2} - 2)$ or $k \int v'(u) du = kv(u)$, where k is a nonzero constant, with or without appropriate limits of integration displayed on either integral.

· $\int_4^{12} v'(\frac{t}{2} - 2) dt = kv(\frac{t^2}{4} - 2t)$ where k is a nonzero constant is a common response. This does not earn the point.

Likewise, similar mishandled antiderivatives do not earn the point.

Third point: Answer

8 or equivalent, with credible supporting work, earns this point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{2} - 2$. Then $du = \frac{dt}{2}$.

When $t = 4$, $u = 0$.

When $t = 12$, $u = 4$.

$$\begin{aligned} \int_4^{12} v' \left(\frac{t}{2} - 2 \right) dt &= 2 \int_0^4 v'(u) du \\ &= 2(v(4) - v(0)) \\ &= 2(6 - 2) = 8 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-2

First point: Fundamental Theorem of Calculus

Substitutes $4x$ into $v(3t)$. We must see either $v(3 \cdot 4x)$ or $v(12x)$.

The expression $v(3 \cdot 4x) - v(3(2))$ or $v(3 \cdot 4x) \cdot 4 - v(3(2))$ will earn this point. The error of subtracting $v(3(2))$ will be viewed as an error in applying the chain rule.

Beware of possible substitutions prior to using FTC:

· For example, $u = 3t$, $\frac{1}{3}du = dt$ and rewrites: $\int_2^{4x} v(3t) dt = \frac{1}{3} \int_6^{12x} v(u) du$.

· Or $u = 4x$, $\frac{du}{dx} = 4$ and rewrites: $\frac{d}{dx}[h(x)] = \frac{d}{du}[h(u)] \cdot \frac{du}{dx}$

· Such approaches can earn this point if done properly.

Second point: Chain rule

Multiplies by 4

It is not possible to earn this point without having earned the first point.

Depending on substitutions, this may look more like $\frac{1}{3}$ and 12 multiplied separately.

Third point: Answer

80 or equivalent

It is not possible to earn this point without having earned both the first and second points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_2^{4x} v(3t) dt = v(3 \cdot 4x) \cdot 4$$

$$h'(1) = v(12) \cdot 4 = 20 \cdot 4 = 80$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Narrows consideration to g' and g''

You must be convinced the response considers only the functions g' and g'' as being critical to the reasoning.

6-2

Second point: Answer with reason

A correct response must:

- Conclude “increasing”
- Appeal to the signs of g' and g''

It is not possible to earn this point without having earned the first point.

A response that mentions velocity and acceleration must tie velocity to g' and acceleration to g'' . A response that says only “increasing because velocity and acceleration are both negative” earns neither of the two points. It is not clear that the response is considering g' and g'' . A response that calls out all three given values ($g(4)$, $g'(4)$, and $g''(4)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g' and g'' only, both points can be earned.

The phrase “speeding up” will be interpreted as “increasing.”



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g' and g''
- ☐ Answer with reason

Solution:

Because $g'(4) = -3 < 0$, the velocity of the particle is negative. Because $g''(4) = -2 < 0$, the acceleration of the particle is negative. Therefore, because both the velocity and acceleration are negative, the speed of the particle is increasing at time $t = 4$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point: Tangent line equation

This point is earned only for the correct tangent line equation written in any form.

Some common forms:

- $y - 1 = (3(2)^2 - (1)^3)(x - 2)$
- $y - 1 = 11(x - 2)$
- $y = 11x - 21$

f does not earn the point. This is not the equation of a line. The response can go on present one of the correct linear forms, and then would earn the point.

$f(x) = 11(x - 2) + 1$ or equivalent does not earn the point. $f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.

Some responses have the correct tangent line equation written initially, with subsequent errors in simplification. Such responses do not earn the point.

6-2

The point can be earned after pursuing a less fruitful approach (such as solving the differential equation) and then restarting.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{(2,1)} = 3 \cdot 2^2 - 1^3 = 11$$

$$y = 11(x - 2) + 1$$

71. What is the approximation of $\int_2^5 \ln x \, dx$ found by using a left Riemann sum with three subintervals of equal length?

(A) $\ln 2 + \ln 3 + \ln 4$

(B) $\ln 3 + \ln 4 + \ln 5$

(C) $\ln 2 + \ln 3 + \ln 4 + \ln 5$

(D) $3(\ln 2 + \ln 3 + \ln 4)$

**Answer A**

Correct. A Riemann sum with three subintervals of equal length over the interval $[2, 5]$ would have $\Delta x = \frac{5-2}{3} = 1$. The three subintervals would therefore be $[2, 3]$, $[3, 4]$, and $[4, 5]$. A left Riemann sum would use the endpoints $x = 2$, $x = 3$, and $x = 4$. The value of the left Riemann sum would be $(\ln 2 + \ln 3 + \ln 4)\Delta x = \ln 2 + \ln 3 + \ln 4$.

6-2

72.

The function f is continuous and increasing on the interval $[0, 6]$ and $\int_0^6 f(x) \, dx = 8$. Consider Riemann sums with three subintervals of equal length for $\int_0^6 f(x) \, dx$. Which of the following statements must be true?

- I. $2f(0) + 2f(2) + 2f(4) < 8$
- II. $2f(1) + 2f(3) + 2f(5) > 8$
- III. $2f(2) + 2f(4) + 2f(6) > 8$

(A) I and II only

(B) I and III only

(C) II and III only

(D) I, II, and III

**Answer B**

Correct. Since the function is increasing on the interval $[0, 6]$, a left Riemann sum underapproximates $\int_0^6 f(x) \, dx$ and a right Riemann sum overapproximates $\int_0^6 f(x) \, dx$. The sum in statement I is a left Riemann sum and so statement I must be true. The sum in statement III is a right Riemann sum and so statement III must be true. The sum in statement II is a midpoint Riemann sum, but since no information is given about the concavity of the graph of f , it is not possible to determine whether that midpoint Riemann sum is an underapproximation or an overapproximation of the definite integral. Therefore, statement II could be false.

73.

t	0	2	3	6
$f(t)$	12	36	48	42

The function f is continuous on the closed interval $[0, 6]$ and has values given in the table above. What is the approximation to $\int_0^6 f(t) \, dt$ obtained by using a right Riemann sum with the subintervals $[0, 2]$, $[2, 3]$, and $[3, 6]$?

(A) 204

(B) 225

(C) 246

(D) 252

**Answer C**

Correct. A right Riemann sum that approximates the value of the definite integral uses the values of $f(t)$ at the endpoints $t = 2$, $t = 3$, and $t = 6$, with corresponding interval widths 2, 1, and 3.

$$\int_0^6 f(t) \, dt \approx (36)(2) + (48)(1) + (42)(3) = 72 + 48 + 126 = 246$$

6-2

74.  Let f be the function given by $f(x) = x^2e^{-x}$. It is known that $\int_0^1 f(x) \, dx = 0.160603$. If a midpoint Riemann sum with two intervals of equal length is used to approximate $\int_0^1 f(x) \, dx$, what is the absolute difference between the approximation and $\int_0^1 f(x) \, dx$?

(A) 0.003

(B) 0.007

(C) 0.009

(D) 0.154

Answer A

Correct. With two intervals of equal length on $[0, 1]$, the midpoints are $x = \frac{1}{4}$ and $x = \frac{3}{4}$ with $\Delta x = \frac{1}{2}$. The midpoint Riemann sum approximation is $\frac{1}{2} \left[\left(\frac{1}{4}\right)^2 e^{-\frac{1}{4}} + \left(\frac{3}{4}\right)^2 e^{-\frac{3}{4}} \right] = 0.157191$.

The absolute difference is $|0.160603 - 0.157191| = 0.003412$.

75.

x	1	3	4	7	10
$f(x)$	-0.2	-0.1	0.5	0.6	0.8

The continuous function f is known to be increasing for all x . Selected values of f are given in the table above. Let L be the left Riemann sum approximation for $\int_1^{10} f(x) \, dx$ using the four subintervals indicated by the table. Which of the following statements is true?

(A) $L = 1.8$ and is an underestimate for $\int_1^{10} f(x) \, dx$.

(B) $L = 1.8$ and is an overestimate for $\int_1^{10} f(x) \, dx$.

(C) $L = 2.8$ and is an underestimate for $\int_1^{10} f(x) \, dx$.

(D) $L = 2.8$ and is an overestimate for $\int_1^{10} f(x) \, dx$.

Answer C

Correct. The left Riemann sum uses the endpoints $x = 1$, $x = 3$, $x = 4$, and $x = 7$, with corresponding interval lengths 2, 1, 3, and 3. The left Riemann sum is calculated as follows.

$$L = (-0.2)(2) + (-0.1)(1) + (0.5)(3) + (0.6)(3) = 2.8$$

Because f is increasing, a left Riemann sum is an underestimate for $\int_1^{10} f(x) \, dx$.

6-2

76.

x	0	a^2	$3a$	$6a$	$7a$
$f(x)$	1	-1	-3	-7	-9

The continuous function f is decreasing for all x . Selected values of f are given in the table above, where a is a constant with $0 < a < 3$. Let R be the right Riemann sum approximation for $\int_0^{7a} f(x) \, dx$ using the four subintervals indicated by the data in the table. Which of the following statements is true?

(A) $R = (a^2 - 0) \cdot 1 + (3a - a^2) \cdot (-1) + (6a - 3a) \cdot (-3) + (7a - 6a) \cdot (-7)$ and is an underestimate for $\int_0^{7a} f(x) \, dx$.

(B) $R = (a^2 - 0) \cdot 1 + (3a - a^2) \cdot (-1) + (6a - 3a) \cdot (-3) + (7a - 6a) \cdot (-7)$ and is an overestimate for $\int_0^{7a} f(x) \, dx$.

(C) $R = (a^2 - 0) \cdot (-1) + (3a - a^2) \cdot (-3) + (6a - 3a) \cdot (-7) + (7a - 6a) \cdot (-9)$ and is an underestimate ✓
for $\int_0^{7a} f(x) \, dx$.

(D) $R = (a^2 - 0) \cdot (-1) + (3a - a^2) \cdot (-3) + (6a - 3a) \cdot (-7) + (7a - 6a) \cdot (-9)$ and is an overestimate for $\int_0^{7a} f(x) \, dx$.

Answer C

Correct. In a right Riemann sum, the value of the function at the right endpoint of each subinterval and the width of each subinterval is used. Because f is decreasing, a right Riemann sum approximation would be an underestimate for $\int_0^{7a} f(x) \, dx$.

6-2

77.

t	1	4	6	13
$R(t)$	8	20	26	34

The differentiable function R is increasing, and the graph of R is concave down. Values of $R(t)$ at selected values of t are given in the table above. The trapezoidal sum $(4 - 1) \left(\frac{8+20}{2} \right) + (6 - 4) \left(\frac{20+26}{2} \right) + (13 - 6) \left(\frac{26+34}{2} \right)$ approximates

$\int_1^{13} R(t) \, dt$. Which of the following statements is true?

- (A) The trapezoidal sum is an underestimate for $\int_1^{13} R(t) \, dt$ because R is increasing.
- (B) The trapezoidal sum is an overestimate for $\int_1^{13} R(t) \, dt$ because R is increasing.
- (C) The trapezoidal sum is an underestimate for $\int_1^{13} R(t) \, dt$ because the graph of R is concave down. ✓
- (D) The trapezoidal sum is an overestimate for $\int_1^{13} R(t) \, dt$ because the graph of R is concave down.

Answer C

Correct. Since the graph of R is concave down, the area under the curve for $1 \leq t \leq 13$ will be greater than the sum of the areas of the trapezoids.

78.

Which of the following is the midpoint Riemann sum approximation of $\int_4^6 \sqrt{x^3 + 1} \, dx$ using 4 subintervals of equal width?

- (A) $\frac{1}{4} \left(\sqrt{4.25^3 + 1} + \sqrt{4.75^3 + 1} + \sqrt{5.25^3 + 1} + \sqrt{5.75^3 + 1} \right)$
- (B) $\frac{1}{2} \left(\sqrt{4.25^3 + 1} + \sqrt{4.75^3 + 1} + \sqrt{5.25^3 + 1} + \sqrt{5.75^3 + 1} \right)$ ✓
- (C) $\frac{1}{4} \left(\frac{\sqrt{4^3+1} + \sqrt{4.5^3+1}}{2} + \frac{\sqrt{4.5^3+1} + \sqrt{5^3+1}}{2} + \frac{\sqrt{5^3+1} + \sqrt{5.5^3+1}}{2} + \frac{\sqrt{5.5^3+1} + \sqrt{6^3+1}}{2} \right)$
- (D) $\frac{1}{2} \left(\frac{\sqrt{4^3+1} + \sqrt{4.5^3+1}}{2} + \frac{\sqrt{4.5^3+1} + \sqrt{5^3+1}}{2} + \frac{\sqrt{5^3+1} + \sqrt{5.5^3+1}}{2} + \frac{\sqrt{5.5^3+1} + \sqrt{6^3+1}}{2} \right)$

Answer B

Correct. The midpoint sum is formed using the midpoint of each of the four intervals. The width of each interval is $\Delta x = \frac{6-4}{4} = \frac{1}{2}$. The four intervals are therefore $[4, 4.5]$, $[4.5, 5]$, $[5, 5.5]$, and $[5.5, 6]$, with the midpoints at 4.25, 4.75, 5.25, and 5.75. The midpoint sum is found by evaluating $\sqrt{x^3 + 1}$ at each midpoint, adding the four values, and then multiplying the sum by Δx .

6-2

79. Let f be a continuous function such that f changes from increasing to decreasing, and the graph of f changes from concave up to concave down. Which of the following is true about the midpoint Riemann sum approximation for

$$\int_1^3 f(x^2 + x) \, dx \text{ using 4 subintervals of equal width?}$$

(A) $\frac{1}{2} \left(\frac{f(1^2+1)+f(1.5^2+1.5)}{2} + \frac{f(1.5^2+1.5)+f(2^2+2)}{2} + \frac{f(2^2+2)+f(2.5^2+2.5)}{2} + \frac{f(2.5^2+2.5)+f(3^2+3)}{2} \right)$

is the midpoint Riemann sum approximation and underestimates $\int_1^3 f(x^2 + x) \, dx$.

(B) $\frac{1}{2} \left(\frac{f(1^2+1)+f(1.5^2+1.5)}{2} + \frac{f(1.5^2+1.5)+f(2^2+2)}{2} + \frac{f(2^2+2)+f(2.5^2+2.5)}{2} + \frac{f(2.5^2+2.5)+f(3^2+3)}{2} \right)$

is the midpoint Riemann sum approximation and overestimates $\int_1^3 f(x^2 + x) \, dx$.

(C) $\frac{1}{2} (f(1.25^2 + 1.25) + f(1.75^2 + 1.75) + f(2.25^2 + 2.25) + f(2.75^2 + 2.75))$ is the midpoint Riemann sum approximation and underestimates $\int_1^3 f(x^2 + x) \, dx$.

(D) $\frac{1}{2} (f(1.25^2 + 1.25) + f(1.75^2 + 1.75) + f(2.25^2 + 2.25) + f(2.75^2 + 2.75))$ is the midpoint Riemann sum approximation. There is not enough information to determine whether the approximation underestimates or overestimates $\int_1^3 f(x^2 + x) \, dx$. ✓

Answer D

Correct. The midpoint sum is formed using the midpoint of each of the four intervals. The width of each interval is $\Delta x = \frac{3-1}{4} = \frac{1}{2}$. The four intervals are therefore $[1, 1.5]$, $[1.5, 2]$, $[2, 2.5]$, and $[2.5, 3]$, with the midpoints at 1.25, 1.75, 2.25, and 2.75. The midpoint sum is found by evaluating $f(x^2 + x)$ at each midpoint, adding the four values, and then multiplying the sum by Δx .

Because the concavity of the graph of $y = f(x^2 + x)$ on the interval $1 < x < 3$ is unknown, it cannot be determined whether the approximation underestimates or overestimates $\int_1^3 f(x^2 + x) \, dx$.

6-2

80.

t	1	5	8	13
$R(t)$	1	10	17	32

The differentiable function R is increasing, and the graph of R is concave down. Values of $R(t)$ at selected values of t are given in the table above. The trapezoidal sum $(5 - 1) \left(\frac{10+1}{2} \right) + (8 - 5) \left(\frac{17+10}{2} \right) + (13 - 8) \left(\frac{17+32}{2} \right)$ approximates

$\int_1^{13} R(t) \, dt$. Which of the following statements is true?

- (A) The trapezoidal sum is an underestimate for $\int_1^{13} R(t) \, dt$ because R is increasing.
- (B) The trapezoidal sum is an overestimate for $\int_1^{13} R(t) \, dt$ because R is increasing.
- (C) The trapezoidal sum is an underestimate for $\int_1^{13} R(t) \, dt$ because the graph of R is concave down. ✓
- (D) The trapezoidal sum is an overestimate for $\int_1^{13} R(t) \, dt$ because the graph of R is concave down.

Answer C

Correct. Since the graph of R is concave down, the area under the curve for $1 \leq t \leq 13$ will be greater than the sum of the areas of the trapezoids.

6-2

81.

x	a	a^2	a^3	$10a$	$12a$
$f(x)$	5	4	2	1	-1

The continuous function f is decreasing for all x . Selected values of f are given in the table above, where a is a constant with $1 < a < 3$. Let R be the right Riemann sum approximation for $\int_a^{12a} f(x) \, dx$ using the four subintervals indicated by the data in the table. Which of the following statements is true?

- (A) $R = (a^2 - a) \cdot 5 + (a^3 - a^2) \cdot 4 + (10a - a^3) \cdot 2 + (12a - 10a) \cdot 1$ and is an underestimate for

$$\int_a^{12a} f(x) \, dx.$$

- (B) $R = (a^2 - a) \cdot 5 + (a^3 - a^2) \cdot 4 + (10a - a^3) \cdot 2 + (12a - 10a) \cdot 1$ and is an overestimate for

$$\int_a^{12a} f(x) \, dx.$$

- (C) $R = (a^2 - a) \cdot 4 + (a^3 - a^2) \cdot 2 + (10a - a^3) \cdot 1 + (12a - 10a) \cdot (-1)$ and is an underestimate for

$$\int_a^{12a} f(x) \, dx.$$



- (D) $R = (a^2 - a) \cdot 4 + (a^3 - a^2) \cdot 2 + (10a - a^3) \cdot 1 + (12a - 10a) \cdot (-1)$ and is an overestimate for

$$\int_a^{12a} f(x) \, dx.$$

Answer C

Correct. In a right Riemann sum, the value of the function at the right endpoint of each subinterval and the width of each subinterval is used. Because f is decreasing, a right Riemann sum approximation would be an underestimate for $\int_a^{12a} f(x) \, dx$.

6-2

82.

x	2	4	6	7	11
$f(x)$	0.7	1.1	1.9	2.5	5.3

The continuous function f is known to be increasing for all x . Selected values of f are given in the table above. Let L be the left Riemann sum approximation for $\int_2^{11} f(x) \, dx$ using the four subintervals indicated by the table. Which of the following statements is true?

- (A) $L = 13.95$ and is an underestimate for $\int_2^{11} f(x) \, dx$.
- (B) $L = 13.95$ and is an overestimate for $\int_2^{11} f(x) \, dx$.
- (C) $L = 15.5$ and is an underestimate for $\int_2^{11} f(x) \, dx$. ✓
- (D) $L = 15.5$ and is an overestimate for $\int_2^{11} f(x) \, dx$.

Answer C

Correct. The left Riemann sum uses the endpoints $x = 2$, $x = 4$, $x = 6$, and $x = 7$, with corresponding interval lengths 2, 2, 1, and 4. The left Riemann sum is calculated as follows.

$$L = (0.7)(2) + (1.1)(2) + (1.9)(1) + (2.5)(4) = 15.5$$

Because f is increasing, a left Riemann sum is an underestimate for $\int_2^{11} f(x) \, dx$.

83.

Let f be the function given by $f(x) = \sin^2\left(\frac{x}{4}\right)e^{-x^2}$. It is known that $\int_0^{\frac{\pi}{2}} f(x) \, dx = 0.0223$. If a midpoint Riemann sum with two intervals of equal length is used to approximate $\int_0^{\frac{\pi}{2}} f(x) \, dx$, what is the absolute difference between the approximation and $\int_0^{\frac{\pi}{2}} f(x) \, dx$?

- (A) 0.0007 ✓
- (B) 0.0013
- (C) 0.0100
- (D) 0.0107

Answer A

Correct. With two intervals of equal length on $[0, \frac{\pi}{2}]$, the midpoints are $x = \frac{\pi}{8}$ and $x = \frac{3\pi}{8}$ with $\Delta x = \frac{\pi}{4}$. The midpoint Riemann sum approximation is $\frac{\pi}{4} \left[\sin^2\left(\frac{\pi}{8}\right)e^{-\left(\frac{\pi}{8}\right)^2} + \sin^2\left(\frac{3\pi}{8}\right)e^{-\left(\frac{3\pi}{8}\right)^2} \right] = 0.0230$.

The absolute difference is $|0.0223 - 0.0230| = 0.0007$.

6-2

84.  Let f be the function given by $f(x) = \frac{x^3+1}{\sqrt{x^2+x}}$. It is known that f is increasing on the interval $[2, 8]$. Let R_3 be the value of the right Riemann sum approximation for $\int_2^8 f(x) \, dx$ using 3 intervals of equal length. Which of the following statements is true?
- (A) $R_3 = 103.385$ is an underestimate for $\int_2^8 f(x) \, dx$.
- (B) $R_3 = 103.385$ is an overestimate for $\int_2^8 f(x) \, dx$.
- (C) $R_3 = 216.952$ is an underestimate for $\int_2^8 f(x) \, dx$.
- (D) $R_3 = 216.952$ is an overestimate for $\int_2^8 f(x) \, dx$. ✓

Answer D

Correct. The right Riemann sum using 3 intervals of equal length would be $R_3 = 2 \cdot (f(4) + f(6) + f(8)) = 216.952$. Because f is increasing on the interval $[2, 8]$, a right Riemann sum approximation would be an overestimate for $\int_2^8 f(x) \, dx$.

85.  Let f be the function given by $f(x) = \frac{x^2+1}{\sqrt{x+x+5}}$. It is known that f is increasing on the interval $[1, 7]$. Let R_3 be the value of the right Riemann sum approximation for $\int_1^7 f(x) \, dx$ using 3 intervals of equal length. Which of the following statements is true?
- (A) $R_3 = 6.876$ and is an underestimate for $\int_1^7 f(x) \, dx$.
- (B) $R_3 = 6.876$ and is an overestimate for $\int_1^7 f(x) \, dx$.
- (C) $R_3 = 13.133$ and is an underestimate for $\int_1^7 f(x) \, dx$.
- (D) $R_3 = 13.133$ and is an overestimate for $\int_1^7 f(x) \, dx$. ✓

Answer D

Correct. The right Riemann sum using 3 intervals of equal length would be $R_3 = 2 \cdot (f(3) + f(5) + f(7)) = 13.132715$. Because f is increasing on the interval $[1, 7]$, a right Riemann sum approximation would be an overestimate for $\int_1^7 f(x) \, dx$.

6-2

86. The definite integral $\int_0^4 \sqrt{x} \, dx$ is approximated by a left Riemann sum, a right Riemann sum, and a trapezoidal sum, each with 4 subintervals of equal width. If L is the value of the left Riemann sum, R is the value of the right Riemann sum, and T is the value of the trapezoidal sum, which of the following inequalities is true?

(A) $L < \int_0^4 \sqrt{x} \, dx < T < R$

(B) $L < T < \int_0^4 \sqrt{x} \, dx < R$ ✓

(C) $R < \int_0^4 \sqrt{x} \, dx < T < L$

(D) $R < T < \int_0^4 \sqrt{x} \, dx < L$

Answer B

Correct. Because the function $f(x) = \sqrt{x}$ is increasing on the interval $[0, 4]$, the left Riemann sum is an underestimate and the right Riemann sum is an overestimate of the definite integral, so $L < \int_0^4 \sqrt{x} \, dx < R$. Since the graph of f is concave down, the trapezoidal sum is also an underestimate, but it is a closer approximation to the definite integral than the left Riemann sum. Therefore,

$$L < T < \int_0^4 \sqrt{x} \, dx < R.$$

6-2

87.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be the function defined by $f(x) = \cos(\sqrt{x})$.

- (a) Approximate the definite integral $\int_1^3 f(x) \, dx$ using a midpoint Riemann sum with the subintervals $[1, 1.6]$, $[1.6, 2]$, and $[2, 3]$. Show the work that leads to your answer.
- (b) Approximate the definite integral $\int_1^3 f(x) \, dx$ using a trapezoidal sum with the subintervals $[1, 1.6]$, $[1.6, 2]$, and $[2, 3]$. Show the work that leads to your answer.
- (c) It is known that $f'(x) = -\frac{\sin(\sqrt{x})}{2\sqrt{x}}$ and $f''(x) = \frac{\sin(\sqrt{x})}{4x\sqrt{x}} - \frac{\cos(\sqrt{x})}{4x}$. Would a trapezoidal sum approximation for $\int_1^3 f(x) \, dx$ overestimate or underestimate the value of $\int_1^3 f(x) \, dx$? Give a reason for your answer.
- (d) Write $\int_1^3 f(x) \, dx$ as the limit of a right Riemann sum with n subintervals of equal length.

Part A

The first point is earned for correctly presenting the form of the midpoint sum.

The second point is earned for correctly substituting the x -values into the expression for f in the sum. Therefore,

$(0.6) \cdot \cos(\sqrt{1.3}) + (0.4) \cdot \cos(\sqrt{1.8}) + (1) \cdot \cos(\sqrt{2.5})$ earns both points. Numerical answers must be accurate to three places after the decimal point and may be rounded or truncated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ midpoint sum
- ☐ approximation

6-2

Solution:

$$\begin{aligned}\int_1^3 f(x) \, dx &\approx (1.6 - 1) \cdot f(1.3) + (2 - 1.6) \cdot f(1.8) + (3 - 2) \cdot f(2.5) \\ &= (0.6) \cdot \cos(\sqrt{1.3}) + (0.4) \cdot \cos(\sqrt{1.8}) + (1) \cdot \cos(\sqrt{2.5}) \\ &= 0.331 \text{ (or } 0.330)\end{aligned}$$

Part B

The first point is earned for correctly presenting the form of the trapezoidal approximation. The second point is earned for correctly substituting the x -values into the expression for f in the sum. Therefore,

$$\begin{aligned}0.3 \cdot (\cos(\sqrt{1}) + \cos(\sqrt{1.6})) + 0.2 \cdot (\cos(\sqrt{1.6}) + \cos(\sqrt{2})) \\ + 0.5 \cdot (\cos(\sqrt{2}) + \cos(\sqrt{3}))\end{aligned}$$

earns both points. Numerical answers must be accurate to three places after the decimal point and may be rounded or truncated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ trapezoidal approximation
- ☐ approximation

Solution:

$$\begin{aligned}\int_1^3 f(x) \, dx &\approx (1.6 - 1) \cdot \frac{f(1) + f(1.6)}{2} + (2 - 1.6) \cdot \frac{f(1.6) + f(2)}{2} \\ &\quad + (3 - 2) \cdot \frac{f(2) + f(3)}{2} \\ &= 0.3 \cdot (\cos(\sqrt{1}) + \cos(\sqrt{1.6})) + 0.2 \cdot (\cos(\sqrt{1.6}) + \cos(\sqrt{2})) \\ &\quad + 0.5 \cdot (\cos(\sqrt{2}) + \cos(\sqrt{3})) \\ &= 0.342 \text{ (or } 0.341)\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ considers $f''(x)$

6-2

☐ overestimate with reason

Solution:

$$f''(x) > 0 \text{ for } 1 < x < 3.$$

$\Rightarrow$ The graph of f is concave up for $1 < x < 3$.

A trapezoidal sum is an overestimate because the graph of f is concave up for $1 < x < 3$.

(Note: Since the increasing/decreasing behavior of a function does not affect whether a trapezoidal sum is an overestimate or an underestimate, the first derivative f' is not taken into consideration.)

Part D

The first point may be earned as presented in the solution or by using that information to determine that the subintervals are of length $\frac{2}{n}$. A response that produces the correct expression earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\Delta x = \frac{2}{n}$
- ☐ summation notation with limit as $n \rightarrow \infty$
- ☐ summand

Solution:

$$\Delta x = \frac{3-1}{n} = \frac{2}{n}$$

$$\int_1^3 f(x) dx = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n f(1 + k\Delta x) \cdot \Delta x \right) = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \cos \sqrt{1 + \frac{2k}{n}} \cdot \frac{2}{n} \right)$$

Note: The definite integral can be written as any limit of a Riemann sum of the form $\lim_{n \rightarrow \infty} \left(\sum_{k=a}^{n+a-1} \cos \sqrt{1 + \frac{2}{n}(k-a+1)} \cdot \frac{2}{n} \right)$.

6-2

88. The table shown gives the speed $s(t)$, in miles per hour, of a car at selected times t , in hours.

t (hours)	1	1.25	1.5	2.25	3	3.5	4
$s(t)$ (miles per hour)	55	50	65	70	55	60	55

Using a midpoint Riemann sum with subintervals $[1, 1.5]$, $[1.5, 3]$, and $[3, 4]$, what is the approximate distance, in miles, the car traveled from time $t = 1$ to time $t = 4$?

- (A) 170
(B) 175
(C) 180
(D) 190

**Answer D**

Correct. The midpoint Riemann sum is formed using the midpoint of each of the three intervals. The respective midpoints are 1.25, 2.25, and 3.5. The midpoint sum is therefore the following.

$$f(1.25)(1.5 - 1) + f(2.25)(3 - 1.5) + f(3.5)(4 - 3) = 50(0.5) + 70(1.5) + 60(1) = 25 + 105 + 60 = 190$$

6-2

89. Let f be a continuous function defined on the interval $0 \leq x \leq 7$. The function f and its derivatives have the properties indicated in the table shown, where DNE indicates the x -values at which the derivatives of f do not exist.

x	$x = 0$	$0 < x < 2$	$x = 2$	$2 < x < 4$	$x = 4$	$4 < x < 7$	$x = 7$
$f(x)$	-1	Negative	-4	Negative	-2	Negative	0
$f'(x)$	DNE	Negative	DNE	Positive	0	Positive	DNE
$f''(x)$	DNE	Positive	DNE	Negative	0	Positive	DNE

Part A

On what open intervals contained in $0 < x < 7$, if any, is the graph of f both increasing and concave down? Give a reason for your answer.

Part B

Find the average rate of change of f over the interval $0 \leq x \leq 4$. Can the Mean Value Theorem be applied to f on the interval $0 \leq x \leq 4$ to guarantee a value of c , for $0 < c < 4$, such that $f'(c)$ is equal to this average rate of change? Give a reason for your answer.

Part C

Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^7 f(x) dx$.

Part D

Let g be the function defined by $g(x) = x \cdot f(x)$. Is g increasing or decreasing at $x = 1$? Give a reason for your answer.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned only by an answer of $2 < x < 4$ or this interval including one or both endpoints.
- Special case: A response of “ f is increasing on $2 < x < 4$ and $4 < x < 7$ because $f'(x) > 0$ there” earns **P1**.
- Special case: A response of “ f is concave down on $2 < x < 4$ because $f''(x) < 0$ there” earns **P1**.

Model Solution:

f is increasing and concave down on the interval $2 < x < 4$.

6-2



0	1
---	---

The response accurately includes the criteria below.

- Answer

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response must have earned **P1** to be eligible for **P2**.
- To earn **P2**, a response must correctly cite both $f'(x) > 0$ and $f''(x) < 0$.
- Special case: A response of “ f is increasing on $2 < x < 4$ and $4 < x < 7$ because $f'(x) > 0$ there” does not earn **P2**.
- Special case: A response of “ f is concave down on $2 < x < 4$ because $f''(x) < 0$ there” does not earn **P2**.

Model Solution:

The graph of function f is increasing on an interval when $f'(x) > 0$ and concave down on an interval when $f''(x) < 0$.



0	1
---	---

The response accurately includes the criteria below.

- Reason

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P3**, a response must include a difference and a quotient. Any of the following is sufficient to earn **P3**:

$$\frac{f(4)-f(0)}{4-0} = -\frac{1}{4}, \frac{-2-(-1)}{4}, \text{ or } \frac{-1}{4-0}.$$

Model Solution:

$$\frac{f(4)-f(0)}{4-0} = \frac{-2-(-1)}{4} = -\frac{1}{4}$$



0	1
---	---

The response accurately includes the criteria below.

- Average rate of change

6-2

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response does not need to have earned **P3** to be eligible for **P4**.
- To earn **P4**, a response must note that f is not differentiable and conclude “no.”
- To earn **P4**, a response need not explicitly name the Mean Value Theorem, but if a theorem is named, it must be correct.

Model Solution:

No, the Mean Value Theorem cannot be applied to f on the interval $0 \leq x \leq 4$ because f is not differentiable at $x = 2$.



0	1
---	---

The response accurately includes the criteria below.

- Answer with reason

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- Read “=” as “ $\approx$ ” for **P5**.
- The form of a left Riemann sum includes three terms, each of which includes a product of two factors. To earn **P5**, at least five of the six factors must be correct.
- A response of $(f(0))(2) + (f(2))(2) + (f(4))(3)$ earns **P5**.
- A response of $(-1)(2) + (-4)(2) + (-2)(3)$ earns **P5**.
- An unsupported answer of -16 does not earn **P5**.
- A completely correct right Riemann sum with accompanying work (e.g., $(-4)(2) + (-2)(2) + (0)(3) = -12$) earns **P5**. A response with any errors or missing factors in the right Riemann sum does not earn **P5**.

Model Solution:

$$\int_0^7 f(x)dx \approx f(0)(2-0) + f(2)(4-2) + f(4)(7-4)$$



0	1
---	---

The response accurately includes the criteria below.

- Form of left Riemann sum

Part C Point 6

6-2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response of $(f(0))(2) + (f(2))(2) + (f(4))(3)$ is eligible to earn **P6** with correct values for $f(0)$, $f(2)$, and $f(4)$ imported from the table. If any of the six factors is incorrect, the response does not earn **P6**.
- A response of $(-1)(2) + (-4)(2) + (-2)(3)$ earns **P6**.
- A response that includes $(-1)(2) + (-4)(2) + (-2)(3)$ or equivalent banks **P6** (i.e., subsequent errors in simplification will not be considered in scoring for **P6**).
- An unsupported answer of -16 does not earn **P6**.
- A right Riemann sum (e.g., $(-4)(2) + (-2)(2) + (0)(3) = -12$) does not earn **P6**.

Model Solution:

$$(-1)(2) + (-4)(2) + (-2)(3) = -16$$



0	1
---	---

The response accurately includes the criteria below.

- Answer with supporting work

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P7** is earned for considering either $g'(x)$ or $g'(1)$.

Model Solution:

$$g'(x) = 1 \cdot f(x) + x \cdot f'(x)$$

$$g'(1) = f(1) + 1 \cdot f'(1)$$



0	1
---	---

The response accurately includes the criteria below.

- Considers $g'(x)$

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P8** is earned with either $g'(x) = f(x) + x \cdot f'(x)$ or $g'(1) = f(1) + 1 \cdot f'(1)$.

6-2

Model Solution:

$$g'(x) = 1 \cdot f(x) + x \cdot f'(x)$$

$$g'(1) = f(1) + 1 \cdot f'(1)$$



0	1
---	---

The response accurately includes the criteria below.

- $g'(x)$ or $g'(1)$.

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response must have earned **P7** and **P8** to be eligible for **P9**.
- An eligible response that states either “decreasing because $g'(1) < 0$ ” or “decreasing because $g'(x) < 0$ for $0 < x < 2$ ” earns **P9**.
- A response that states “ $g(1)$ is decreasing” does not earn **P9**.
- A response that estimates negative values for $f(1)$ and/or $f'(1)$ using the table and reaches a correct conclusion earns **P9**.

Model Solution:

Because $g'(1) < 0$ and g' is continuous on $0 < x < 2$, g is decreasing on an interval containing $x = 1$.



0	1
---	---

The response accurately includes the criteria below.

- Answer with reason